

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	20
Flg Calc [Int P]: FLANGE	21
Flg Calc [Int P]: New Flange	27
Internal Pressure Calculations:	33
Element: CH. FRONT CAP	34
Element: CH. SHELL	36
Element: SHELL	38
Element: REAR CH. SHELL	40
Element: CH. REAR CAP	42
Shop Pressure Test Results:	44
ASME TS Calc: Pressure Test	46
External Pressure Calculations:	57
Element and Detail Weights:	58
Nozzle Flange MAWP:	60
Wind Load Calculation:	62
Earthquake Load Calculation:	64
Center of Gravity Calculation:	66
Saddle Calcs: Operating Case:	67
Saddle Calcs: Test Case:	83
Saddle Calcs: Ext. Pressure:	97
Nozzle Calcs.: CH. IN Noz N3	109
Nozzle Calcs.: CH. OUT Noz N4	121
Nozzle Calcs.: SHELL IN Noz N1	133
Nozzle Calcs.: SHELL OUT Noz N	144
Nozzle Calcs.: Vent Noz V	155
Nozzle Calcs.: Drain Noz D	158
ASME TS Calc: DESIGN CASE	161
ASME TS Calc: 100% EOR	179
ASME TS Calc: 110% EOR	185
ASME TS Calc: 10% SOR	191
ASME TS Calc: START-UP	197
ASME TS Calc: SHUT-DOWN	203
ASME TS Calc: NO SHELL SIDE P	209



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-31011.pvdb

Date of Analysis : May 21, 2026 4:41pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 273.000 mm OD
 Vessel Design Length, Tangent to Tangent 296.10 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 180 °C
 Channel Side Design Temperature 65 °C
 Shell Side Design Pressure 5.609 kgf/cm²
 Channel Side Design Pressure 10.197 kgf/cm²
 Shop Shell Side Test Pressure 8.413 kgf/cm²
 Shop Channel Side Test Pressure 15.296 kgf/cm²
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-106 B	...	...	K03006	
Head	SA-234 WPB	...	...	K03006	
Flange	SA-266 4	...	...	K03017	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-106 B	...	...	K03006	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 4	...	...	K03017	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH. FRONT CAP	10.197	0.00	59.000	3.0000	No	No
CH. SHELL	10.223	0.00	56.000	3.0000	N/A	No
LEFT CH. FLG	10.223	0.00	38.000	3.0000	N/A	No
SHELL	5.634	1.05	56.000	3.0000	N/A	No
RIGHT CH. FLG	10.223	1.05	37.000	3.0000	N/A	No
REAR CH. SHELL	10.223	1.05	56.000	3.0000	N/A	No
CH. REAR CAP	10.197	1.05	59.000	3.0000	No	No

Liquid Level: 25.45 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	0.0	9.3	9.3	4.5	...	1.00	1.00
Cylinder	23.2	23.2	9.3	9.3	4.5	...	1.00	1.00
Body Flg	28.5	4.8	28.0	28.0	20.6	19.2	1.00	1.00
Cylinder	263.6	231.2	9.3	9.3	4.5	5.2	1.00	1.00



Body Flg	272.9	4.8	28.0	28.0	20.7	19.9	1.00	1.00
Cylinder	296.1	23.2	9.3	9.3	4.5	3.8	1.00	1.00
Ellipse	296.1	0.0	9.3	9.3	4.5	3.7	1.00	1.00

Local Stress Analysis Results:

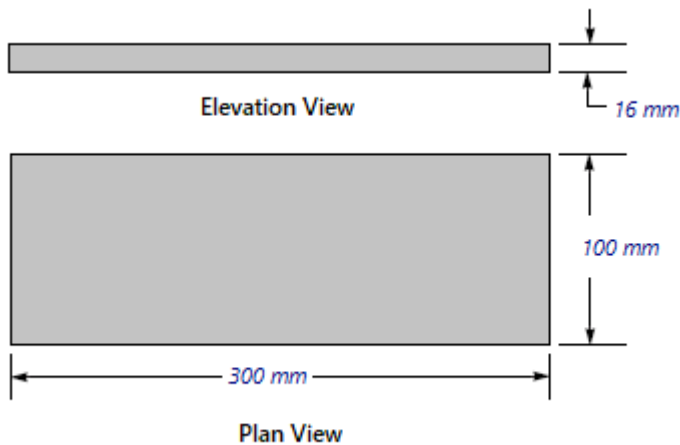
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
CH. IN Noz N3	WRC-297	0.180	Shell	Passed
CH. OUT Noz N4	WRC-297	0.180	Shell	Passed
SHELL IN Noz N	WRC-297	0.099	Shell	Passed
SHELL OUT Noz	WRC-297	0.099	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	100.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	390.000 mm
Wear Pad Width	165.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	490.000 mm
Baseplate Length	300.000 mm
Baseplate Thickness	16.000 mm
Baseplate Width	100.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	2

Baseplate Sketch



Maximum Tensile Bolt Load 30. kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	806.20 kgf
Maximum Transverse Saddle Shear Load	98.72 kgf
Maximum Longitudinal Saddle Shear Load	55.49 kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	518.82 kgf
Maximum Transverse Saddle Shear Load	32.58 kgf
Maximum Longitudinal Saddle Shear Load	6.19 kgf

Weights:



Fabricated	- Bare w/o removable internals	565.7	kg
Shop Test	- Fabricated + water (full)	703.4	kg
Shipping	- Fab. + rem. intls.+ shipping app.	565.7	kg
Erected	- Fab. + rem. intls.+ insul. (etc)	565.7	kg
Empty	- Fab. + intls. + details + wghts.	565.7	kg
Operating	- Empty + operating liquid (no ca)	699.1	kg

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
CH. IN Noz N3	56	...	...	OK	4.50	OK	No Calc[*]
CH. OUT Noz N4	56	...	...	OK	4.50	OK	No Calc[*]
SHELL IN Noz N1	56	...	...	OK	4.50	OK	No Calc[*]
SHELL OUT Noz N	56	...	...	OK	4.50	OK	No Calc[*]
Vent Noz V	56	...	...	OK	4.50	OK	No Calc[*]
Drain Noz D	56	...	...	OK	4.50	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 56 Nozzle : SHELL IN Noz N1 [Shellside]
 Minimum MAWP Nozzles : 56 Nozzle : CH. IN Noz N3 [Tubside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	CH. IN Noz N3	116.000	-61.563	97.700
20	CH. OUT Noz N4	116.000	61.563	97.700
50	SHELL IN Noz N1	2488.500	90.000	97.700
50	SHELL OUT Noz N2	454.500	-90.000	97.700
80	Vent Noz V	2845.000	90.000	97.700
80	Drain Noz D	2845.000	-90.000	97.700

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Thk mm	Cut Length mm	Flg Class
CH. IN Noz N3	2.000 in	160	WNF	2.375	8.738	160.32	10.00	239.08	300
CH. OUT Noz N4	2.000 in	160	WNF	2.375	8.738	160.32	10.00	239.08	300
SHELL IN Noz N1	2.000 in	160	WNF	2.375	8.738	160.32	10.00	216.40	600
SHELL OUT Noz N2	2.000 in	160	WNF	2.375	8.738	160.32	10.00	216.40	600
Vent Noz V	2.000 in	160	WNF	2.375	8.738	160.32	10.00	216.40	300
Drain Noz D	2.000 in	160	WNF	2.375	8.738	160.32	10.00	216.40	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
CH. IN Noz	SA-106 B	3.912	10.000	8.000	10.000	...
CH. OUT Noz	SA-106 B	3.912	10.000	8.000	10.000	...
SHELL IN No	SA-106 B	3.912	10.000	8.000	10.000	...
SHELL OUT N	SA-106 B	3.912	10.000	8.000	10.000	...
Vent Noz V	SA-106 B	3.912	10.000	8.000	10.000	...
Drain Noz D	SA-106 B	3.912	10.000	8.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
CH. IN Noz N3	11.600	-61.6	203.50	0.00	CH. SHELL
CH. OUT Noz N4	11.600	61.6	203.50	0.00	CH. SHELL
SHELL IN Noz N1	248.850	90.0	203.50	0.00	SHELL
SHELL OUT Noz N2	45.450	-90.0	203.50	0.00	SHELL
Vent Noz V	284.500	90.0	203.50	0.00	REAR CH. SHELL
Drain Noz D	284.500	-90.0	203.50	0.00	REAR CH. SHELL

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 9.3mm THK X 254.5mm ID X 0.0cm	SA-234 WPB
2	CYLINDER: 9.3mm THK X 254.5mm ID X 23.2cm	SA-106 B
2	WELD NECK FLANGE: 28.0mm THK X 400.0mm OD	SA-266 4
1	CYLINDER: 9.3mm THK X 254.5mm ID X 231.2cm	SA-106 B
12	BAFFLES: 248.0mm DIA X 6.0mm THK	SA-516 70
4	TIE RODS: 12mm DIA X 2000mm	SA-193 B7
2	TUBESHEET: 400.0mm X 40.0mm THK	SA-266 4
52	TUBES: 2394.0mm X 19.0mm DIA X 2.1mm THK	SA-179
2	SADDLE: 100.0mm X 120 DEG	SA-516 70
2	CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)	SA-105
2	GASKET: 306.5mm OD X 260.5mm ID	...
32	BODY FLANGE BOLTS: 20.0mm DIA	SA-193 B7
32	NUTS FOR BODY FLANGE BOLTS: 20.0mm DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
SHELL	[8]	B	-29	-48	-29	0.481	9.270	1.00	No
Left Saddle	[24]	B	-29	-29	-29	1.000	9.270	1.00	No
Right Saddle	[24]	B	-29	-29	-29	1.000	9.270	1.00	No
SHELL IN Noz N	[1]	B	-8	-104	-29	0.025	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
SHELL OUT Noz	[1]	B	-8	-104	-29	0.025	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Tubesheet: SS	[13]	B	-29	-104	-29	0.228	10.000	1.00	No

Warmest MDMT: -8 -29

LEFT CH. FLG	[11]	B	-29	-48	-29	0.655	9.270	1.00	No
RIGHT CH. FLG	[11]	B	-29	-48	-29	0.655	9.270	1.00	No
CH. FRONT CAP	[10]	B	-29	-48	-29	0.631	9.270	1.00	No
CH. FRONT CAP	[7]	B	-29	-48	-29	0.660	9.270	1.00	No
CH. SHELL	[8]	B	-29	-104	-29	0.317	9.270	1.00	No
REAR CH. SHELL	[8]	B	-29	-104	-29	0.317	9.270	1.00	No
CH. REAR CAP	[10]	B	-29	-48	-29	0.631	9.270	1.00	No
CH. REAR CAP	[7]	B	-29	-48	-29	0.660	9.270	1.00	No
CH. IN Noz N3	[1]	B	-8	-104	-29	0.045	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
CH. OUT Noz N4	[1]	B	-8	-104	-29	0.045	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Vent Noz V	[1]	B	-8	-104	-29	0.045	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Drain Noz D	[1]	B	-8	-104	-29	0.045	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Tubesheet: CS	[14]	B	-29	-104	-29	0.228	10.000	1.00	No
Bolting	[21]		-48						
Bolting	[21]		-48						

Warmest MDMT: -8 -48

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	-28.9	Pass
Channel/Tube	-48.0	-28.9	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	5.6086	kgf/cm ²
Channel Side Design Pressure	10.197	kgf/cm ²
Shell Side Design Temperature	180.0	°C
Channel Side Design Temperature	65.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	8.4129	kgf/cm ²
User defined Test Pres., Channel Side	15.296	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	User Entered Pressure
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
-------------	------



0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y

Consider MAP New and Cold in Noz. Design N

Consider External Loads for Nozzle Des. Y

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No

Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	CH. FRONT CAP
Distance "FROM" to "TO"	0 cm
Element Outside Diameter	273 mm
Element Thickness	9.27 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	9.27 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	10.197 kgf/cm ²
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-234 WPB
Allowable Stress, Ambient	1203.2 kgf/cm ²
Allowable Stress, Operating	1203.2 kgf/cm ²



Allowable Stress, Hydrotest	2202.5	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2314.7	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. & wld. fittings	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element#	2/7	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	CH. SHELL	
Distance "FROM" to "TO"	23.2	cm
Element Outside Diameter	273	mm
Element Thickness	9.27	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	9.27	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-106 B	
Allowable Stress, Ambient	1203.2	kgf/cm ²
Allowable Stress, Operating	1203.2	kgf/cm ²
Allowable Stress, Hydrotest	2202.5	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2314.7	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	25.446	cm
Liquid Density	999.54	kg/m ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	CH. IN Noz N3	
Dist. from "FROM" Node / Offset dist	11.6	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	-61.563099	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7.8059	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	CH. OUT Noz N4	
Dist. from "FROM" Node / Offset dist	11.6	cm



Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	61.563099	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7.8059	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	3/7	
Element From Node	30	
Element To Node	50	
Element Type	Flange	
Description	LEFT CH. FLG	
Distance "FROM" to "TO"	4.8	cm
Flange Inside Diameter	254.46	mm
Element Thickness	28	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 4	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2294.2	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2375.9	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03017	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	25.446	cm
Liquid Density	999.54	kg/m ³

Element#	4/7	
Element From Node	50	
Element To Node	70	
Element Type	Cylinder	
Description	SHELL	
Distance "FROM" to "TO"	231.2	cm
Element Outside Diameter	273	mm
Element Thickness	9.27	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	9.27	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5.6086	kgf/cm ²
Design Temperature Internal Pressure	180	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	180	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-106 B	
Allowable Stress, Ambient	1203.2	kgf/cm ²
Allowable Stress, Operating	1203.2	kgf/cm ²
Allowable Stress, Hydrotest	2202.5	kgf/cm ²



Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2135.2	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	49	cm
Width of Saddle	100	mm
Height of Saddle at Bottom	390	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	165	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	50	
Detail Type	Saddle	
Detail ID	Right Saddle	
Dist. from "FROM" Node / Offset dist	191	cm
Width of Saddle	100	mm
Height of Saddle at Bottom	390	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	165	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	25.446	cm
Liquid Density	999.54	kg/m ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	SHELL IN Noz N1	
Dist. from "FROM" Node / Offset dist	216.4	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7.8059	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	SHELL OUT Noz N2	



Dist. from "FROM" Node / Offset dist	13	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7.8059	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	5/7	
Element From Node	70	
Element To Node	80	
Element Type	Flange	
Description	RIGHT CH. FLG	
Distance "FROM" to "TO"	4.8	cm
Flange Inside Diameter	254.46	mm
Element Thickness	28	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 4	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2294.2	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2375.9	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03017	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid 70	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	25.446	cm
Liquid Density	999.54	kg/m ³

Element#	6/7	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	REAR CH. SHELL	
Distance "FROM" to "TO"	23.2	cm
Element Outside Diameter	273	mm
Element Thickness	9.27	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	9.27	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-106 B	
Allowable Stress, Ambient	1203.2	kgf/cm ²
Allowable Stress, Operating	1203.2	kgf/cm ²



Allowable Stress, Hydrotest	2202.5	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2314.7	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid 80	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	25.446	cm
Liquid Density	999.54	kg/m ³

Element From Node	80	
Detail Type	Nozzle	
Detail ID	Vent Noz V	
Dist. from "FROM" Node / Offset dist	11.6	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	9.1856	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	Drain Noz D	
Dist. from "FROM" Node / Offset dist	11.6	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	-90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	9.1856	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	7/7	
Element From Node	90	
Element To Node	100	
Element Type	Elliptical	
Description	CH. REAR CAP	
Distance "FROM" to "TO"	0	cm
Element Outside Diameter	273	mm
Element Thickness	9.27	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	9.27	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-234 WPB	
Allowable Stress, Ambient	1203.2	kgf/cm ²
Allowable Stress, Operating	1203.2	kgf/cm ²
Allowable Stress, Hydrotest	2202.5	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2314.7	kgf/cm ²

PV Elite 28 Licensee: SL User
FileName : 401-E-31011
Input Echo:

Page 19 of 214
Step: 1 4:41pm May 21,2026



HEXAGON

UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03006
Product Form	Smls. & wld. fittings
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
CH. FRONT CAP		...	...	...	...
CH. SHELL		23.2	...	23.2	...
LEFT CH. FLG		28.45	...	4.8	...
SHELL		263.65	...	231.2	...
RIGHT CH. FLG		272.9	...	4.8	...
REAR CH. SHELL		296.1	...	23.2	...
CH. REAR CAP		296.1	...	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

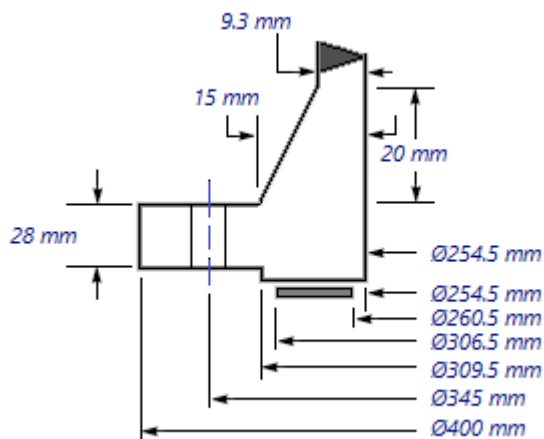


Flange Input Data Values

Description: FLANGE :

LEFT CH. FLG

Flange Type	Integral Weld Neck		
Design Pressure	P	10.22	kgf/cm ²
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	254.460	mm
Flange Outside Diameter	A	400.000	mm
Flange Thickness	t	28.0000	mm
Thickness of Hub at Small End	go	9.2700	mm
Thickness of Hub at Large End	gl	15.0000	mm
Length of Hub	h	20.0000	mm
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	345.000	mm
Nominal Bolt Diameter	a	20.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		16	
Flange Face Outside Diameter	Fod	309.460	mm
Flange Face Inside Diameter	Fid	254.460	mm
Flange Facing Sketch		Sketch 1b	
Gasket Outside Diameter	Go	306.460	mm
Gasket Inside Diameter	Gi	260.460	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	26.4600	mm
Width of Partition Gasket	tp	6.0000	mm
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	534.33	kgf/cm ²





ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (10.22 \cdot (254.46/2 + 3.0)) / (1407.13 \cdot 1.0 - 0.6 \cdot 10.22) + Ca \\ &= 3.9503 \text{ mm} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 6.27) / (130.23 + 0.6 \cdot 6.27) \\ &= 65.845 \text{ kgf/cm}^2 \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 9.27) / (127.23 + 0.6 \cdot 9.27) \\ &= 98.230 \text{ kgf/cm}^2 \end{aligned}$$

Corroded Flange ID,	Bcor = B+2*ci	260.460 mm
Corroded Large Hub,	g1Cor = g1-ci	12.000 mm
Corroded Small Hub,	g0Cor = go-ci	6.270 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	30.270 mm

Gasket Contact Width,	N = Self-Energizing	0.000 mm
Basic Gasket Width,	bo = N/2	0.000 mm
Effective Gasket Width,	b = bo	0.000 mm
Gasket Reaction Diameter,	G = Go (Self-Energizing)	306.460 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 306.46^2 \cdot 10.223 \\ &= 7540.641 \text{ kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P \\ &= 2 \cdot 0.0 \cdot \pi \cdot 306.46 \cdot 0.0 \cdot 10.22 \\ &\quad + 2 \cdot 26.46 \cdot 3.0 \cdot 3.75 \cdot 10.2228 \\ &= 60.862 \text{ kgf} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 260.46^2 \cdot 10.2228 \\ &= 5446.816 \text{ kgf} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 7541 - 5447 \\ &= 2093.824 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(7541 + 61 + 0, 0) \\ &= 7601.503 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 0.0 \cdot 0.0 \cdot \pi \cdot 306.46 + 534.33 \cdot 3.0 \cdot 26.46 \\ &= 424.153 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\ &= \max((7602 + 0 + 4 \cdot 0 / 306.46) / 1754, 424 / 1754) \\ &= 4.334 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t / (m + 0.5) \\ &= 2 \cdot 20.0 + 6 \cdot 28.0 / (0.0 + 0.5) \\ &= 376.000 \text{ mm} \end{aligned}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 345.0 \cdot \sin(\pi / 16)$$

$$= 67.306 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{67.306 / (2 \cdot 20.0 + 28.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4.334	34.728	
Radial Distance between Hub and Bolts:	28.200	30.270	
Radial Distance between Bolts and Edge:	22.500	27.500	
Circ. Spacing between the Bolts:	46.800	67.306	376.000

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(A_m + A_b) / 2$$

$$= 1753.82(4.3343 + 34.7282) / 2$$

$$= 34254.17 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 7602 - 7541$$

$$= 60.86 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (345.0 - 306.46) / 2$$

$$= 19.2700 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((345.0 - 260.46) / 2 + 19.27)$$

$$= 30.7700 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (345.0 - 260.46 - 12.0) / 2$$

$$= 36.2700 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	5447.	36.2700	1.0000	198.
Face Pressure, Mt	2094.	30.7700	1.0000	64.
Gasket Load, Mg	61.	19.2700	1.0000	1.
Gasket Seating, Matm	34254.	19.2700	1.0000	660.

Total Moment for Operation, Mo 263. kgf-m.
 Total Moment for Gasket seating, Mg 660. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 40.411 mm
 Hub Ratio, h/h0 = HL / H0 0.495
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.914

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.827
 Factor V per 4.16.5 0.239
 Factor f per 4.16.5 1.123

Stress factor from Table 4.16.4 [K]:

$$= A / B$$



$$= 400.0/260.46$$

$$= 1.536$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K)) / (K^2 - 1)) / (K - 1))$$

$$= (0.66845 + (5.7169(1.536^2 \cdot \log_{10}(1.536)) / (1.536^2 - 1)) / (1.536 - 1))$$

$$= 4.699$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.536^2(1 + 8.55246 \cdot \log_{10}(1.536)) - 1) / ((1.0472 + 1.9448 \cdot 1.536^2) \cdot (1.536 - 1))$$

$$= 1.695$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.536^2(1 + 8.55246 \cdot \log_{10}(1.536)) - 1) / (1.36136(1.536^2 - 1) \cdot (1.536 - 1))$$

$$= 5.164$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.536^2 + 1) / (1.536^2 - 1)$$

$$= 2.472$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 5.164 \cdot 6.27^2 \cdot 40.411 / 0.239$$

$$= 34261.242 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.827 / 40.411$$

$$= 0.020 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d)$$

$$= (28.0 \cdot 0.02 + 1) / (1.695 + 28.0^3 / 34261.242)$$

$$= 1.569$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.1229 \cdot 263 / 260.46) / (1.5687 \cdot 12.0^2)$$

$$= 502.26 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.1229 \cdot 660 / 260.46) / (1.5687 \cdot 12.0^2)$$

$$= 1259.82 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.7641 \cdot 263 / 260.46) / (1.5687 \cdot 28.0^2)$$

$$= 144.92 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.7641 \cdot 660 / 260.46) / (1.5687 \cdot 28.0^2)$$

$$= 363.52 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.6994 \cdot 263 / (28.0^2 \cdot 260.46)) - 2.4722 \cdot 145$$

$$= 247.34 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.6994 \cdot 660 / (28.0^2 \cdot 260.46)) - 2.4722 \cdot 364$$

$$= 620.40 \text{ kgf/cm}^2$$



Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (502 + \max(145, 247)) / 2$$

$$= 374.80 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1260 + \max(364, 620)) / 2$$

$$= 940.11 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 7602 / 34.7282$$

$$= 218.89 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (424 / 34.7282)$$

$$= 12.21 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	502.26	2110.70	1259.82 2110.70
Radial Flange	144.92	1407.13	363.52 1407.13
Tangential Flange	247.34	1407.13	620.40 1407.13
Maximum Average	374.80	1407.13	940.11 1407.13
Bolting	218.89	1753.82	12.21 1753.82

Minimum Required Flange Thickness 20.574 mm

Estimated M.A.W.P. (Operating) 38 kgf/cm²

Estimated Finished Weight of Flange at given Thk. 17.8 kg

Estimated Unfinished Weight of Forging at given Thk 27.8 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 695.8 / 1.0541 \cdot 100000.023 \cdot 0.239 / (1.091 \cdot 2067026 \cdot 6.27^2 \cdot 40.411 \cdot 0.3)$$

$$= 0.766 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 277.4 / 1.0541 \cdot 100000.023 \cdot 0.239 / (1.091 \cdot 2041395 \cdot 6.27^2 \cdot 40.411 \cdot 0.3)$$

$$= 0.309 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 660.1 / 1.0 \cdot 100000.023 \cdot 0.239 / (1.569 \cdot 2067026 \cdot 6.27^2 \cdot 40.411 \cdot 0.3)$$

$$= 0.533 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 263.2 / 1.0 \cdot 100000.023 \cdot 0.239 / (1.569 \cdot 2041395 \cdot 6.27^2 \cdot 40.411 \cdot 0.3)$$

$$= 0.215 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.655, Temperature Reduction per Fig. UCS 66.1 = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:



Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	390.69	2110.70	944.29	2110.70
Radial Flange	185.68	1407.13	448.77	1407.13
Tangential Flange	173.39	1407.13	419.08	1407.13
Maximum Average	288.18	1407.13	696.53	1407.13
Bolting	218.89	1753.82	12.21	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 18.085 mm

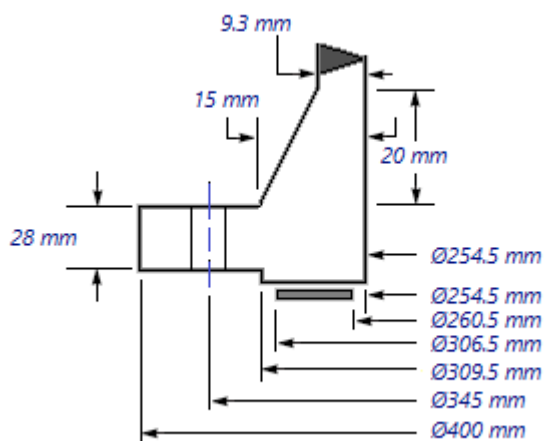
PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Flange Input Data Values

Description: New Flange :

RIGHT CH. FLG

Flange Type	Integral Weld Neck		
Design Pressure	P	10.22	kgf/cm ²
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	254.460	mm
Flange Outside Diameter	A	400.000	mm
Flange Thickness	t	28.0000	mm
Thickness of Hub at Small End	go	9.2700	mm
Thickness of Hub at Large End	gl	15.0000	mm
Length of Hub	h	20.0000	mm
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	345.000	mm
Nominal Bolt Diameter	a	20.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		16	
Flange Face Outside Diameter	Fod	309.460	mm
Flange Face Inside Diameter	Fid	254.460	mm
Flange Facing Sketch		Sketch 1b	
Gasket Outside Diameter	Go	306.460	mm
Gasket Inside Diameter	Gi	260.460	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	260.3600	mm
Width of Partition Gasket	tp	6.0000	mm
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	534.33	kgf/cm ²





ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (10.22 \cdot (254.46/2 + 3.0)) / (1407.13 \cdot 1.0 - 0.6 \cdot 10.22) + Ca \\ &= 3.9503 \text{ mm} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 6.27) / (130.23 + 0.6 \cdot 6.27) \\ &= 65.845 \text{ kgf/cm}^2 \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 9.27) / (127.23 + 0.6 \cdot 9.27) \\ &= 98.230 \text{ kgf/cm}^2 \end{aligned}$$

Corroded Flange ID,	Bcor = B+2*ci	260.460 mm
Corroded Large Hub,	g1Cor = g1-ci	12.000 mm
Corroded Small Hub,	g0Cor = go-ci	6.270 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	30.270 mm

Gasket Contact Width,	N = Self-Energizing	0.000 mm
Basic Gasket Width,	bo = N/2	0.000 mm
Effective Gasket Width,	b = bo	0.000 mm
Gasket Reaction Diameter,	G = Go (Self-Energizing)	306.460 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 306.46^2 \cdot 10.223 \\ &= 7540.641 \text{ kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P \\ &= 2 \cdot 0.0 \cdot \pi \cdot 306.46 \cdot 0.0 \cdot 10.22 \\ &\quad + 2 \cdot 260.36 \cdot 3.0 \cdot 3.75 \cdot 10.2228 \\ &= 598.863 \text{ kgf} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 260.46^2 \cdot 10.2228 \\ &= 5446.816 \text{ kgf} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 7541 - 5447 \\ &= 2093.824 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(7541 + 599 + 0, 0) \\ &= 8139.504 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 0.0 \cdot 0.0 \cdot \pi \cdot 306.46 + 534.33 \cdot 3.0 \cdot 260.36 \\ &= 4173.564 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\ &= \max((8140 + 0 + 4 \cdot 0 / 306.46) / 1754, 4174 / 1754) \\ &= 4.641 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t / (m + 0.5) \\ &= 2 \cdot 20.0 + 6 \cdot 28.0 / (0.0 + 0.5) \\ &= 376.000 \text{ mm} \end{aligned}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 345.0 \cdot \sin(\pi / 16)$$

$$= 67.306 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{67.306 / (2 \cdot 20.0 + 28.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4.641	34.728	
Radial Distance between Hub and Bolts:	28.200	30.270	
Radial Distance between Bolts and Edge:	22.500	27.500	
Circ. Spacing between the Bolts:	46.800	67.306	376.000

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1753.82(4.641 + 34.7282) / 2$$

$$= 34523.17 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 8140 - 7541$$

$$= 598.86 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (345.0 - 306.46) / 2$$

$$= 19.2700 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((345.0 - 260.46) / 2 + 19.27)$$

$$= 30.7700 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (345.0 - 260.46 - 12.0) / 2$$

$$= 36.2700 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	5447.	36.2700	1.0000	198.
Face Pressure, Mt	2094.	30.7700	1.0000	64.
Gasket Load, Mg	599.	19.2700	1.0000	12.
Gasket Seating, Matm	34523.	19.2700	1.0000	665.

Total Moment for Operation, Mo 274. kgf-m.
 Total Moment for Gasket seating, Mg 665. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 40.411 mm
 Hub Ratio, h/h0 = HL / H0 0.495
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.914

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.827
 Factor V per 4.16.5 0.239
 Factor f per 4.16.5 1.123

Stress factor from Table 4.16.4 [K]:

$$= A / B$$



$$= 400.0/260.46$$
$$= 1.536$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K)) / (K^2 - 1)) / (K - 1))$$
$$= (0.66845 + (5.7169(1.536^2 \cdot \log_{10}(1.536)) / (1.536^2 - 1)) / (1.536 - 1))$$
$$= 4.699$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$
$$= (1.536^2(1 + 8.55246 \cdot \log_{10}(1.536)) - 1) / ((1.0472 + 1.9448 \cdot 1.536^2) \cdot (1.536 - 1))$$
$$= 1.695$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$
$$= (1.536^2(1 + 8.55246 \cdot \log_{10}(1.536)) - 1) / (1.36136(1.536^2 - 1) \cdot (1.536 - 1))$$
$$= 5.164$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$
$$= (1.536^2 + 1) / (1.536^2 - 1)$$
$$= 2.472$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$
$$= 5.164 \cdot 6.27^2 \cdot 40.411 / 0.239$$
$$= 34261.242 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$
$$= 0.827 / 40.411$$
$$= 0.020 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d)$$
$$= (28.0 \cdot 0.02 + 1) / (1.695 + 28.0^3/34261.242)$$
$$= 1.569$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$
$$= (1.1229 \cdot 274 / 260.46) / (1.5687 \cdot 12.0^2)$$
$$= 522.04 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$
$$= (1.1229 \cdot 665 / 260.46) / (1.5687 \cdot 12.0^2)$$
$$= 1269.72 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$
$$= (1.7641 \cdot 274 / 260.46) / (1.5687 \cdot 28.0^2)$$
$$= 150.63 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$
$$= (1.7641 \cdot 665 / 260.46) / (1.5687 \cdot 28.0^2)$$
$$= 366.37 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$
$$= (4.6994 \cdot 274 / (28.0^2 \cdot 260.46)) - 2.4722 \cdot 151$$
$$= 257.08 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$
$$= (4.6994 \cdot 665 / (28.0^2 \cdot 260.46)) - 2.4722 \cdot 366$$
$$= 625.28 \text{ kgf/cm}^2$$



Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (522 + \max(151, 257)) / 2$$

$$= 389.56 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1270 + \max(366, 625)) / 2$$

$$= 947.50 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 8140 / 34.7282$$

$$= 234.38 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (4174 / 34.7282)$$

$$= 120.18 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	522.04	2110.70	1269.72	2110.70
Radial Flange	150.63	1407.13	366.37	1407.13
Tangential Flange	257.08	1407.13	625.28	1407.13
Maximum Average	389.56	1407.13	947.50	1407.13
Bolting	234.38	1753.82	120.18	1753.82

Minimum Required Flange Thickness 20.650 mm

Estimated M.A.W.P. (Operating) 37 kgf/cm²

Estimated Finished Weight of Flange at given Thk. 17.8 kg

Estimated Unfinished Weight of Forging at given Thk 27.8 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 700.8 / 1.0534 \cdot 100000.023 \cdot 0.239 / (1.096 \cdot 2067026 \cdot 6.27^2 \cdot 40.411 \cdot 0.3)$$

$$= 0.769 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 288.1 / 1.0534 \cdot 100000.023 \cdot 0.239 / (1.096 \cdot 2041395 \cdot 6.27^2 \cdot 40.411 \cdot 0.3)$$

$$= 0.320 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 665.3 / 1.0 \cdot 100000.023 \cdot 0.239 / (1.569 \cdot 2067026 \cdot 6.27^2 \cdot 40.411 \cdot 0.3)$$

$$= 0.537 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 273.5 / 1.0 \cdot 100000.023 \cdot 0.239 / (1.569 \cdot 2041395 \cdot 6.27^2 \cdot 40.411 \cdot 0.3)$$

$$= 0.223 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.655, Temperature Reduction per Fig. UCS 66.1 = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:



Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	405.52	2110.70	951.71	2110.70
Radial Flange	192.72	1407.13	452.30	1407.13
Tangential Flange	179.97	1407.13	422.37	1407.13
Maximum Average	299.12	1407.13	702.00	1407.13
Bolting	234.38	1753.82	120.18	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 18.186 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
CH. FRONT CAP		10.197	9.27	3	273	1203.2	1
CH. SHELL		10.223	9.27	3	273	1203.2	2
LEFT CH. FLG		10.223	28	3	254.46	1407.1	3
SHELL		5.6343	9.27	3	273	1203.2	4
RIGHT CH. FLG		10.223	28	3	254.46	1407.1	5
REAR CH. SHELL		10.223	9.27	3	273	1203.2	6
CH. REAR CAP		10.197	9.27	3	273	1203.2	7

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
CH. FRONT CAP		10.1974	59.4063	87.0314	9.27	4.5	1
CH. SHELL		10.1974	56.2767	83.9936	9.27	4.5	2
LEFT CH. FLG		10.1974	38.3549	49.9155	28	20.574	3
SHELL		5.60857	56.2767	83.9936	9.27	4.5	4
RIGHT CH. FLG		10.1974	36.9002	48.09	28	20.6502	5
REAR CH. SHELL		10.1974	56.2767	83.9936	9.27	4.5	6
CH. REAR CAP		10.1974	59.4063	87.0314	9.27	4.5	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 56 kgf/cm²
 Shell Side MAPnc = 84 kgf/cm²
 Channel Side MAWP = 37 kgf/cm²
 Channel Side MAPnc = 45 kgf/cm²

Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: CH. FRONT CAP

Elliptical Head Design Information:					
Design Pressure	P	10.197 kgf/cm ²	Int. Design Temperature	65 °C	
Material		SA-234 WPB	Ext. Design Temperature	0 °C	
Length		0.000 cm	External Pressure Chart	CS-2	
			UNS Number	K03006	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1203.2 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1203.2 kgf/cm ²
Outside Diameter	Do	273.000 mm	Actual Stress	206.5 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		9.2700 mm	Spec. Nominal Thk.	9.270 mm	
Surface Area		807.9 cm ²	SG of Contents	0.000	
Element Volume		2156.7 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		6.0 kgf	Operating Weight	6.0 kgf	
			Comp. All. Ope. Stress	1443.8 kgf/cm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 3.297 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.648 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D_o \cdot K_{cor}) / (2 \cdot S \cdot E + 2 \cdot P \cdot (K_{cor} - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (10.197 \cdot 273.0 \cdot 0.97) / (2 \cdot 1203.2 \cdot 1.0 + 2 \cdot 10.197 \cdot (0.97 - 0.1))$$

$$= 1.1143 + 3.0000 = 4.1143 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1203.2 \cdot 1.0 \cdot 6.27) / (0.97 \cdot 273.0 - 2 \cdot 6.27 \cdot (0.97 - 0.1))$$

$$= 59 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D_o - 2 \cdot t \cdot (K - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1203.2 \cdot 1.0 \cdot 9.27) / (1.0 \cdot 273.0 - 2 \cdot 9.27 \cdot (1.0 - 0.1))$$

$$= 87 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1))) / (2 \cdot E \cdot t)$$

$$= (10.197 \cdot (0.97 \cdot 273.0 - 2 \cdot 6.27 \cdot (0.97 - 0.1))) / (2 \cdot 1.0 \cdot 6.27)$$

$$= 206.536 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) + c_a \text{ per Appendix 1-1 (a)(1)}$$

$$= (10.197 \cdot 136.5) / (1203.2 \cdot 1.0 + 0.4 \cdot 10.197) + 3.0$$

$$= 4.153 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1203.2 \cdot 1.0 \cdot 6.27) / (136.5 - 0.4 \cdot 6.27)$$

$$= 56 \text{ kgf/cm}^2$$

Factor K, corroded condition [Kcor]:



$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (260.46 / (2 \cdot 66.615))^2) / 6 \\ &= 0.970315 \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 9.27, tr = 3.956, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.631, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 9.27, tr = 4.136, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.66, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	36.900 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	9.270 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.660	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 3.297 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.648 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: CH. SHELL

Cylindrical Shell Design Information:				
Design Pressure	P	10.223 kgf/cm²	Int. Design Temperature	65 °C
Material		SA-106 B	Ext. Design Temperature	0 °C
Length		23.200 cm	External Pressure Chart	CS-2
			UNS Number	K03006
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S 1203.2 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa 1203.2 kgf/cm²
Outside Diameter	Do	273.000 mm	Actual Stress	218.5 kgf/cm²
Spec. Min. Thk t _e , or s _a		9.2700 mm	Spec. Nominal Thk.	9.270 mm
Surface Area		1989.8 cm²	SG of Contents	1.000
Element Volume		11798.3 cm³	Weight of Contents	11.8 kgf
Empty Weight		13.8 kgf	Operating Weight	25.6 kgf
			Comp. All. Ope. Stress	1443.8 kgf/cm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (10.223 \cdot 136.5) / (1203.2 \cdot 1.0 + 0.4 \cdot 10.223)$
 $= 1.1559 + 3.0000 = 4.1559 \text{ mm}$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 kgf/cm²
 $= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t)$ per Appendix 1-1 (a)(1)
 $= (1203.2 \cdot 1.0 \cdot 6.27) / (136.5 - 0.4 \cdot 6.27)$
 $= 56 - 0 = 56 \text{ kgf/cm}^2$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t)$ per Appendix 1-1 (a)(1)
 $= (1203.2 \cdot 1.0 \cdot 9.27) / (136.5 - 0.4 \cdot 9.27)$
 $= 84 \text{ kgf/cm}^2$

Actual stress at given pressure and thickness, corroded [Sact]:

$= (P \cdot (Ro - 0.4 \cdot t)) / (E \cdot t)$
 $= (10.223 \cdot (136.5 - 0.4 \cdot 6.27)) / (1.0 \cdot 6.27)$
 $= 218.472 \text{ kgf/cm}^2$

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 9.27, tr = 1.985, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.317, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	36.900 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	9.270 mm	Basic MDMT	-29 °C

PV Elite 28 Licensee: SL User
FileName : 401-E-31011
Element: CH. SHELL

Page 37 of 214

Step: 7 4:41pm May 21,2026



HEXAGON

Stress/Thk. Ratio	0.317	Temperature Difference	75 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	5.634 kgf/cm²	Int. Design Temperature	180 °C	
Material		SA-106 B	Ext. Design Temperature	180 °C	
Length		231.200 cm	External Pressure Chart	CS-2	
			UNS Number	K03006	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1203.2 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1203.2 kgf/cm²
Outside Diameter	Do	273.000 mm	Actual Stress	120.4 kgf/cm²	
Spec. Min. Thk t _e , or s _a		9.2700 mm	Spec. Nominal Thk.	9.270 mm	
Surface Area		19829.0 cm²	SG of Contents	1.000	
Element Volume		83309.1 cm³	Weight of Contents	83.3 kgf	
Empty Weight		137.6 kgf	Operating Weight	220.9 kgf	
			Comp. All. Ope. Stress	1388.5 kgf/cm²	

Radiography Information:			
User Defined			User Defined
Circ. Joint Efficiency	100 %		Long. Joint Eff. E or v 100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (5.634 \cdot 136.5) / (1203.2 \cdot 1.0 + 0.4 \cdot 5.634)$
 $= 0.6380 + 3.0000 = 3.6380 \text{ mm}$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 kgf/cm²
 $= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t)$ per Appendix 1-1 (a)(1)
 $= (1203.2 \cdot 1.0 \cdot 6.27) / (136.5 - 0.4 \cdot 6.27)$
 $= 56 - 0 = 56 \text{ kgf/cm}^2$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t)$ per Appendix 1-1 (a)(1)
 $= (1203.2 \cdot 1.0 \cdot 9.27) / (136.5 - 0.4 \cdot 9.27)$
 $= 84 \text{ kgf/cm}^2$

Actual stress at given pressure and thickness, corroded [Sact]:

$= (P \cdot (Ro - 0.4 \cdot t)) / (E \cdot t)$
 $= (5.634 \cdot (136.5 - 0.4 \cdot 6.27)) / (1.0 \cdot 6.27)$
 $= 120.407 \text{ kgf/cm}^2$

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 9.27, tr = 3.017, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.481, Temp. Reduction = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	56.277 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	9.270 mm	Basic MDMT	-29 °C



Stress/Thk. Ratio	0.481	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: SHELL

Cylindrical Shell From 50 to 70 Ext. Chart: CS-2 at 180 °C

Elastic Modulus from Chart: CS-2 at 180 °C: 1999551 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.270	273.00	2312.00	43.54	8.4689	0.0006260	625.89

$$MAEP = (4*B)/(3*(Do/t)) = (4*625.886)/(3*43.5407) = 19.1663 \text{ kgf/cm}^2$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.202	273.00	2312.00	123.98	8.4689	0.0000981	98.06

$$MAEP = (4*B)/(3*(Do/t)) = (4*98.0649)/(3*123.9797) = 1.0546 \text{ kgf/cm}^2$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.270	273.00	168981.50	43.54	50.0000	0.0005903	590.21

$$MAEP = (4*B)/(3*(Do/t)) = (4*590.2106)/(3*43.5407) = 18.0738 \text{ kgf/cm}^2$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: REAR CH. SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	10.223 kgf/cm²	Int. Design Temperature	65 °C	
Material		SA-106 B	Ext. Design Temperature	40 °C	
Length		23.200 cm	External Pressure Chart	CS-2	
			UNS Number	K03006	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1203.2 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1203.2 kgf/cm²
Outside Diameter	Do	273.000 mm	Actual Stress	218.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		9.2700 mm	Spec. Nominal Thk.	9.270 mm	
Surface Area		1989.8 cm²	SG of Contents	1.000	
Element Volume		11798.3 cm³	Weight of Contents	11.8 kgf	
Empty Weight		13.8 kgf	Operating Weight	25.6 kgf	
			Comp. All. Ope. Stress	1479.2 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (10.223 \cdot 136.5) / (1203.2 \cdot 1.0 + 0.4 \cdot 10.223)$
 $= 1.1559 + 3.0000 = 4.1559 \text{ mm}$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 kgf/cm²
 $= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t)$ per Appendix 1-1 (a)(1)
 $= (1203.2 \cdot 1.0 \cdot 6.27) / (136.5 - 0.4 \cdot 6.27)$
 $= 56 - 0 = 56 \text{ kgf/cm}^2$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t)$ per Appendix 1-1 (a)(1)
 $= (1203.2 \cdot 1.0 \cdot 9.27) / (136.5 - 0.4 \cdot 9.27)$
 $= 84 \text{ kgf/cm}^2$

Actual stress at given pressure and thickness, corroded [Sact]:

$= (P \cdot (Ro - 0.4 \cdot t)) / (E \cdot t)$
 $= (10.223 \cdot (136.5 - 0.4 \cdot 6.27)) / (1.0 \cdot 6.27)$
 $= 218.472 \text{ kgf/cm}^2$

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 9.27, tr = 1.985, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.317, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	36.900 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	9.270 mm	Basic MDMT	-29 °C



Stress/Thk. Ratio	0.317	Temperature Difference	75 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: REAR CH. SHELL

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 40 °C

Elastic Modulus from Chart: CS-2 at 40 °C: 2038903 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.270	273.00	253.20	43.54	0.9275	0.0052057	1222.10

$$MAEP = (4*B)/(3*(Do/t)) = (4*1222.0978)/(3*43.5407) = 37.4239 \text{ kgf/cm}^2$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
0.847	273.00	253.20	322.26	0.9275	0.0002500	254.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*254.9081)/(3*322.256) = 1.0547 \text{ kgf/cm}^2$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.270	273.00	7110.68	43.54	26.0465	0.0005922	603.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*603.7551)/(3*43.5407) = 18.4886 \text{ kgf/cm}^2$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: CH. REAR CAP

Elliptical Head Design Information:					
Design Pressure	P	10.197 kgf/cm ²	Int. Design Temperature	65 °C	
Material		SA-234 WPB	Ext. Design Temperature	40 °C	
Length		0.000 cm	External Pressure Chart	CS-2	
			UNS Number	K03006	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1203.2 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1203.2 kgf/cm ²
Outside Diameter	Do	273.000 mm	Actual Stress	206.5 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		9.2700 mm	Spec. Nominal Thk.	9.270 mm	
Surface Area		807.9 cm ²	SG of Contents	0.000	
Element Volume		2156.7 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		6.0 kgf	Operating Weight	6.0 kgf	
			Comp. All. Ope. Stress	1443.8 kgf/cm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 3.297 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.648 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D_o \cdot K_{cor}) / (2 \cdot S \cdot E + 2 \cdot P \cdot (K_{cor} - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (10.197 \cdot 273.0 \cdot 0.97) / (2 \cdot 1203.2 \cdot 1.0 + 2 \cdot 10.197 \cdot (0.97 - 0.1))$$

$$= 1.1143 + 3.0000 = 4.1143 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1203.2 \cdot 1.0 \cdot 6.27) / (0.97 \cdot 273.0 - 2 \cdot 6.27 \cdot (0.97 - 0.1))$$

$$= 59 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D_o - 2 \cdot t \cdot (K - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1203.2 \cdot 1.0 \cdot 9.27) / (1.0 \cdot 273.0 - 2 \cdot 9.27 \cdot (1.0 - 0.1))$$

$$= 87 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1))) / (2 \cdot E \cdot t)$$

$$= (10.197 \cdot (0.97 \cdot 273.0 - 2 \cdot 6.27 \cdot (0.97 - 0.1))) / (2 \cdot 1.0 \cdot 6.27)$$

$$= 206.536 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) + c_a \text{ per Appendix 1-1 (a)(1)}$$

$$= (10.197 \cdot 136.5) / (1203.2 \cdot 1.0 + 0.4 \cdot 10.197) + 3.0$$

$$= 4.153 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1203.2 \cdot 1.0 \cdot 6.27) / (136.5 - 0.4 \cdot 6.27)$$

$$= 56 \text{ kgf/cm}^2$$

Factor K, corroded condition [Kcor]:



$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (260.46 / (2 \cdot 66.615))^2) / 6$$

$$= 0.970315$$

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 9.27, tr = 3.956, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.631, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 9.27, tr = 4.136, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.66, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	36.900 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	9.270 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.660	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 3.297 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.648 mm

EXTERNAL PRESSURE RESULTS for: CH. REAR CAP

Elliptical Head From 90 to 100 Ext. Chart: CS-2 at 40 °C

Elastic Modulus from Chart: CS-2 at 40 °C: 2038903 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.270	273.00	43.54	0.0031899	1151.92

$$\text{MAEP} = B / (K0 \cdot \text{Do}/t) = 1151.9241 / (0.9 \cdot 43.5407) = 29.3959 \text{ kgf/cm}^2$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
0.707	273.00	386.22	0.0003596	366.60

$$\text{MAEP} = B / (K0 \cdot \text{Do}/t) = 366.6027 / (0.9 \cdot 386.2233) = 1.0547 \text{ kgf/cm}^2$$



Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	73.160 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7.291 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	109.166 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	61.904 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	8.020 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8.020 kgf/cm ²
App 27-4	= M.A.W.P.	56.277 kgf/cm ²

User defined Hydrostatic Test Pressures:

Shell Side Test Pressure at High Point	8.413 kgf/cm ²
Channel Side Test Pressure at High Point	15.296 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	47.970 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.257 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	58.137 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	40.590 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	14.582 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.582 kgf/cm ²
App 27-4	= M.A.W.P.	36.900 kgf/cm ²

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
CH. FRONT CAP	310.4	2202.5	0.141	15.32
CH. SHELL	327.5	2202.5	0.149	15.32
SHELL	180.4	2202.5	0.082	8.44
REAR CH. SHELL	327.5	2202.5	0.149	15.32
CH. REAR CAP	310.4	2202.5	0.141	15.32

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
CH. IN Noz N3	Nozzle	1203.20	1203.20	1.000
CH. IN Noz N3	Pad	1203.20	1203.20	1.000
CH. OUT Noz N4	Nozzle	1203.20	1203.20	1.000
CH. OUT Noz N4	Pad	1203.20	1203.20	1.000
SHELL IN Noz N1	Nozzle	1203.20	1203.20	1.000
SHELL IN Noz N1	Pad	1203.20	1203.20	1.000
SHELL OUT Noz N2	Nozzle	1203.20	1203.20	1.000
SHELL OUT Noz N2	Pad	1203.20	1203.20	1.000
Vent Noz V	Nozzle	1203.20	1203.20	1.000
Vent Noz V	Pad	1203.20	1203.20	1.000
Drain Noz D	Nozzle	1203.20	1203.20	1.000
Drain Noz D	Pad	1203.20	1203.20	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
CH. FRONT CAP	1203.20	1203.20	1.000
CH. SHELL	1203.20	1203.20	1.000



LEFT CH. FLG	1407.13	1407.13	1.000
SHELL	1203.20	1203.20	1.000
RIGHT CH. FLG	1407.13	1407.13	1.000
REAR CH. SHELL	1203.20	1203.20	1.000
CH. REAR CAP	1203.20	1203.20	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	942.17	942.17	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
CH. IN Noz N3	74.43	1564.16	0.048
CH. OUT Noz N4	74.43	1564.16	0.048
SHELL IN Noz N1	40.99	1564.16	0.026
SHELL OUT Noz N2	40.99	1564.16	0.026
Vent Noz V	74.43	1564.16	0.048
Drain Noz D	74.43	1564.16	0.048

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input Echo, Tubesheet Number 1, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Test Pressure		8.41 kgf/cm ²
Shell Thickness	ts	9.2700 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	254.460 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	1203.20 kgf/cm ²
Shell Allowable Stress at Ambient		1203.20 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Test Pressure		15.30 kgf/cm ²
Channel Thickness	tc	9.2700 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	254.460 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	1203.20 kgf/cm ²
Channel Allowable Stress at Ambient		1203.20 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	52
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2394.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2314.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1835.39 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	118.170 mm
Largest Center-to-Center Tube Distance	Ul	31.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	21.11 °C
------------------------------------	---	----------



Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03017
Tubesheet Allowable Stress at Temperature	S	1407.13 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.13 kgf/cm ²
Thickness of Tubesheet	h	40.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	400.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	260.46 mm
Step 1 Depth on the Tubeside	ht1	5.00 mm
Is the Tubesheet Flat on Shellside?		No
Calculated Tubesheet Diameter per Div 2, 4.18.4		400.00 mm

Note: Tubesheet diameter is now: 400.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	168.1 cm ²
---------------------------	----	-----------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	419.000 mm
Tube End condition corresponding to Span (l)	k	1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes

Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	400.000 mm
Flange Inside Diameter	B	254.460 mm
Flange Face Outside Diameter	Fod	309.460 mm
Flange Face Inside Diameter	Fid	254.460 mm
Gasket Outside Diameter	Go	306.460 mm
Gasket Inside Diameter	Gi	260.460 mm
Small end Hub thk.	g0	9.2700 mm
Large end Hub thk.	g1	15.0000 mm
Gasket Factor,	m	0.00
Gasket Design Seating Stress	y	0.00 kgf/cm ²
Flange Facing Sketch		Code Sketch 1b
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	26.460 mm
Width of Partition Gasket	wp	6.0000 mm
Partition Gasket Factor,	mPart	3.7500
Partition Gasket Design Seating Stress	yPart	534.33 kgf/cm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	345.000 mm
Nominal Bolt Diameter	dB	20.0000 mm
Thread Series		TEMA Metric Thread
Number of Bolts	n	16
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	1753.82 kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82 kgf/cm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.



Tubesheet Integral with Shell
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 35.0000 mm
 Is Bolt Load Transferred to the Tubesheet Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (15.3 \cdot (19.05/2 - 0)) / (942.17 \cdot 1.0 + 0.4 \cdot 15.3)$$

$$= 0.1536 + 0.0000 = 0.1536 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2314.00	9.03	50.0000	0.0134948	977.27

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 153.1568 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3388	19.05	2314.00	56.22	50.0000	0.0003480	354.76

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 354.7606) / (3 \cdot 56.2227) = 8.4132 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3388 mm
Allowable Internal Pressure at Corroded thickness	229.00 kgf/cm ²
Required Internal Design Pressure	15.30 kgf/cm ²
Allowable External Pressure at Corroded thickness	153.16 kgf/cm ²
Required External Design Pressure	8.41 kgf/cm ²
Required Thickness due to Shell Side pressure	0.3388 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)**Intermediate Calculations For Tubesheets Extended As Flanges:****ASME Code, Section VIII Division 1, 2025**

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	23.000 mm
Basic Gasket Width,	$b_0 = N / 2$	11.500 mm
Effective Gasket Width,	$b = \sqrt{b_0} * 2.5$	8.542 mm
Gasket Reaction Diameter,	$G = G_o$ (Self-Energizing)	306.460 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:**Results for Tubesheet Calculations Original Thickness:****ASME Section VIII-2 Part 4.18.8.4 Step 1:**

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$
$$= 0/40.0$$
$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d^*]:

$$= \max(dt - 2 * tt(E_{tT}/E) * (StT/S) * (\rho); dt - 2 * tt)$$
$$= \max(19.05 - 2 * 2.11 (2067025/2067025) * (942/1407) * (0); 19.05 - 2 * 2.11)$$
$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 * r_o + dt$$
$$= 2 * 118.17 + 19.05$$
$$= 255.390 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$
$$= (25.0 - 19.05) / 25.0$$
$$= 0.238$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$
$$= 255.39/2$$
$$= 127.6950 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 * \min(AL * CNVfactor; 4 * Do * p) / (\pi * Do^2)) }$$
$$= 25.0 / \sqrt{ (1 - 4 * \min(168.06 * 100.0 ; 4 * 255.39 * 25.0) / (\pi * 255.39^2)) }$$
$$= 30.4985 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$
$$= (30.4985 - 19.05) / 30.4985$$
$$= 0.3754$$

Ratio [ρ_s]:

$$= a_s / a_o$$
$$= 127.23/127.695$$
$$= 0.9964$$



Ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 153.23/127.695 \\ &= 1.2000 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\ &= 1 - 52 ((19.05 - 2 \cdot 2.11) / (2 \cdot 127.695))^2 \\ &= 0.8247 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot ao))^2 \\ &= 1 - 52 (19.05 / (2 \cdot 127.695))^2 \\ &= 0.7107 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max((hg - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((6.0 - 0); 0) \\ &= 6.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned} &= \pi \cdot ts(Ds + ts)Es/L \\ &= \pi \cdot 9.27 (254.46 + 9.27)2067025/2314.0 \\ &= 68607.4297 \text{ kgf/mm} \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned} &= \pi \cdot tt(Dt - tt)Et/L \\ &= \pi \cdot 2.11 (19.05 - 2.11)2067025/2314.0 \\ &= 1003.0618 \text{ kgf/mm} \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned} &= Ks / (Nt \cdot Kt) \\ &= 68607.43 / (52 \cdot 1003.062) \\ &= 1.31535 \end{aligned}$$

Rigidity Ratio [J]:

$$\begin{aligned} &= 1 / (1 + Ks/Kt) \\ &= 1 / (1 + 68607.43/0) \\ &= 1.0000 (= 1 \text{ if No Exp. Jt.}) \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts) ts) \\ &= ((12(1 - 0.3^2))^{1/4}) / ((254.46 + 9.27) 9.27) \\ &= 0.0368 \text{ 1/mm} \end{aligned}$$

Shell coefficient [ks]:

$$\begin{aligned} &= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2)) \\ &= 0.037 \cdot 2067025 \cdot 9.27^3 / (6(1 - 0.3^2)) \\ &= 11087369.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned} &= 6(Ds + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2) \\ &= 6(254.46 + 9.27) \cdot 11087369/40.0^3 \cdot (1 + 40.0 \cdot 0.037 + (40.0 \cdot 0.037)^2/2) \\ &= 973700.6875 \text{ kgf/cm}^2 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned} &= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts))) \\ &= 254.46^2 / (4 \cdot 2067025 \cdot 9.27) \cdot (1 - 254.46 \cdot 0.3 / (2(254.46 + 9.27))) \\ &= 0.72253\text{E-}03 \text{ mm/kgf/cm}^2 \end{aligned}$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$\begin{aligned} h/p &= 1.600000 ; \mu^* = 0.375378 ; E^* = 820033 \text{ kgf/cm}^2 \\ E^*/E &= 0.396721 ; \nu^* = 0.307135 \end{aligned}$$



Tube Bundle Stiffness Factor [Xa]:

$$= \left(\frac{24(1 - \nu^2) N t \cdot E t \cdot t t (d t - t t) a o^2}{E \cdot L \cdot H^3} \right)^{1/4}$$

$$= \left(\frac{24(1 - 0.307^2) 52 \cdot 2067025 \cdot 2.11}{(19.05 - 2.11) 127.695^2} \right)^{1/4}$$

$$= 1.8299$$

Values from Table 4.18.3

$$Z_d = 0.229966 ; Z_v = 0.171315 ; Z_m = 0.704850$$

$$Z_a = 0.687159E+00 ; Z_w = 0.171315$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 400.0/255.39$$

$$= 1.5662$$

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.31) / 820033 \cdot (973700 + 0 + 2067025 \cdot \ln(1.57))$$

$$= 1.6063$$

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.3071) 1.6063$$

$$= 2.0996$$

Compute parameter [Q1]:

$$= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (0.9964 - 1 - 2.0996 \cdot 0.1713) / (1 + 2.0996 \cdot 0.7048)$$

$$= -0.146513$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.22997 + -0.14651 \cdot 0.17131) / 2 \cdot 1.82989^4$$

$$= 1.1485$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.17131 + -0.14651 \cdot 0.70485) / 2 \cdot 1.82989^4$$

$$= 0.3815$$

Compute parameter [U]:

$$= (Z_w + (\psi_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.1713 + (0.9964 - 1) 0.7048) 1.82989^4 / (1 + 2.0996 \cdot 0.7048)$$

$$= 0.7630$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= (G_c - C) / D_o \text{ (config b)}$$

$$= (306.46 - 345.0) / 255.39 = -0.15091$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= \psi_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 0.9964 \cdot 11087369 \cdot 0.0368 \cdot 0.000723 (1 + 40.0 \cdot 0.0368)$$

$$= 725.0055 \text{ mm}^2$$

Calculate parameter [ω_s^*]:

$$= a o^2 (\psi_s^2 - 1) \cdot (\psi_s - 1) / 4 - \omega_s$$

$$= 127.695^2 (0.996^2 - 1) \cdot (0.996 - 1) / 4 - 725.005$$

$$= -724.8976 \text{ mm}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta_c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$



$$= 1.2 \cdot 0 [0 (0 + 0 \cdot 0 \cdot 40.0) - 0 (0/40.0 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$= a_o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - \omega c$$

$$= 127.695^2 [(1.19997^2 + 1) \cdot (1.19997 - 1) / 4 - (0.99636 - 1) / 2] - 0$$

$$= 2018.6500 \text{ mm}^2$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= P_s \{ x_s + 2 (1 - x_s) v_t + [2 / K_{st} \cdot (D_s / D_o)^2] v_s - [(p_s^2 - 1) / (J \cdot K_{st})] - [(1 - J) / (2 J \cdot K_{st})] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 8.413 \{ 0.711 + 2 (1 - 0.711) 0.3 + [2 / 1.315 \cdot (254.46 / 255.39)^2] 0.3 - [(0.996^2 - 1) / (1.0 \cdot 1.315)] - [(1 - 1.0) / (2 \cdot 1.0 \cdot 1.315)] [(0^2 - (254.46)^2) / 255.39^2] \}$$

$$= 11.2954 \text{ kgf/cm}^2$$

Pressure [$P't$]:

$$= [x_t + 2 (1 - x_t) v_t + 1 / (J \cdot K_{st})] P_t$$

$$= [0.825 + 2 (1 - 0.825) 0.3 + 1 / (1. \cdot 1.315)] 15.296$$

$$= 25.8523 \text{ kgf/cm}^2$$

Pressure [P_Y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 52 \cdot 1003.062 \cdot 0 / (3.142 \cdot 127.695^2)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot a_o^2)$$

$$= -0.151 \cdot 0.763 \cdot 60906.84 / (2 \cdot \pi \cdot 127.695^2)$$

$$= 6.8445 \text{ kgf/cm}^2$$

Calculate the pressure [$Prim$]:

$$= - (U / a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= - (0.763 / 127.695^2) \cdot (-724.898 \cdot 8.413 - 2018.65 \cdot 15.296)$$

$$= 1.7301 \text{ kgf/cm}^2$$

Calculate the pressure [P_w]:

$$= (U / a_o^2) \cdot (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)$$

$$= (0.763 / 127.695^2) \cdot (725.005 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (p_s - 1) Q_{z2})) \cdot (P's - P't + P_Y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 1.315 / (1 + 1.0 \cdot 1.315 (1.149 + (0.996 - 1) 0.381)) \cdot (11.295 - 25.852 + 0 + 0 + 6.845 + 1.73)$$

$$= -3.1364 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-724.898 \cdot 8.413 - 2018.65 \cdot 15.296) - (725.005 \cdot 0 - 0 \cdot 0)) \cdot 0.01 + 60906.8 \cdot -0.151 / (2 \cdot \pi)] / (1 + 2.09965 \cdot 0.70485)$$

$$= -738.9667 \text{ kgf}$$

Calculate factor [$Q3$]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2)$$

$$= -0.147 + 2 \cdot -738.967 / (-3.136 \cdot 127.695^2)$$



$$= 2.74336$$

Fm Value from Table 4.18.3 = 1.419563

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e \\ &= (1.5 \cdot 1.4196 / 0.3754) \cdot (2 \cdot 127.695 / (40.0 - 6.0))^2 \cdot -3.14 \\ &= -1003.8188 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 2549.16 = 3632.55 \text{ kgf/cm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ f]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e \\ &= (1.5 \cdot 0.8273 / 0.3754) \cdot (2 \cdot 127.695 / (21.564 - 6.0))^2 \cdot 4.08 \\ &= 3632.2778 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 21.5641 + 0 + 0 \\ &= 21.5641 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [Tlimit]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y) \\ &= (2 \cdot 0.238 \cdot 40.0 / 127.695) \cdot 1937.359 \\ &= 288.8705 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Tau limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned} \text{Tubesheet thickness (Incl. Corr.)} &= 6.2407 \text{ mm} \\ \text{Tubesheet Shear Stress} &= 160.4357 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(21.5641; 6.2407) \\ &= 21.5641 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -1.8698, F_{tmax} = 3.924$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s) \\ &= ((8.41 \cdot 0.7107 - 15.3 \cdot 0.8247) - (-3.136) \cdot -1.87) / \\ &\quad (0.8247 - 0.7107) \\ &= -109.6592 \text{ kgf/cm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\ &= ((8.41 \cdot 0.7107 - 15.3 \cdot 0.8247) - (-3.136) \cdot 3.924) / \\ &\quad (0.8247 - 0.7107) \\ &= 49.7599 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 109.659 \text{ kgf/cm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta T \cdot S_{y,t} = 1743.6229 \text{ kgf/cm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ &= \max((3.25 - 0.25(0.23 + 2.743 \cdot 0.171)1.83^4); 1.25) \\ &= 2.0000 \text{ (Should be } \leq 2) \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{(dt^2 + (dt - 2 \cdot tt)^2)})/4 \\ &= (\sqrt{(19.05^2 + (19.05 - 2 \cdot 2.11)^2)})/4 \\ &= 6.0355 \text{ mm} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot Et/Syt)} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 2067025/1835)} \\ &= 156.7104 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L/r \\ &= 1.0 \cdot 419.0/6.035 \\ &= 69.4229 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (Sy/t/Fs) \cdot (1 - Ft/(2 \cdot Ct)) \\ &= (1835/2.0) \cdot (1 - 69.423/(2 \cdot 156.71)) \\ &= 1743.623 \text{ kgf/cm}^2 \text{ (Never greater than } Sot) \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 109.66 \cdot 1.1229 \\ &= 123.14 \text{ kgf} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (do - t) t \\ &= \pi (19.05 - 2.11) 2.11 \\ &= 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (do - t) S \\ &= \pi \cdot 2.11 (19.05 - 2.11) 1743.623 \\ &= 1957.932 \text{ kgf} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft) \\ &= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 1743.62 \\ &= 1957.9325 \text{ kgf} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / Sw \text{ where } Sw = \min(S \text{ and } St) \\ &= 1743.623/1743.623 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft) \\ &= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 1743.62 \\ &= 1300.8619 \text{ kgf} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd)} - 0.75 \cdot do) \\ &= 2(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11)1.0 \cdot 1.0)} - 0.75 \cdot 19.05) \\ &= 2.562 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Minimum Required Fillet Weld Leg [afr]:} & \quad 1.2810 \text{ mm} \\ \text{Minimum Required Groove Weld Leg [agr]:} & \quad 1.2810 \text{ mm} \end{aligned}$$

Tube-Tubesheet Jt allowable, 1957.93 is $\geq$ tube strength 1957.93 kgf

Note: This tube-tubesheet joint is a Full Strength joint



ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 2447.19 = 3487.25 \text{ kgf/cm}^2$

Axial Membrane Stress in Shell $[\sigma_s, m]$:
 $= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 127.695^2 / ((254.46 + 9.27) 9.27) \cdot [-3.14 + (0.996^2 - 1)$
 $(8.41 - 15.3)] + 127.23^2 \cdot 15.3 / ((254.46 + 9.27) 9.27)$
 $= 80.6944 \text{ kgf/cm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 9.27 mm, the minimum Shell length adjacent to the tubesheet should be 87.422 mm

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	1003.8	$\leq 3632.5 \text{ kgf/cm}^2$	Pass
Tubesheet Shear Stress	21.0	$\leq 1937.4 \text{ kgf/cm}^2$	Pass
Maximum Tube Stress	109.7	$\leq 1743.6 \text{ kgf/cm}^2$	Pass
Minimum Tube Stress (Buckling)	-109.7	$\geq -1743.6 \text{ kgf/cm}^2$	Pass
Maximum Force on any one Tube	123.1	$\leq 1957.9 \text{ kgf}$	Pass
Axial Membrane Stress in Shell	80.7	$\leq 3487.2 \text{ kgf/cm}^2$	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	21.564	40.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
			Bend	Allwd	Shear	Allwd		
D1uc	20.149	17.334	1051	3633	62	1937	Ps+Pt-Th	D1 Ok
D2uc	10.471	...	134	3633	41	1937	Ps+Pt-Th	D2 Ok
D3uc	21.564	...	1004	3633	21	1937	Ps+Pt-Th	D3 Ok
Max:	21.5641	17.334 mm		0.289		0.032 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:



Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	Shell Stresses				:	Shell Band Stress				:	Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd		
D1uc	41	3487	...	...	:	...	...	...	...	:	Ok
D2uc	40	3487	...	...	:	...	...	...	...	:	Ok
D3uc	81	3487	...	...	:	...	...	...	...	:	Ok
Max RATIO	0.023				:	...				:	...

Tube, Shell and Channel Stress Summary:

Case#	Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	117	1743	-117	-1743	131	1957	...	3487	...	...	Ok
D2uc	7	1743	-6	-1743	8	1957	...	3487	...	...	Ok
D3uc	110	1743	-110	-1743	123	1957	...	3487	...	...	Ok
Max RATIO	0.067		0.067		0.067		...		...		...

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	21.564	40.000	Ok
Tubesheet thickness flanged extension :	17.334	35.000	Ok
Tube thickness :	0.339	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (9.270) adj. to tubesheet: 87.422 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.011 mm
 Due to Pressure + Thermal Shell Compresses by : -0.011 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0057	...	...	...	...	...
2	-0.0112	...	...	...	...	...
3	-0.0054	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	273	6.27	0.0031899	1151.92
20	30	25.3205	273	6.27	0.0052057	1222.1
30	50	No Calc	...	25	No Calc	No Calc
50	70	231.2	273	6.27	0.00062603	625.886
70	80	No Calc	...	25	No Calc	No Calc
80	90	25.3205	273	6.27	0.0052057	1222.1
90	100	No Calc	273	6.27	0.0031899	1151.92

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	9.27	No Calc	...	29.3959
20	30	9.27	No Calc	...	37.4239
30	50	28	19.177	...	No Calc
50	70	9.27	5.20197	1.05461	19.1663
70	80	28	19.9136	1.05461	No Calc
80	90	9.27	3.84715	1.05461	37.4239
90	100	9.27	3.70685	1.05461	29.3959

Minimum

19

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	25.3205	No Calc	No Calc	No Calc
30	50	No Calc	No Calc	No Calc	No Calc
50	70	231.2	16898.2	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	25.3205	711.068	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	5.98083	2156.74	4.04529	2312.93	0.59808
20	30	13.8102	11798.3	9.44713	12361.2	1.38102
30	50	17.8098	2508.96	16.9071	2557.49	1.78098
50	70	137.626	83309.1	94.1456	88919.2	13.7626
70	80	17.8098	2508.96	16.9071	2557.49	1.78098
80	90	13.8102	11798.3	9.44713	12361.2	1.38102
90	100	5.98083	2156.74	4.04529	2312.93	0.59808
Total		212	116237.00	154	123382.42	21



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset, Dtl. Cent. cm	Description
20	Liqd	11.7929	11.6	...	...	Liquid 20
20	Nozl	8.58649	11.6	-13.0718	7.07881	CH. IN Noz N3
20	Nozl	8.58649	11.6	13.0719	7.07881	CH. OUT Noz N4
30	Liqd	2.50783	2.4	...	...	Liquid 30
50	Sadl	22.0821	49	24.898	...	Left Saddle
50	Sadl	22.0821	191	24.898	...	Right Saddle
50	Liqd	83.2715	115.6	...	...	Liquid 50
50	Nozl	8.58649	216.4	14.8655	...	SHELL IN Noz N1
50	Nozl	8.58649	13	-14.8655	...	SHELL OUT Noz N2
70	Liqd	2.50783	2.4	...	...	Liquid 70
80	Liqd	11.7929	11.6	...	...	Liquid 80
80	Nozl	10.1042	11.6	14.8655	...	Vent Noz V
80	Nozl	10.1042	11.6	-14.8655	...	Drain Noz D
30	FTsh	37.7991	7.25	...	...	Fixed TS
30	Tube	119.176	124.95	...	...	Tubes
30	RTsh	37.7991	242.65	...	...	Tubesheet
50	Baf1	3.16968	25.1	...	...	Baffles
50	Baf1	3.16968	41.9	...	...	Baffles
50	Baf1	3.16968	58.7	...	...	Baffles
50	Baf1	3.16968	75.5	...	...	Baffles
50	Baf1	3.16968	92.3	...	...	Baffles
50	Baf1	3.16968	109.1	...	...	Baffles
50	Baf1	3.16968	125.9	...	...	Baffles
50	Baf1	3.16968	142.7	...	...	Baffles
50	Baf1	3.16968	159.5	...	...	Baffles
50	Baf1	3.16968	176.3	...	...	Baffles
50	Baf1	3.16968	193.1	...	...	Baffles
50	Baf1	3.23318	209.9	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	44.2
Liquid	111.9
Nozzles	54.6
Exchanger Components	232.9
Liquid in Tubes	21.5
Sum of the Detail Weights	465.0 kg

Weight Summation Results: (kg)



	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	234.1	234.1	234.1	234.1	234.1	234.1
Saddles	44.2	44.2	44.2	44.2	44.2	44.2
Nozzles	54.6	54.6	54.6	54.6	54.6	54.6
Exchanger	232.9	232.9	232.9	232.9	232.9	232.9
Ope. Liquid	...	...	...	...	...	111.9
Tube Ope Lqd	...	...	...	...	...	21.5
Test Liquid	...	116.2	...	...	...	...
Tube Tst Lqd	...	21.5	...	...	...	...
Totals	565.7	703.4	565.7	565.7	565.7	699.1

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	565.7 kg
Shop Test Wt.	- Fabricated weight + water (full)	703.4 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	565.7 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	565.7 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	565.7 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	699.1 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	635.4 kg

Exchanger Tube Data

Volume of Exchanger tubes :	21503.1 cm ³
Weight of Ope Liq in tubes :	21.5 kg
Weight of Water in tubes :	21.5 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	807.884	701.879
20	30	1989.76	1854.63
30	50	1278.68	383.717
50	70	19829	18482.4
70	80	1278.68	383.717
80	90	1989.76	1854.63
90	100	807.884	701.879

Total	27981.6	24362.8 cm ²
Total	30.1	26.2 ft ²

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
CH. IN Noz N3	49.84	52.01	65	300	GR 1.1	2.840	49.835	49.835	49.835
CH. OUT Noz N4	49.84	52.01	65	300	GR 1.1	2.840	49.835	49.835	49.835
SHELL IN Noz N	90.35	104.01	180	600	GR 1.1	2.840	90.350	90.350	90.350
SHELL OUT Noz	90.35	104.01	180	600	GR 1.1	2.840	90.350	90.350	90.350
Vent Noz V	49.84	52.01	65	300	GR 1.1	...	...	...	...
Drain Noz D	49.84	52.01	65	300	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 90.350 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 104.014 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 49.835 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 52.007 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
CH. IN Noz N3	77.851	0.50	0	133	36.900	49.835	Pass
CH. OUT Noz N4	77.851	0.50	0	133	36.900	49.835	Pass
SHELL IN Noz N1	77.851	0.50	0	133	56.277	90.350	Pass
SHELL OUT Noz N2	77.851	0.50	0	133	56.277	90.350	Pass

Evaluating Flange on Nozzle [CH. IN Noz N3]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(0) + 4 \cdot \text{abs}(133) \cdot 77.851 \leq \pi \cdot 77.851^3 \cdot [(49.835 - 36.900) + .500 \cdot 49.835]$$

$$= 0.42098E+02 \leq 0.56110E+03 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle CH. IN Noz N3 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(49.835, 0.500 \cdot 49.835 + 49.835 - 42.1 / (\pi \cdot 77.851^3))$$

$$= \min(49.835, 71.913)$$

$$= 49.835 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [CH. OUT Noz N4]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(0) + 4 \cdot \text{abs}(133) \cdot 77.851 \leq \pi \cdot 77.851^3 \cdot [(49.835 - 36.900) + .500 \cdot 49.835]$$

$$= 0.42098E+02 \leq 0.56110E+03 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle CH. OUT Noz N4 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(49.835, 0.500 \cdot 49.835 + 49.835 - 42.1 / (\pi \cdot 77.851^3))$$

$$= \min(49.835, 71.913)$$

$$= 49.835 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [SHELL IN Noz N1]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(0) + 4 \cdot \text{abs}(133) \cdot 77.851 \leq \pi \cdot 77.851^3 \cdot [(90.350 - 56.277) + .500 \cdot 90.350]$$

$$= 0.42098E+02 \leq 0.11747E+04 \text{ Check Passes}$$



Maximum Pressure for Flange on Nozzle SHELL IN Noz N1 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(90.350, 0.500 \cdot 90.350 + 90.350 - 42.1 / (\pi \cdot 77.851^3)) \\ &= \min(90.350, 132.684) \\ &= 90.350 \text{ kgf/cm}^2 \end{aligned}$$

Evaluating Flange on Nozzle [SHELL OUT Noz N2]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(0) + 4 \cdot \text{abs}(133) 77.851 \leq \pi \cdot 77.851^3 [(90.350 - 56.277) + 0.500 \cdot 90.350] \\ &= 0.42098\text{E}+02 \leq 0.11747\text{E}+04 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle SHELL OUT Noz N2 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(90.350, 0.500 \cdot 90.350 + 90.350 - 42.1 / (\pi \cdot 77.851^3)) \\ &= \min(90.350, 132.684) \\ &= 90.350 \text{ kgf/cm}^2 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 11.484

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 1.0 • 1.0 • 0.6 • 57.5²)

= 1983.5 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.0 • 0.915 • 1983.503

= 28.4 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	0.69	1.0000	1.0000	1.0000	57.50	0.9148
20	0.69	1.0000	1.0000	1.0000	57.50	0.9148
30	0.69	1.0000	1.0000	1.0000	57.50	0.9148
50	0.69	1.0000	1.0000	1.0000	57.50	0.9148
70	0.69	1.0000	1.0000	1.0000	57.50	0.9148
80	0.69	1.0000	1.0000	1.0000	57.50	0.9148
90	0.69	1.0000	1.0000	1.0000	57.50	0.9148

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm²	Wind Pressure kgf/m²	Element Wind Load kgf
10	20	69	32.76	156.375	202.344	3.183
20	30	69	32.76	760.032	202.344	15.4704
30	50	69	30.5352	146.569	202.344	2.9834
50	70	69	32.76	7574.11	202.344	154.17
70	80	69	30.5352	146.569	202.344	2.9834
80	90	69	32.76	760.032	202.344	15.4704
90	100	69	32.76	156.375	202.344	3.183

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
= (Sa/g)ss • DampVal) • (I / R)
= (0.0939 • 1.0) • 1.0
= 0.094

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
= Earthquake Weight • Ah • Scalar
= 65.69 • 0.0939 • 1.0
= 6.2 kgf

Vertical load per 8.4:
= 2/3 • Ah • W • Scalar
= 2/3 • 0.0939 • 591 • 1.0
= 37.0 kgf

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------



From	To	Height cm	Weight kgf	Ope Load kgf
10	20	12.723	65.6926	6.16565
20	30	12.723	65.6926	6.16565
30	50	12.723	65.6926	6.16565
50	Sad1	12.723	65.6926	6.16565
Sad1	70	12.723	65.6926	6.16565
50	70	12.723	65.6926	6.16565
70	80	12.723	65.6926	6.16565
80	90	12.723	65.6926	6.16565
90	100	12.723	65.6926	6.16565

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	152.450 cm
Center of Gravity of the Liquid	148.060 cm
Center of Gravity of the Nozzles	155.359 cm
Center of Gravity of the Tubesheet(s)	148.150 cm
Center of Gravity of the Tubes	148.150 cm
Center of Gravity of the Baffles	150.104 cm
Center of Gravity of Bare Shell New and Cold	148.050 cm
Center of Gravity of Bare Shell Corroded	148.050 cm
Vessel CG in the Operating Condition	149.142 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	149.271 cm
Vessel CG in the Test Condition	149.035 cm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

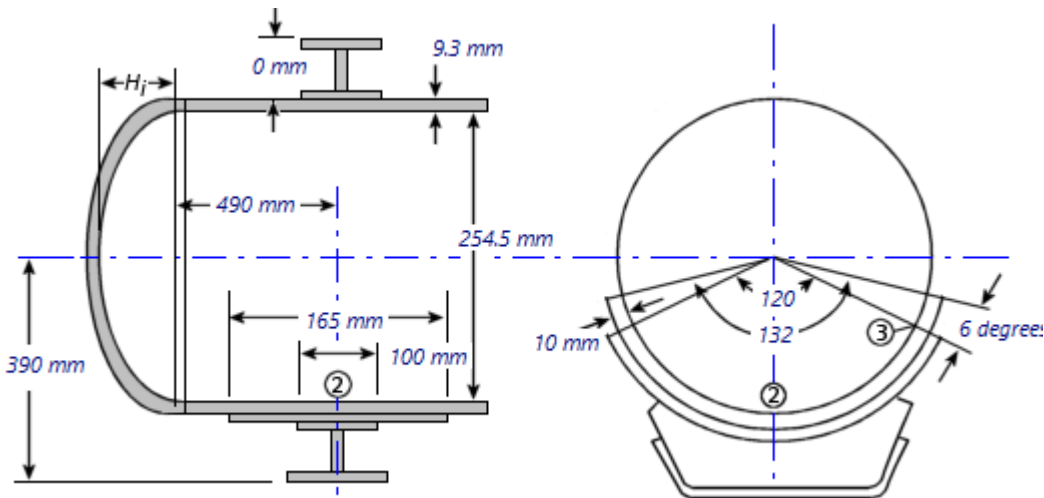


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	133.36	mm
Shell Thickness used in this Case	t	9.270	mm
Stiffened Vessel Length per 4.15.6	L	231.20	cm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	100.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	165.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1203.20	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	390.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		784.11	kgf
Pressure used in Analysis	P	5.621	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		39.23	1203.20
Long. Stress at Bottom of Midspan		80.34	1203.20
Long. Stress at Top of Saddles		269.72	1203.20
Long. Stress at Bottom of Saddles		-56.58	-1921.67
Long. Stress at Bottom of Saddles		56.58	1203.20
Tangential Shear in Shell		63.25	962.56
Circ. Stress at Horn of Saddle		31.79	1504.00
Circ. Compressive Stress in Shell		2.52	1203.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(197.4/2 + 0) \cdot 390.0/253.7455$$

$$= 455.2 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(18.76, 0) \cdot 390.0/1420.0002$$

$$= 5.2 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(55.49, 0) \cdot 390.0/1420.0002$$

$$= 15.2 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(55/2 + 0) \cdot 390.0/253.7455$$

$$= 127.9 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 329 + \max(5, 455, 15, 128)$$

$$= 784.1 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.273 \cdot 1.2)^2)$$

$$= 183.985 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	806.20	kgf
Transverse Shear Load	98.72	kgf
Longitudinal Shear Load	55.49	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(254.46/2 + 9.27 + 10.0)\sin(120.0/2), 2 \cdot 133.365)$$

$$= 253.745 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/(3\text{L}))]$$

$$= -784 \cdot 49.0 [1 - (1 - 49.0/231.2 + (13.336^2 - 0.0^2)/$$

$$(2 \cdot 49.0 \cdot 231.2)) / (1 + (4 \cdot 0.0)/(3 \cdot 231.2))]$$

$$= -78.4 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(3\text{L})) - 4a/L)$$

$$= 784 \cdot 231/4((1 + 2(13^2 - 0^2)/(231^2)) / (1 + (4 \cdot 0)/$$

$$(3 \cdot 231)) - 4 \cdot 49/231)$$

$$= 72.0 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$$

$$= 5.621 \cdot 133.365/(2 \cdot 6.27) - 72.0/(\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 39.23 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 5.621 \cdot 133.365 / (2 \cdot 6.27) + 72.0 / (\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 80.34 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 5.621 \cdot 133.365 / (2 \cdot 6.27) - 78.4 / (0.1066 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= 269.72 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 5.621 \cdot 133.365 / (2 \cdot 6.27) + 78.4 / (0.1923 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= -56.58 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm / 3))$$

$$= 784(231.2 - 2 \cdot 49.0) / (231.2 + (4 \cdot 0.0 / 3))$$

$$= 451.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 451.75 / (133.365 \cdot 6.27)$$

$$= 63.25 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(133.365 \cdot 6.27)}$$

$$= 22.555 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(100.0 + 1.56 \cdot \sqrt{(133.365 \cdot 6.27)}, 2 \cdot 49.0)$$

$$= 145.11 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(1407.131/1203.202, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma6, r$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 784 \cdot 0.1 / (145.111(6.27 + 1.0 \cdot 10.0))$$

$$= -2.52 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [$\sigma7, r$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -784 / (4(6.27 + 1.0 \cdot 10.0)145.111) -$$

$$3 \cdot 0.053 \cdot 784 / (2(6.27 + 1.0 \cdot 10.0)^2)$$

$$= -31.79 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 1420.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 1420.0(180.0 - 21.1)$$

$$= 2.847 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	300.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	100.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		180.0	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(130.23 \cdot 6.27)} + 165.0)^3) / 12$$

$$= 0.430 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 0.43 + 13.14 (105.474 - 3.135)^2$$

$$= 1376.672 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	209.6	6.3	3.1	13.1	4119.5	0.138E+04
Wearplate	165.0	10.0	11.3	16.5	18595.5	0.147E+04
Web	10.0	227.5	130.0	22.8	295795.5	0.112E+04
BasePlate	100.0	16.0	251.8	16.0	402832.0	0.343E+04
Totals	...	...	...	68.4	721342.6	0.739E+04

Distance to Centroid [C1]:

$$= AY / A$$

$$= 721343 / 68.391$$

$$= 105.474 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 784.113$$

$$= 159.5841 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.8884 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1391.2390 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 390.0 - 130.23 \cdot \sin(120.0/2) / 1.0472$$

$$= 282.301 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 45.0507 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 6.4313 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1545.8213 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(784 + 22) \cdot 100.0 / (2 \cdot 300.0 \cdot 1545.821)}$$

$$= 5.107 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (127.23 + 9.27 + 10.0)$$

$$= 253.745 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 253.7454 / (3 - 1)$$



= 126.873 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 126.8727 • 100.0/2
 = 63.436 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 806.195/(300.0 • 100.0)
 = 2.687 kgf/cm²

Axial Load [P]:

= Ap • Bp
 = 63.4 • 2.69
 = 170.474 kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 9.0 + 6.344
 = 15.344 cm²

Compressive Stress [Sc]:

= P/Ar
 = 170.5/15.3436
 = 11.110 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	99.2
Web	63.4	10.0	5.0	6.3	3171.8	55.1
Totals	...	...	...	15.3	52671.8	154.

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 100.0 - 10.0
 = 90.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 52671.812/15.344
 = 34.328 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(34.328, 100.0 - 34.328)
 = 65.672 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{154.3/15.344}$
 = 31.712 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 390.0 - cos(120.0/2)(127.23 + 9.27 + 10.0) - 16.0
 = 300.750 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 1973952/2318.7}$
 = 129.631

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 300.75/31.712



= 9.484

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 55.5 / (2 \cdot 300.0) \cdot 126.873 \cdot 300.75 / 2$$

$$= 1.764 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (9.4838 < 129.6307) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.48)^2 / (2 \cdot 129.63^2)) 2319 /$$

$$(5/3 + 3 \cdot (9.48) / (8 \cdot 129.63) - (9.48^3) / (8 \cdot 129.63^3))$$

$$= 1365 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 11.11 / 1365.09 + (1.76 \cdot 65.672 / 1543021) / 1545.82$$

$$= 0.013$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	138.
Web	126.9	10.0	5.0	12.7	6343.6	55.7
Totals	...	...	...	21.7	55843.6	193.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 55843.629 / 21.687$$

$$= 25.749 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(25.749, \text{abs}(100.0 - 25.749))$$

$$= 74.251 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 390.0 - \cos(146.5/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 227.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(193.4 / 21.687)}$$

$$= 29.865 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 227.5 / 29.865$$

$$= 7.618$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 55.491 / (2 \cdot 300.0)$$

$$= 0.092 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.092 \cdot 126.873 \cdot 227.5$$

$$= 2.669 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2.669 / 26051.607$$

$$= 10.247 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (7.6176 < 129.6307) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.62)^2 / (2 \cdot 129.63^2)) 2319 /$$



$$\begin{aligned} & (5/3+3 \cdot (7.62)/(8 \cdot 129.63) - (7.62^3)/(8 \cdot 129.63^3)) \\ & = 1371 \text{ kgf/cm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} & = S_c/S_{c_a} + (M_{base,c} \cdot C1/I) / S_{b_a} \\ & = 17.81/1370.73 + (2.67 \cdot 74.251/193.434) / 1545.82 \\ & = 0.020 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		25.0 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	30.0 mm
Nominal Bolt Diameter	Bnd	20.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	0.8874 cm ²
Saddle Load QO (Weight)	QO	351.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	15.2 kgf
Maximum Transverse Force	Ft	98.7 kgf
Maximum Longitudinal Force	Fl	55.5 kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$\begin{aligned} & = F_l / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ & = 55.5 / (0.887 \cdot 2) \\ & = 31.3 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2. \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$\begin{aligned} & = F_t / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ & = 98.7 / (0.887 \cdot 2) \\ & = 55.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2. \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned} & = F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\ & = 55.49 / (1052.29 \cdot 2.0) \\ & = 0.0264 \text{ cm}^2 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned} & = B \cdot F_t + \text{Sum of X Moments} \\ & = 390.0 \cdot 98.72 + 0.0 \\ & = 38.50 \text{ kgf-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} & = R_{mom} / QO \\ & = 38.5/351.0 \\ & = 109.69 \text{ mm} > B_{plen}/6 \text{ --> Uplift in Transverse direction} \end{aligned}$$

f = B_{plen} / 2 - Edgedis

$$\begin{aligned} & = 300.0/2 - 30.0 \\ & = 120.00 \text{ mm} \end{aligned}$$

K1 = 3 (e - 0.5 • B_{plen})

$$\begin{aligned} & = 3 (109.69 - 0.5 \cdot 300.0) \\ & = -120.92 \text{ mm} \end{aligned}$$

K2 = 6 • n1 • A_t / B_{pwid} • (f + e)

$$\begin{aligned} & = 6 \cdot 1.0 \cdot 0.89/100.0 \cdot (120.0 + 109.69) \\ & = 1223.00 \text{ mm}^2 \end{aligned}$$



$$\begin{aligned}K3 &= -K2 \cdot (0.5 \cdot Bplen + f) \\&= -1223.0 \cdot (0.5 \cdot 300.0 + 120.0) \\&= -330214.10 \text{ mm}^3\end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned}Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\Y^3 + -120.92 \cdot Y^2 + 1223.0 \cdot Y + -330214.12 &= 0 \\Y &= 130.86 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Num} &= (Bplen / 2 - Y / 3 - e) \\&= (300.0/2 - 130.86/3 - 109.69) \\&= -3.31\end{aligned}$$

$$\begin{aligned}\text{Denom} &= (Bplen / 2 - Y / 3 + f) \\&= (300.0/2 - 130.86/3 + 120.0) \\&= 226.38\end{aligned}$$

$$\begin{aligned}\text{Total Bolt Tension Force [Tforce]:} \\&= -QO \cdot \text{Num} / \text{Denom} \\&= -351.0 \cdot -3.31/226.38 \\&= 5.14 \text{ kgf}\end{aligned}$$

$$\begin{aligned}\text{Bolt Area Required due to Transverse Load [Bltareart]:} \\&= \text{Tforce} / (\text{Stba} \cdot \text{Nbt}) \\&= 5.14 / (1753.82 \cdot 1.0) \\&= 0.0029 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Required Area of a Single Bolt [Bltarear]:} \\&= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\&= \max[0.0, 0.0264, 0.0029] \\&= 0.0264 \text{ cm}^2\end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned}\text{Bearing Pressure (fc)} \\&= 2(QO + \text{Tforce}) / (Y \cdot \text{Bpwid}) \\&= 2(351.0 + 5.14) / (130.86 \cdot 100.0) \\&= 5.44 \text{ kgf/cm}^2\end{aligned}$$

$$\begin{aligned}\text{Distance from Baseplate Edge to the Web [ADIST]:} \\&= (Bplen - \text{WebLength}) / 2 \\&= (300.0 - 249.2) / 2 \\&= 25.4000 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Overturning Moment due To Bolt Tension [Mt]:} \\&= \text{Tforce} \cdot \text{Adist} \\&= 5.14 \cdot 2.54 \\&= 0.13 \text{ kgf-m.}\end{aligned}$$

$$\begin{aligned}\text{Equivalent Bearing Pressure (f1):} \\&= fc \cdot (Y - \text{Adist}) / Y \\&= 5.44 \cdot (130.86 - 25.4) / 130.86 \\&= 4.39 \text{ kgf/cm}^2\end{aligned}$$

$$\begin{aligned}\text{Overturning Moment due to Bearing Pressure [Mc]:} \\&= (\text{Adist}^2 \cdot \text{Bpwid} / 6) \cdot (f1 + 2 \cdot fc) \\&= (25.4^2 \cdot 100.0 / 6) \cdot (4.39 + 2 \cdot 5.44) \\&= 1.64 \text{ kgf-m.}\end{aligned}$$

$$\begin{aligned}\text{Baseplate Required Thickness [Treq]:} \\&= \sqrt{(6 \cdot \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} \cdot \text{Sba}))} \\&= \sqrt{(6 \cdot \max(0.13, 1.64) / (100.0 \cdot 2110.7))} \\&= 2.1606 \text{ mm}\end{aligned}$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

$$\text{Govrn. thk, tg} = 9.27, \text{ tr} = 6.267, \text{ c} = 3.0 \text{ mm, } E^* = 1.0$$



Thickness Ratio = $tr * E * (tg - c) = 1.$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	133.36	mm
Shell Thickness used in this Case	t	9.270	mm
Stiffened Vessel Length per 4.15.6	L	231.20	cm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	100.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	165.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm²
Shell Allowable Stress used in Calculation		1203.20	kgf/cm²
Head Allowable Stress used in Calculation		1407.13	kgf/cm²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	390.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		754.51	kgf
Pressure used in Analysis	P	5.621	kgf/cm²

Horizontal Vessel Analysis Results:		Actual kgf/cm²	Allowable kgf/cm²
Long. Stress at Top of Midspan		40.00	1203.20
Long. Stress at Bottom of Midspan		79.56	1203.20
Long. Stress at Top of Saddles		261.79	1203.20
Long. Stress at Bottom of Saddles		-52.18	-1921.67
Long. Stress at Bottom of Saddles		52.18	1203.20
Tangential Shear in Shell		60.86	962.56
Circ. Stress at Horn of Saddle		30.59	1504.00
Circ. Compressive Stress in Shell		2.43	1203.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(197.4/2 + 0) \cdot 390.0/253.7455$
 $= 455.2 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(18.76, 0) \cdot 390.0/1420.0002$
 $= 5.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(55.49, 0) \cdot 390.0/1420.0002$
 $= 15.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(55/2 + 0) \cdot 390.0/253.7455$
 $= 127.9 \text{ kgf}$



Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 299 + \max(5, 455, 15, 128)$$

$$= 754.5 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.273 \cdot 1.2)^2)$$

$$= 183.985 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	776.59	kgf
Transverse Shear Load	98.72	kgf
Longitudinal Shear Load	55.49	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(254.46/2 + 9.27 + 10.0)\sin(120.0/2), 2 \cdot 133.365)$$

$$= 253.745 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -755 \cdot 49.0 [1 - (1 - 49.0/231.2 + (13.336^2 - 0.0^2)/(2 \cdot 49.0 \cdot 231.2))/(1 + (4 \cdot 0.0)/(3 \cdot 231.2))]$$

$$= -75.5 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 755 \cdot 231.4((1 + 2(13^2 - 0^2)/(231^2))/(1 + (4 \cdot 0)/(3 \cdot 231)) - 4 \cdot 49/231)$$

$$= 69.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 5.621 \cdot 133.365/(2 \cdot 6.27) - 69.3/(\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 40.00 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 5.621 \cdot 133.365/(2 \cdot 6.27) + 69.3/(\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 79.56 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 5.621 \cdot 133.365/(2 \cdot 6.27) - 75.5/(0.1066 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= 261.79 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 5.621 \cdot 133.365/(2 \cdot 6.27) + 75.5/(0.1923 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= -52.18 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 755(231.2 - 2 \cdot 49.0)/(231.2 + (4 \cdot 0.0/3))$$

$$= 434.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:



$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 434.69 / (133.365 \cdot 6.27)$$

$$= 60.86 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(133.365 \cdot 6.27)}$$

$$= 22.555 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(100.0 + 1.56 \cdot \sqrt{(133.365 \cdot 6.27)}, 2 \cdot 49.0)$$

$$= 145.11 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(1407.131/1203.202, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 755 \cdot 0.1 / (145.111(6.27 + 1.0 \cdot 10.0))$$

$$= -2.43 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.28) [σ7,r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -755 / (4(6.27 + 1.0 \cdot 10.0)145.111) -$$

$$3 \cdot 0.053 \cdot 755 / (2(6.27 + 1.0 \cdot 10.0)^2)$$

$$= -30.59 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	300.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	100.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(130.23 \cdot 6.27)} + 165.0)6.27^3) / 12$$

$$= 0.430 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 0.43 + 13.14(105.474 - 3.135)^2$$

$$= 1376.672 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	209.6	6.3	3.1	13.1	4119.5	0.138E+04
Wearplate	165.0	10.0	11.3	16.5	18595.5	0.147E+04
Web	10.0	227.5	130.0	22.8	295795.5	0.112E+04
BasePlate	100.0	16.0	251.8	16.0	402832.0	0.343E+04
Totals	...	...	...	68.4	721342.6	0.739E+04

Distance to Centroid [C1]:

$$= AY / A$$

$$= 721343/68.391$$

$$= 105.474 \text{ mm}$$



Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120.0/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 754.512 \\ &= 153.5596 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 2.7794 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1391.2390 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 390.0 - 130.23 \cdot \sin(120.0/2) / 1.0472 \\ &= 282.301 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 43.3500 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 6.1885 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1545.8213 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{3(755 + 22)100.0 / (2 \cdot 300.0 \cdot 1545.821)} \\ &= 5.012 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (127.23 + 9.27 + 10.0) \\ &= 253.745 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 253.7454 / (3 - 1) \\ &= 126.873 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 126.8727 \cdot 100.0/2 \\ &= 63.436 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 776.594 / (300.0 \cdot 100.0) \\ &= 2.589 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 63.4 \cdot 2.59 \\ &= 164.214 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 9.0 + 6.344 \\ &= 15.344 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P/Ar \\ &= 164.2/15.3436 \\ &= 10.702 \text{ kgf/cm}^2 \end{aligned}$$



Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	99.2
Web	63.4	10.0	5.0	6.3	3171.8	55.1
Totals	...	...	...	15.3	52671.8	154.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 100.0 - 10.0$$

$$= 90.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 52671.812 / 15.344$$

$$= 34.328 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(34.328, 100.0 - 34.328)$$

$$= 65.672 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 154.3 / 15.344 }$$

$$= 31.712 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 390.0 - \cos(120.0 / 2) (127.23 + 9.27 + 10.0) - 16.0$$

$$= 300.750 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 \cdot \pi^2 \cdot 1973952 / 2318.7 }$$

$$= 129.631$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 300.75 / 31.712$$

$$= 9.484$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 55.5 / (2 \cdot 300.0) \cdot 126.873 \cdot 300.75 / 2$$

$$= 1.764 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.4838 < 129.6307$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.48)^2 / (2 \cdot 129.63^2)) 2319 / (5/3 + 3 \cdot (9.48) / (8 \cdot 129.63) - (9.48^3) / (8 \cdot 129.63^3))$$

$$= 1365 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 10.7 / 1365.09 + (1.76 \cdot 65.672 / 1543021) / 1545.82$$

$$= 0.013$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	138.
Web	126.9	10.0	5.0	12.7	6343.6	55.7
Totals	...	...	...	21.7	55843.6	193.



Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 55843.629/21.687$$

$$= 25.749 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(25.749, \text{abs}(100.0 - 25.749))$$

$$= 74.251 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 390.0 - \cos(146.5/2)(16.0 + 0.0 + 0.0) - 0.0$$

$$= 227.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 193.4/21.687 }$$

$$= 29.865 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 227.5/29.865$$

$$= 7.618$$

Unit Force [Force,u]:

$$= F1 / (2 \cdot \text{Baseplate Length})$$

$$= 55.491/(2 \cdot 300.0)$$

$$= 0.092 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.092 \cdot 126.873 \cdot 227.5$$

$$= 2.669 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2.669/26051.607$$

$$= 10.247 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (7.6176 < 129.6307) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.62)^2 / (2 \cdot 129.63^2)) 2319 /$$

$$(5/3 + 3 \cdot (7.62) / (8 \cdot 129.63) - (7.62^3) / (8 \cdot 129.63^3))$$

$$= 1371 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 17.15/1370.73 + (2.67 \cdot 74.251/193.434) / 1545.82$$

$$= 0.019$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1396.96 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	30.0 mm
Nominal Bolt Diameter	Bnd	20.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	0.8874 cm ²
Saddle Load Q0 (Weight)	Q0	321.4 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	15.2 kgf
Maximum Transverse Force	Ft	98.7 kgf
Maximum Longitudinal Force	F1	55.5 kgf
Saddle Bolted to Steel Foundation	No	



Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 55.5 / (0.887 \cdot 2)$$

$$= 31.3 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 98.7 / (0.887 \cdot 2)$$

$$= 55.6 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_O > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 55.49 / (1101.26 \cdot 2.0)$$

$$= 0.0252 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 390.0 \cdot 98.72 + 0.0$$

$$= 38.50 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_O$$

$$= 38.5 / 321.39$$

$$= 119.80 \text{ mm} > B_{plen} / 6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 300.0 / 2 - 30.0$$

$$= 120.00 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= E_S / E_C$$

$$= 2074057 / 219499.77$$

$$= 9.45$$

K1 = 3 (e - 0.5 • Bplen)

$$= 3 (119.8 - 0.5 \cdot 300.0)$$

$$= -90.61 \text{ mm}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$= 6 \cdot 9.45 \cdot 0.89 / 100.0 \cdot (120.0 + 119.8)$$

$$= 12064.47 \text{ mm}^2$$

K3 = -K2 • (0.5 • Bplen + f)

$$= -12064.47 \cdot (0.5 \cdot 300.0 + 120.0)$$

$$= -3257437.50 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K_1 \cdot Y^2 + K_2 \cdot Y + K_3 = 0$$

$$Y^3 + -90.61 \cdot Y^2 + 12064.47 \cdot Y + -3257437.50 = 0$$

$$Y = 152.10 \text{ mm}$$

Num = (Bplen / 2 - Y / 3 - e)

$$= (300.0 / 2 - 152.1 / 3 - 119.8)$$

$$= -20.50$$

Denom = (Bplen / 2 - Y / 3 + f)

$$= (300.0 / 2 - 152.1 / 3 + 120.0)$$

$$= 219.30$$

Total Bolt Tension Force [Tforce]:

$$= - Q_O \cdot \text{Num} / \text{Denom}$$

$$= - 321.39 \cdot -20.5 / 219.3$$

$$= 30.04 \text{ kgf}$$



Bolt Area Required due to Transverse Load [Bltareart]:
= Tforce / (Stba • Nbt)
= 30.04/(1396.96 • 1.0)
= 0.0215 cm²

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.0252, 0.0215]
= 0.0252 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
= 2(QO + Tforce) / (Y • Bpwid)
= 2(321.39 + 30.04)/(152.1 • 100.0)
= 4.62 kgf/cm²

Distance from Baseplate Edge to the Web [ADIST]:
= (Bplen - Weblngh) / 2
= (300.0 - 249.2)/2
= 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
= Tforce • Adist
= 30.04 • 2.54
= 0.76 kgf-m.

Equivalent Bearing Pressure (f1):
= fc • (Y - Adist) / Y
= 4.62 • (152.1 - 25.4)/152.1
= 3.85 kgf/cm²

Overturning Moment due to Bearing Pressure [Mc]:
= (Adist² • Bpwid / 6) • (f1 + 2 • fc)
= (25.4² • 100.0/6) • (3.85 + 2 • 4.62)
= 1.41 kgf-m.

Baseplate Required Thickness [Treq]:
= $\sqrt{6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba)}$
= $\sqrt{6 \cdot \max(0.76, 1.41) / (100.0 \cdot 2110.7)}$
= 2.0004 mm

MDMT Calculations for: Right Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 9.27, tr = 6.267, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 1., Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

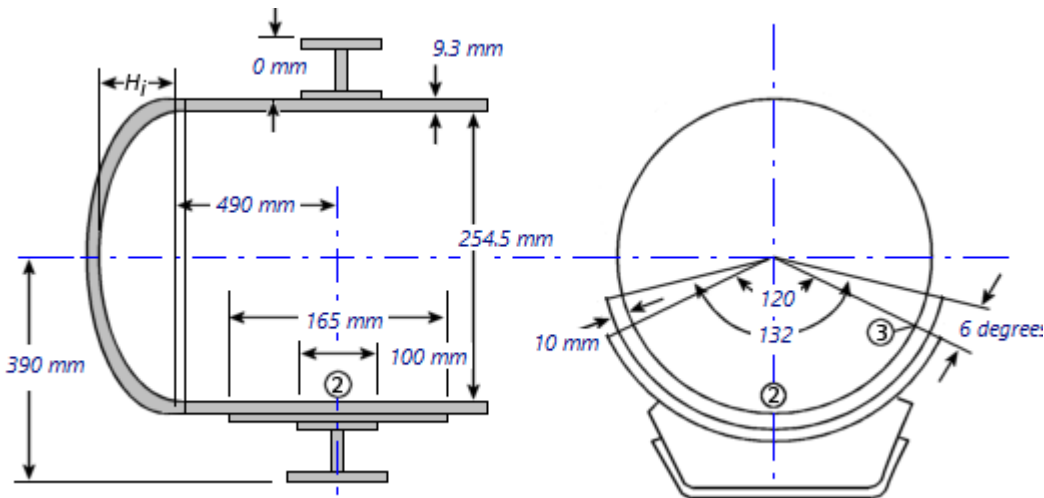


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	133.36	mm
Shell Thickness used in this Case	t	9.270	mm
Stiffened Vessel Length per 4.15.6	L	231.20	cm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	100.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	165.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2324.83	kgf/cm ²
Head Allowable Stress used in Calculation		2324.83	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	390.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		496.74	kgf
Pressure used in Analysis	P	8.426	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		76.59	2324.83
Long. Stress at Bottom of Midspan		102.64	2324.83
Long. Stress at Top of Saddles		222.61	2324.83
Long. Stress at Bottom of Saddles		15.90	2324.83
Tangential Shear in Shell		40.07	1304.35
Circ. Stress at Horn of Saddle		32.88	3487.25
Circ. Compressive Stress in Shell		2.11	2324.83

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(65.2/2 + 0) \cdot 390.0/253.7455$$

$$= 150.2 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(6.19, 0) \cdot 390.0/1420.0002$$

$$= 1.7 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 347 + \max(2, 150, 0, 0)$$

$$= 496.7 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.273 \cdot 1.2)^2)$$

$$= 183.985 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	518.82	kgf
Transverse Shear Load	32.58	kgf
Longitudinal Shear Load	6.19	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(254.46/2 + 9.27 + 10.0)\sin(120.0/2), 2 \cdot 133.365)$$

$$= 253.745 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -497 \cdot 49.0 [1 - (1 - 49.0/231.2 + (13.336^2 - 0.0^2)/(2 \cdot 49.0 \cdot 231.2))/(1 + (4 \cdot 0.0)/(3 \cdot 231.2))]$$

$$= -49.7 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 497 \cdot 231.4((1 + 2(13^2 - 0^2)/(231^2))/(1 + (4 \cdot 0)/(3 \cdot 231)) - 4 \cdot 49/231)$$

$$= 45.6 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 8.426 \cdot 133.365/(2 \cdot 6.27) - 45.6/(\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 76.59 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 8.426 \cdot 133.365/(2 \cdot 6.27) + 45.6/(\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 102.64 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 8.426 \cdot 133.365/(2 \cdot 6.27) - 49.7/(0.1066 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= 222.61 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$



$$= 8.426 \cdot 133.365 / (2 \cdot 6.27) + 49.7 / (0.1923 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= 15.90 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 497(231.2 - 2 \cdot 49.0) / (231.2 + (4 \cdot 0.0 / 3))$$

$$= 286.2 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 286.18 / (133.365 \cdot 6.27)$$

$$= 40.07 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(133.365 \cdot 6.27)}$$

$$= 22.555 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(100.0 + 1.56 \cdot \sqrt{(133.365 \cdot 6.27)}, 2 \cdot 49.0)$$

$$= 145.11 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1407.131 / 2324.831, 1)$$

$$= 0.6053$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 497 \cdot 0.1 / (145.111(6.27 + 0.605 \cdot 10.0))$$

$$= -2.11 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -497 / (4(6.27 + 0.605 \cdot 10.0)145.111) -$$

$$3 \cdot 0.053 \cdot 497 / (2(6.27 + 0.605 \cdot 10.0)^2)$$

$$= -32.88 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	300.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	100.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(130.23 \cdot 6.27)} + 165.0)6.27^3) / 12$$

$$= 0.430 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 0.43 + 13.14(105.474 - 3.135)^2$$

$$= 1376.672 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	209.6	6.3	3.1	13.1	4119.5	0.138E+04
Wearplate	165.0	10.0	11.3	16.5	18595.5	0.147E+04
Web	10.0	227.5	130.0	22.8	295795.5	0.112E+04
BasePlate	100.0	16.0	251.8	16.0	402832.0	0.343E+04

Totals	...	...	...	68.4	721342.6	0.739E+04
--------	-----	-----	-----	------	----------	-----------

Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 721343/68.391 \\ &= 105.474 \text{ mm} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120.0/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 496.739 \\ &= 101.0971 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 1.8298 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 390.0 - 130.23 \cdot \sin(120.0/2) / 1.0472 \\ &= 282.301 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 28.5398 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 4.0742 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{(3(497 + 22) \cdot 100.0) / (2 \cdot 300.0 \cdot 1767.415)} \\ &= 3.831 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (127.23 + 9.27 + 10.0) \\ &= 253.745 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 253.7454 / (3 - 1) \\ &= 126.873 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 126.8727 \cdot 100.0/2 \\ &= 63.436 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 518.821 / (300.0 \cdot 100.0) \\ &= 1.729 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 63.4 \cdot 1.73 \\ &= 109.707 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 9.0 + 6.344 \end{aligned}$$



$$= 15.344 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 109.7/15.3436$$

$$= 7.150 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	99.2
Web	63.4	10.0	5.0	6.3	3171.8	55.1
Totals	...	...	...	15.3	52671.8	154.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 100.0 - 10.0$$

$$= 90.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 52671.812/15.344$$

$$= 34.328 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(34.328, 100.0 - 34.328)$$

$$= 65.672 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(154.3/15.344)}$$

$$= 31.712 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 390.0 - \cos(120.0/2) (127.23 + 9.27 + 10.0) - 16.0$$

$$= 300.750 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 1973952/2651.1)}$$

$$= 121.232$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 300.75/31.712$$

$$= 9.484$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 6.2 / (2 \cdot 300.0) \cdot 126.873 \cdot 300.75/2$$

$$= 0.197 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.4838 < 121.2323$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.48)^2 / (2 \cdot 121.23^2)) 2651 / (5/3 + 3 \cdot (9.48) / (8 \cdot 121.23) - (9.48^3) / (8 \cdot 121.23^3))$$

$$= 1558 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 7.15/1558.43 + (0.2 \cdot 65.672/1543021) / 1767.42$$

$$= 0.005$$

Check of Inside Ribs:



Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	138.
Web	126.9	10.0	5.0	12.7	6343.6	55.7
Totals	...	...	...	21.7	55843.6	193.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 55843.629/21.687$$

$$= 25.749 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(25.749, \text{abs}(100.0 - 25.749))$$

$$= 74.251 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 390.0 - \cos(146.5/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 227.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (193.4/21.687) }$$

$$= 29.865 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 227.5/29.865$$

$$= 7.618$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 6.191/(2 \cdot 300.0)$$

$$= 0.010 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.01 \cdot 126.873 \cdot 227.5$$

$$= 0.298 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 0.298/26051.607$$

$$= 1.143 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (7.6176 < 121.2323) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.62)^2 / (2 \cdot 121.23^2)) 2651 /$$

$$(5/3 + 3 \cdot (7.62) / (8 \cdot 121.23) - (7.62^3) / (8 \cdot 121.23^3))$$

$$= 1565 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 11.46/1565.43 + (0.3 \cdot 74.251/193.434) / 1767.42$$

$$= 0.008$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	30.0 mm
Nominal Bolt Diameter	Bnd	20.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	0.8874 cm ²



Saddle Load Q0 (Weight)	Q0	368.6 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1.7 kgf
Maximum Transverse Force	Ft	32.6 kgf
Maximum Longitudinal Force	F1	6.2 kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6.2 / (0.887 \cdot 2)$$

$$= 3.5 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 32.6 / (0.887 \cdot 2)$$

$$= 18.4 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 6.19 / (1052.29 \cdot 2.0)$$

$$= 0.0029 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 390.0 \cdot 32.58 + 0.0$$

$$= 12.71 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 12.71 / 368.61$$

$$= 34.47 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0029, 0.0]$$

$$= 0.0029 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	133.36	mm
Shell Thickness used in this Case	t	9.270	mm
Stiffened Vessel Length per 4.15.6	L	231.20	cm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	100.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	165.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2324.83	kgf/cm ²
Head Allowable Stress used in Calculation		2324.83	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	390.00	mm



Coefficient of Friction mu 0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 462.91 kgf
 Pressure used in Analysis P 8.426 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	77.48	2324.83
Long. Stress at Bottom of Midspan	101.75	2324.83
Long. Stress at Top of Saddles	213.55	2324.83
Long. Stress at Bottom of Saddles	20.92	2324.83
Tangential Shear in Shell	37.34	1304.35
Circ. Stress at Horn of Saddle	30.64	3487.25
Circ. Compressive Stress in Shell	1.97	2324.83

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3.0(65.2/2 + 0) • 390.0/253.7455
 = 150.2 kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(6.19, 0) • 390.0/1420.0002
 = 1.7 kgf

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 313 + max(2, 150, 0, 0)
 = 462.9 kgf

Longitudinal Wind Force [Fl]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 1984.306(0.0 + $\pi/4$ (0.273 • 1.2)²)
 = 183.985 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 484.99 kgf
 Transverse Shear Load 32.58 kgf
 Longitudinal Shear Load 6.19 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(254.46/2 + 9.27 + 10.0)sin(120.0/2), 2•133.365)
 = 253.745 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -463•49.0[1-(1-49.0/231.2+(13.336²-0.0²)/
 (2•49.0•231.2))/(1+(4•0.0)/(3•231.2))]
 = -46.3 kgf-m.



Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L \right)$$

$$= 463 \cdot 231 / 4 \left((1 + 2(13^2 - 0^2) / (231^2)) / (1 + (4 \cdot 0) / (3 \cdot 231)) - 4 \cdot 49 / 231 \right)$$

$$= 42.5 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 8.426 \cdot 133.365 / (2 \cdot 6.27) - 42.5 / (\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 77.48 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 8.426 \cdot 133.365 / (2 \cdot 6.27) + 42.5 / (\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 101.75 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 8.426 \cdot 133.365 / (2 \cdot 6.27) - 46.3 / (0.1066 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= 213.55 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 8.426 \cdot 133.365 / (2 \cdot 6.27) + 46.3 / (0.1923 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= 20.92 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 463(231.2 - 2 \cdot 49.0) / (231.2 + (4 \cdot 0.0 / 3))$$

$$= 266.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 266.69 / (133.365 \cdot 6.27)$$

$$= 37.34 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(133.365 \cdot 6.27)}$$

$$= 22.555 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(100.0 + 1.56 \cdot \sqrt{(133.365 \cdot 6.27)}, 2 \cdot 49.0)$$

$$= 145.11 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr / S, 1)$$

$$= \min(1407.131 / 2324.831, 1)$$

$$= 0.6053$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 463 \cdot 0.1 / (145.11(6.27 + 0.605 \cdot 10.0))$$

$$= -1.97 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -463 / (4(6.27 + 0.605 \cdot 10.0)145.111) - 3 \cdot 0.053 \cdot 463 / (2(6.27 + 0.605 \cdot 10.0)^2)$$

$$= -30.64 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	300.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	100.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm

Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(130.23 \cdot 6.27)} + 165.0)^3) / 12$
 $= 0.430 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 0.43 + 13.14 (105.474 - 3.135)^2$
 $= 1376.672 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	209.6	6.3	3.1	13.1	4119.5	0.138E+04
Wearplate	165.0	10.0	11.3	16.5	18595.5	0.147E+04
Web	10.0	227.5	130.0	22.8	295795.5	0.112E+04
BasePlate	100.0	16.0	251.8	16.0	402832.0	0.343E+04
Totals	...	...	...	68.4	721342.6	0.739E+04

Distance to Centroid [C1]:
 $= AY / A$
 $= 721343 / 68.391$
 $= 105.474 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 462.908$
 $= 94.2119 \text{ kgf}$

Tension Stress, St = (Fh/As) = 1.7052 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1590.6736 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 390.0 - 130.23 \cdot \sin(120.0/2) / 1.0472$
 $= 282.301 \text{ mm}$

Bending Moment, M = Fh * d = 26.5961 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 3.7968 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1767.4150 kgf/cm²

Minimum Thickness of Baseplate per Moss:
 $= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(463 + 22)100.0 / (2 \cdot 300.0 \cdot 1767.415))}$
 $= 3.704 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (127.23 + 9.27 + 10.0)$
 $= 253.745 \text{ mm}$



Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 253.7454/(3 - 1)
 = 126.873 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 126.8727 • 100.0/2
 = 63.436 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 484.99/(300.0 • 100.0)
 = 1.617 kgf/cm²

Axial Load [P]:
 = Ap • Bp
 = 63.4 • 1.62
 = 102.553 kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 9.0 + 6.344
 = 15.344 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 102.6/15.3436
 = 6.684 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	99.2
Web	63.4	10.0	5.0	6.3	3171.8	55.1
Totals	...	...	...	15.3	52671.8	154.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 100.0 - 10.0
 = 90.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 52671.812/15.344
 = 34.328 mm

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(34.328, 100.0 - 34.328)
 = 65.672 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{154.3/15.344}$
 = 31.712 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 390.0 - cos(120.0/2)(127.23 + 9.27 + 10.0) - 16.0
 = 300.750 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 1973952/2651.1}$
 = 121.232



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 300.75/31.712$$

$$= 9.484$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 6.2 / (2 \cdot 300.0) \cdot 126.873 \cdot 300.75/2$$

$$= 0.197 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ (9.4838 < 121.2323) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (9.48)^2 / (2 \cdot 121.23^2)) 2651 /$$

$$(5/3 + 3 \cdot (9.48) / (8 \cdot 121.23) - (9.48^3) / (8 \cdot 121.23^3))$$

$$= 1558 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.68/1558.43 + (0.2 \cdot 65.672/1543021) / 1767.42$$

$$= 0.005$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	138.
Web	126.9	10.0	5.0	12.7	6343.6	55.7
Totals	...	...	...	21.7	55843.6	193.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 55843.629/21.687$$

$$= 25.749 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(25.749, \text{abs}(100.0 - 25.749))$$

$$= 74.251 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 390.0 - \cos(146.5/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 227.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (193.4/21.687) }$$

$$= 29.865 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 227.5/29.865$$

$$= 7.618$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 6.191 / (2 \cdot 300.0)$$

$$= 0.010 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.01 \cdot 126.873 \cdot 227.5$$

$$= 0.298 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 0.298/26051.607$$

$$= 1.143 \text{ kgf/cm}^2$$



Compressive Allowable, $KL/r < C_c$ (7.6176 < 121.2323) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.62)^2 / (2 \cdot 121.23^2)) 2651 / (5/3 + 3 \cdot (7.62) / (8 \cdot 121.23) - (7.62^3) / (8 \cdot 121.23^3))$$

$$= 1565 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 10.71 / 1565.43 + (0.3 \cdot 74.251 / 193.434) / 1767.42$$

$$= 0.007$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	30.0 mm
Nominal Bolt Diameter	Bnd	20.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	0.8874 cm ²
Saddle Load QO (Weight)	QO	334.8 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1.7 kgf
Maximum Transverse Force	Ft	32.6 kgf
Maximum Longitudinal Force	Fl	6.2 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6.2 / (0.887 \cdot 2)$$

$$= 3.5 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 32.6 / (0.887 \cdot 2)$$

$$= 18.4 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 6.19 / (1052.29 \cdot 2.0)$$

$$= 0.0029 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 390.0 \cdot 32.58 + 0.0$$

$$= 12.71 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 12.71 / 334.77$$

$$= 37.95 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0029, 0.0]$$

$$= 0.0029 \text{ cm}^2$$



Review notes about nozzle loadings and supports in the Warnings report.

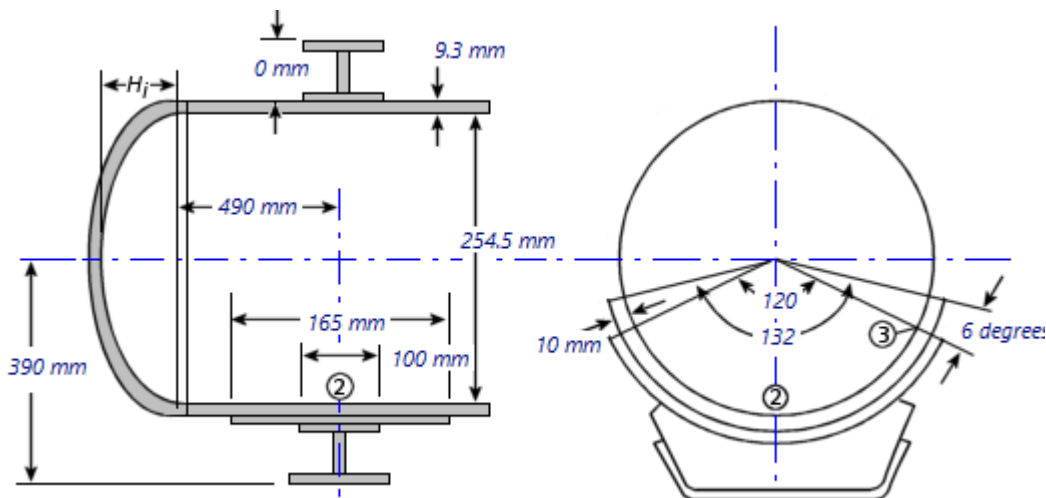
PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	133.36	mm
Shell Thickness used in this Case	t	9.270	mm
Stiffened Vessel Length per 4.15.6	L	231.20	cm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	100.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	165.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1203.20	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	390.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		784.11	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		-31.77	-1921.67
Long. Stress at Top of Midspan		31.77	1203.20
Long. Stress at Bottom of Midspan		9.34	1203.20
Long. Stress at Top of Saddles		198.72	1203.20
Long. Stress at Bottom of Saddles		-127.58	-1921.67
Long. Stress at Bottom of Saddles		127.58	1203.20
Tangential Shear in Shell		63.25	962.56
Circ. Stress at Horn of Saddle		31.79	1504.00
Circ. Compressive Stress in Shell		2.52	1203.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(197.4/2 + 0) \cdot 390.0/253.7455$
 $= 455.2 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(18.76, 0) \cdot 390.0/1420.0002$
 $= 5.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(55.49, 0) \cdot 390.0/1420.0002$
 $= 15.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(55/2 + 0) \cdot 390.0/253.7455$
 $= 127.9 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 329 + \max(5, 455, 15, 128)$
 $= 784.1 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.306(0.0 + \pi/4(0.273 \cdot 1.2)^2)$
 $= 183.985 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	806.20	kgf
Transverse Shear Load	98.72	kgf
Longitudinal Shear Load	55.49	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(254.46/2 + 9.27 + 10.0)\sin(120.0/2), 2 \cdot 133.365)$
 $= 253.745 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$
 $= -784 \cdot 49.0 [1 - (1 - 49.0/231.2 + (13.336^2 - 0.0^2)/(2 \cdot 49.0 \cdot 231.2))/(1 + (4 \cdot 0.0)/(3 \cdot 231.2))]$
 $= -78.4 \text{ kgf-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$
 $= 784 \cdot 231/4((1 + 2(13^2 - 0^2)/(231^2))/(1 + (4 \cdot 0)/(3 \cdot 231)) - 4 \cdot 49/231)$
 $= 72.0 \text{ kgf-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$
 $= -1.055 \cdot 133.365/(2 \cdot 6.27) - 72.0/(\pi \cdot 133.4^2 \cdot 6.27)$
 $= -31.77 \text{ kgf/cm}^2$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 133.365 / (2 \cdot 6.27) + 72.0 / (\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 9.34 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 133.365 / (2 \cdot 6.27) - 78.4 / (0.1066 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= 198.72 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 133.365 / (2 \cdot 6.27) + 78.4 / (0.1923 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= -127.58 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 784(231.2 - 2 \cdot 49.0) / (231.2 + (4 \cdot 0.0 / 3))$$

$$= 451.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 451.75 / (133.365 \cdot 6.27)$$

$$= 63.25 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(133.365 \cdot 6.27)}$$

$$= 22.555 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(100.0 + 1.56 \cdot \sqrt{(133.365 \cdot 6.27)}, 2 \cdot 49.0)$$

$$= 145.11 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1407.131 / 1203.202, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 784 \cdot 0.1 / (145.111(6.27 + 1.0 \cdot 10.0))$$

$$= -2.52 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -784 / (4(6.27 + 1.0 \cdot 10.0)145.111) -$$

$$3 \cdot 0.053 \cdot 784 / (2(6.27 + 1.0 \cdot 10.0)^2)$$

$$= -31.79 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	300.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	100.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		180.0	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(130.23 \cdot 6.27)} + 165.0)^3) / 12$$

$$= 0.430 \text{ cm}^4$$



Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = 0.43 + 13.14(105.474 - 3.135)²
 = 1376.672 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	209.6	6.3	3.1	13.1	4119.5	0.138E+04
Wearplate	165.0	10.0	11.3	16.5	18595.5	0.147E+04
Web	10.0	227.5	130.0	22.8	295795.5	0.112E+04
BasePlate	100.0	16.0	251.8	16.0	402832.0	0.343E+04
Totals	...	...	...	68.4	721342.6	0.739E+04

Distance to Centroid [C1]:
 = AY / A
 = 721343/68.391
 = 105.474 mm

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120.0/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120.0) - 0.5•sin(120.0)²)/(π - 2.094 + sin(120.0)cos(120.0))
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.204 • 784.113
 = 159.5841 kgf

Tension Stress, St = (Fh/As) = 2.8884 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1391.2390 kgf/cm²

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 390.0 - 130.23 • sin(120.0/2)/1.0472
 = 282.301 mm

Bending Moment, M = Fh * d = 45.0507 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 6.4313 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1545.8213 kgf/cm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (2 • BasePlateLength • AllStress))
 = √(3(784 + 22)100.0/(2 • 300.0 • 1545.821))
 = 5.107 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120.0/2)(127.23 + 9.27 + 10.0)
 = 253.745 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 253.7454/(3 - 1)
 = 126.873 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 126.8727 • 100.0/2
 = 63.436 cm²



Bearing Pressure [Bp]:
= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
= 806.195/(300.0 • 100.0)
= 2.687 kgf/cm²

Axial Load [P]:
= Ap • Bp
= 63.4 • 2.69
= 170.474 kgf

Area of the Rib and Web [Ar]:
= Rib Area + Web Area
= 9.0 + 6.344
= 15.344 cm²

Compressive Stress [Sc]:
= P/Ar
= 170.5/15.3436
= 11.110 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	99.2
Web	63.4	10.0	5.0	6.3	3171.8	55.1
Totals	...	...	...	15.3	52671.8	154.

Rib dimension [D]:
= Saddle Width - Web Thickness
= 100.0 - 10.0
= 90.000 mm

Distance to Centroid from Datum [ytot]:
= AY / A
= 52671.812/15.344
= 34.328 mm

Distance to Centroid [C1]:
= max(ytot, Saddle Width - ytot)
= max(34.328, 100.0 - 34.328)
= 65.672 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{154.3/15.344}$
= 31.712 mm

Length of Outer Rib [L]:
= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 390.0 - cos(120.0/2)(127.23 + 9.27 + 10.0) - 16.0
= 300.750 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 1973952/2318.7}$
= 129.631

Slenderness ratio [KL/r]:
= KL/r
= 1 • 300.75/31.712
= 9.484

Bending Moment [Rm]:
= FI / (2 • Bplen) • e • L / 2
= 55.5/(2 • 300.0) • 126.873 • 300.75/2
= 1.764 kgf-m.



Compressive Allowable, $KL/r < Cc$ (9.4838 < 129.6307) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (9.48)^2 / (2 \cdot 129.63^2)) 2319 / (5/3 + 3 \cdot (9.48) / (8 \cdot 129.63) - (9.48^3) / (8 \cdot 129.63^3))$$

$$= 1365 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.11/1365.09 + (1.76 \cdot 65.672/1543021) / 1545.82$$

$$= 0.013$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	138.
Web	126.9	10.0	5.0	12.7	6343.6	55.7
Totals	...	...	...	21.7	55843.6	193.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 55843.629/21.687$$

$$= 25.749 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(25.749, \text{abs}(100.0 - 25.749))$$

$$= 74.251 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 390.0 - \cos(146.5/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 227.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ (193.4/21.687) }$$

$$= 29.865 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 227.5/29.865$$

$$= 7.618$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 55.491 / (2 \cdot 300.0)$$

$$= 0.092 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.092 \cdot 126.873 \cdot 227.5$$

$$= 2.669 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2.669/26051.607$$

$$= 10.247 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ (7.6176 < 129.6307) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.62)^2 / (2 \cdot 129.63^2)) 2319 / (5/3 + 3 \cdot (7.62) / (8 \cdot 129.63) - (7.62^3) / (8 \cdot 129.63^3))$$

$$= 1371 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 17.81/1370.73 + (2.67 \cdot 74.251/193.434) / 1545.82$$

$$= 0.020$$



ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	133.36	mm
Shell Thickness used in this Case	t	9.270	mm
Stiffened Vessel Length per 4.15.6	L	231.20	cm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	100.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	165.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1203.20	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	390.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		754.51	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-31.00	-1921.67
Long. Stress at Top of Midspan	31.00	1203.20
Long. Stress at Bottom of Midspan	8.56	1203.20
Long. Stress at Top of Saddles	190.80	1203.20
Long. Stress at Bottom of Saddles	-123.18	-1921.67
Long. Stress at Bottom of Saddles	123.18	1203.20
Tangential Shear in Shell	60.86	962.56
Circ. Stress at Horn of Saddle	30.59	1504.00
Circ. Compressive Stress in Shell	2.43	1203.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (197.4/2 + 0) \cdot 390.0/253.7455$
 $= 455.2 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(18.76, 0) \cdot 390.0/1420.0002$
 $= 5.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(55.49, 0) \cdot 390.0/1420.0002$
 $= 15.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (55/2 + 0) \cdot 390.0/253.7455$
 $= 127.9 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$



$$= 299 + \max(5, 455, 15, 128)$$

$$= 754.5 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.273 \cdot 1.2)^2)$$

$$= 183.985 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	776.59	kgf
Transverse Shear Load	98.72	kgf
Longitudinal Shear Load	55.49	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(254.46/2 + 9.27 + 10.0)\sin(120.0/2), 2 \cdot 133.365)$$

$$= 253.745 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -755 \cdot 49.0 [1 - (1 - 49.0/231.2 + (13.336^2 - 0.0^2)/(2 \cdot 49.0 \cdot 231.2))/(1 + (4 \cdot 0.0)/(3 \cdot 231.2))]$$

$$= -75.5 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 755 \cdot 231/4((1 + 2(13^2 - 0^2)/(231^2))/(1 + (4 \cdot 0)/(3 \cdot 231)) - 4 \cdot 49/231)$$

$$= 69.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= -1.055 \cdot 133.365/(2 \cdot 6.27) - 69.3/(\pi \cdot 133.4^2 \cdot 6.27)$$

$$= -31.00 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 133.365/(2 \cdot 6.27) + 69.3/(\pi \cdot 133.4^2 \cdot 6.27)$$

$$= 8.56 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= -1.055 \cdot 133.365/(2 \cdot 6.27) - 75.5/(0.1066 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= 190.80 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 133.365/(2 \cdot 6.27) + 75.5/(0.1923 \cdot \pi \cdot 133.4^2 \cdot 6.27)$$

$$= -123.18 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 755(231.2 - 2 \cdot 49.0)/(231.2 + (4 \cdot 0.0/3))$$

$$= 434.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 434.69 / (133.365 \cdot 6.27)$$



$$= 60.86 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(133.365 \cdot 6.27)}$$

$$= 22.555 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(100.0 + 1.56 \cdot \sqrt{(133.365 \cdot 6.27)}, 2 \cdot 49.0)$$

$$= 145.11 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(1407.131/1203.202, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 755 \cdot 0.1 / (145.111(6.27 + 1.0 \cdot 10.0))$$

$$= -2.43 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -755 / (4(6.27 + 1.0 \cdot 10.0)145.111) -$$

$$3 \cdot 0.053 \cdot 755 / (2(6.27 + 1.0 \cdot 10.0)^2)$$

$$= -30.59 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	300.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	100.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(130.23 \cdot 6.27)} + 165.0)6.27^3)/12$$

$$= 0.430 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 0.43 + 13.14(105.474 - 3.135)^2$$

$$= 1376.672 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	209.6	6.3	3.1	13.1	4119.5	0.138E+04
Wearplate	165.0	10.0	11.3	16.5	18595.5	0.147E+04
Web	10.0	227.5	130.0	22.8	295795.5	0.112E+04
BasePlate	100.0	16.0	251.8	16.0	402832.0	0.343E+04
Totals	...	...	...	68.4	721342.6	0.739E+04

Distance to Centroid [C1]:

$$= AY / A$$

$$= 721343/68.391$$

$$= 105.474 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$



$$= 180 - 120.0/2$$
$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$
$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$
$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$
$$= 0.204 \cdot 754.512$$
$$= 153.5596 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.7794 \text{ kgf/cm}^2$$
$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1391.2390 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$
$$= 390.0 - 130.23 \cdot \sin(120.0/2) / 1.0472$$
$$= 282.301 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 43.3500 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 6.1885 \text{ kgf/cm}^2$$
$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1545.8213 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}) }$$
$$= \sqrt{ 3(755 + 22)100.0 / (2 \cdot 300.0 \cdot 1545.821) }$$
$$= 5.012 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$
$$= 2 \cdot \cos(90 - 120.0/2) (127.23 + 9.27 + 10.0)$$
$$= 253.745 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$
$$= 253.7454 / (3 - 1)$$
$$= 126.873 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$
$$= 126.8727 \cdot 100.0/2$$
$$= 63.436 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$
$$= 776.594 / (300.0 \cdot 100.0)$$
$$= 2.589 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$
$$= 63.4 \cdot 2.59$$
$$= 164.214 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$
$$= 9.0 + 6.344$$
$$= 15.344 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$
$$= 164.2 / 15.3436$$
$$= 10.702 \text{ kgf/cm}^2$$

Check of Outside Ribs:



Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	99.2
Web	63.4	10.0	5.0	6.3	3171.8	55.1
Totals	...	...	...	15.3	52671.8	154.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 100.0 - 10.0$$

$$= 90.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 52671.812 / 15.344$$

$$= 34.328 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(34.328, 100.0 - 34.328)$$

$$= 65.672 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(154.3 / 15.344)}$$

$$= 31.712 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 390.0 - \cos(120.0 / 2) (127.23 + 9.27 + 10.0) - 16.0$$

$$= 300.750 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 1973952 / 2318.7)}$$

$$= 129.631$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 300.75 / 31.712$$

$$= 9.484$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 55.5 / (2 \cdot 300.0) \cdot 126.873 \cdot 300.75 / 2$$

$$= 1.764 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($9.4838 < 129.6307$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (9.48)^2 / (2 \cdot 129.63^2)) 2319 / (5/3 + 3 \cdot (9.48) / (8 \cdot 129.63) - (9.48^3) / (8 \cdot 129.63^3))$$

$$= 1365 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 10.7 / 1365.09 + (1.76 \cdot 65.672 / 1543021) / 1545.82$$

$$= 0.013$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	90.0	55.0	9.0	49500.0	138.
Web	126.9	10.0	5.0	12.7	6343.6	55.7
Totals	...	...	...	21.7	55843.6	193.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$



$$= 55843.629/21.687$$
$$= 25.749 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$
$$= \max(25.749, \text{abs}(100.0 - 25.749))$$
$$= 74.251 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$
$$= 390.0 - \cos(146.5/2) (16.0 + 0.0 + 0.0) - 0.0$$
$$= 227.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$
$$= \sqrt{ (193.4/21.687) }$$
$$= 29.865 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$
$$= 1 \cdot 227.5/29.865$$
$$= 7.618$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$
$$= 55.491/(2 \cdot 300.0)$$
$$= 0.092 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$
$$= 0.092 \cdot 126.873 \cdot 227.5$$
$$= 2.669 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$
$$= 2.669/26051.607$$
$$= 10.247 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.6176 < 129.6307$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$
$$= (1 - (7.62)^2 / (2 \cdot 129.63^2)) 2319 /$$
$$(5/3 + 3 \cdot (7.62) / (8 \cdot 129.63) - (7.62^3) / (8 \cdot 129.63^3))$$
$$= 1371 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$
$$= 17.15/1370.73 + (2.67 \cdot 74.251/193.434) / 1545.82$$
$$= 0.019$$

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: CH. IN Noz N3

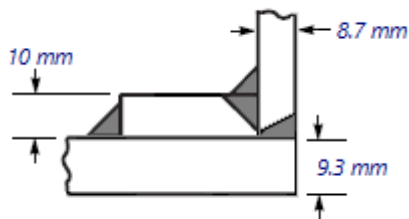
From: 20

Pressure for Reinforcement Calculations	P	10.221	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	1203.20	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1203.20	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	254.46	mm
Shell Finished (Minimum) Thickness	t	9.2700	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	65.0000	mm
Distance from Bottom/Left Tangent		11.60	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-61.56	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	203.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	3.9116	mm
Pad Material		SA-106 B	
Pad Allowable Stress at Temperature	Sp	1203.20	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1203.20	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: CH. IN Noz N3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.22 \cdot 130.23 / (1203 \cdot 1.0 - 0.6 \cdot 10.22)$
 $= 1.1120 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.22 \cdot 24.42 / (1203 \cdot 1.0 - 0.6 \cdot 10.22)$
 $= 0.2086 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.4229 mm
Parallel to Vessel Wall, opening length	d	57.2114 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.6750 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: CH. IN Noz N3.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.645, tr = 0.209, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.645, tr = 0.209, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 7.645, tr = 0.209, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 9.27$, $t_r = 1.112$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.177$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 7.645$, $t_r = 0.209$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $10.22/52.01 = 0.197$

Weld Size Calculations, Description: CH. IN Noz N3

Intermediate Calc. for nozzle/shell Welds T_{min} 5.7376 mm
Intermediate Calc. for pad/shell Welds T_{minPad} 6.2700 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.0163 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$3.1350 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: CH. IN Noz N3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.22 \cdot 130.23 / (1203 \cdot 1.0 - 0.6 \cdot 10.22)$
 $= 1.1120$ mm

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.22 \cdot 24.42 / (1203 \cdot 1.0 - 0.6 \cdot 10.22)$
 $= 0.2086$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	97.6996 mm
Parallel to Vessel Wall, opening length	d	48.8498 mm
Normal to Vessel Wall (Thickness Limit), pad side T_{lwp}		15.6750 mm



The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: CH. IN Noz N3.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.2086$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 4.5000$ mm
 Wall Thickness, shell/head, internal pressure $tr_{b1} = 4.1120$ mm
 Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 4.5000$ mm
 Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000$ mm
 Wall Thickness per table UG-45 $tb_3 = 6.4200$ mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.42, \max(4.5, 4.5)]$
 $= 4.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(3.2086, 4.5)$
 $= 4.5000$ mm

Available Nozzle Neck Thickness = $0.875 * 8.738 = 7.645$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	6.4,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	3001.6 kgf/cm ²	Passed
Occasional	:	19.5,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	65.8,	Allowable	:	842.2 kgf/cm ²	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
 the expansion stress allowable.*

Weld Size Calculations, Description: CH. IN Noz N3

Intermediate Calc. for nozzle/shell Welds	T_{min}	5.7376	mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	6.2700	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.0163 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$3.1350 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 24.9245 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 239.0816 mm

Warning:
*The equations for nozzle local stress analysis pertain to radial nozzles only
 and this nozzle is not specified as radial. Please check the specific
 analysis code for all assumptions and limitations.*

Input Echo, WRC297 Item 1, Description: CH. IN Noz N3 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	254.460	mm
Shell Thickness	9.2700	mm
Shell Stress Concentration Factor	1.000	



Vessel Material		SA-106 B	
Vessel Cold S.I. Allowable	Smc	1203.20	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	42.850	mm	
Nozzle Thickness	8.7376	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-106 B		
Nozzle Cold S.I. Allowable	SNmc	1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	160.325	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	10.22	kgf/cm ²
Radial Load	P	-132.57	kgf
Circumferential Shear	Vc	203.95	kgf
Longitudinal Shear	Vl	203.95	kgf
Circumferential Moment	Mc	0.03	kgf-m.
Longitudinal Moment	Ml	0.04	kgf-m.
Torsional Moment	Mt	0.05	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (6.904) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(138.365(9.27 - 3.0))}$$

$$= 157.524 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	276.730	mm
Nozzle Outside Diameter	(d) =	60.325	mm
Vessel Thickness used	(T) =	16.270	mm
Nozzle Thickness used	(t) =	5.738	mm
T / t	=	2.836	
d / t	=	10.514	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.899$$

Nr/P	=	0.156
Mr/P	=	0.125
M0/P	=	0.126
N0/P	=	0.184
MrD/Mc	=	0.187
NrDL/Mc	=	0.181
M0d/Mc	=	0.194
N0DL/Mc	=	0.074
MrD/Ml	=	0.154
NrDL/Ml	=	0.155
M0D/Ml	=	0.168



NØDL/M1 = 0.149

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	76	87	76	87
Outplane Stress Summary	123	58	124	58

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	83	8	84	8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	13	13	-13	-13
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	13	13	-13	-13

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	127	61	128	61

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	85	9	85	9

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	76	87	76	87

Inplane Stress Summary	122	57	122	57
------------------------	-----	----	-----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-13	-13	13	13
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-13	-13	13	13
----------------------	-----	-----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	126	61	126	60
------------------------	-----	----	-----	----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	46	58	46	58

Hoop Stress Summary	55	68	55	68
---------------------	----	----	----	----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	113	-113	113	-113
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	23	23	23	23

Axial Stress Summary	149	-76	151	-77
----------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	37	37	-37	-37
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	37	37	-37	-37
----------------------	----	----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 162 162 164 163

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction) Cu Cl Du D1
 Left Left Right Right
 Outside Inside Outside Inside

Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	46	58	46	58

Hoop Stress Summary 55 68 55 68

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section) Cu Cl Du D1
 Left Left Right Right
 Outside Inside Outside Inside

Axial Membrane (P)	13	13	13	13
Axial Bending (P)	113	-113	113	-113
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	23	23	23	23

Axial Stress Summary 149 -76 151 -77

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Shear stress) Cu Cl Du D1
 Outside Inside Outside Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-37	-37	37	37
Shear due to Torsion	0	0	0	0

Shear Stress Summary -37 -37 37 37

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities Cu Cl Du D1
 Outside Inside Outside Inside

Two * Max Shear Stress 162 162 164 164

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{133.365(9.27 - 3.0)}$$

$$= 155.751 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	266.730 mm
Nozzle Outside Diameter	(d) =	160.325 mm
Vessel Thickness used	(T) =	6.270 mm
Nozzle Thickness used	(t) =	55.738 mm
T / t	=	0.112
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.920$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.



Nr/P = 0.052
 Mr/P = 0.085
 M0/P = 0.025
 N0/P = 0.037
 MrD/Mc = 0.254
 NrDL/Mc = 0.102
 M0d/Mc = 0.076
 N0DL/Mc = 0.064
 MrD/Ml = 0.063
 NrDL/Ml = 0.060
 M0D/Ml = 0.019
 N0DL/Ml = 0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	207	217	207	217
Outplane Stress Summary	269	180	269	180

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	171	-171	171	-171
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	103	103	103	103
Inplane Stress Summary	292	-50	293	-50

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	12	12	-12	-12

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	298	232	298	232

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0

Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	103	103	103	103

Outplane Stress Summary	165	66	166	66
-------------------------	-----	----	-----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	171	-171	171	-171
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	207	217	207	217

Inplane Stress Summary	396	63	397	62
------------------------	-----	----	-----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-12	-12	12	12
----------------------	-----	-----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	396	78	398	77
------------------------	-----	----	-----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	76	87	76	87	76	87	76	87
Circ. Pl (SUS)	9	9	9	9	7	7	7	7
Circ. Q (SUS)	37	-37	37	-37	37	-37	37	-37
Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	7	7	7	7	9	9	9	9
Long. Q (SUS)	37	-37	37	-37	37	-37	37	-37
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	13	13	-13	-13	-13	-13	13	13
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	76.0	87.0	76.0	87.0	76.0	87.0	76.0	87.0
Pm+Pl (SUS)	88.9	99.1	88.9	99.1	87.2	97.4	87.2	97.4
Pm+Pl+Q (Total)	125.9	62.1	125.9	62.1	124.2	60.4	124.2	60.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of	Max. S.I.	S.I. Allowable	Result
---------	-----------	----------------	--------

Stress Int.			
Pm (SUS)	87.00	1203.20	Passed
Pm+Pl (SUS)	99.12	1804.80	Passed
Pm+Pl+Q (TOTAL)	125.85	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	207	217	207	217	207	217	207	217
Circ. Pl (SUS)	12	12	12	12	17	17	17	17
Circ. Q (SUS)	49	-49	49	-49	171	-171	171	-171
Long. Pm (SUS)	103	103	103	103	103	103	103	103
Long. Pl (SUS)	17	17	17	17	12	12	12	12
Long. Q (SUS)	171	-171	171	-171	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	207.0	217.0	207.0	217.0	207.0	217.0	207.0	217.0
Pm+Pl (SUS)	220.4	230.3	220.4	230.3	225.3	235.2	225.3	235.2
Pm+Pl+Q (Total)	296.1	232.2	296.1	232.2	395.6	76.6	395.6	76.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	217.00	1203.20	Passed
Pm+Pl (SUS)	235.20	1804.80	Passed
Pm+Pl+Q (TOTAL)	395.62	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	58	46	58	46	58	46	58
Circ. Pl (SUS)	9	9	9	9	9	9	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	13	13	13	13	13	13	13	13
Long. Q (SUS)	113	-113	113	-113	113	-113	113	-113
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	37	37	-37	-37	-37	-37	37	37
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	46.0	58.0	46.0	58.0	46.0	58.0	46.0	58.0
Pm+Pl (SUS)	83.7	91.6	83.7	91.6	83.7	91.6	83.7	91.6
Pm+Pl+Q (Total)	161.8	161.9	161.8	161.9	161.8	161.9	161.8	161.9



Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.00	1203.20	Passed
Pm+Pl (SUS)	91.62	1804.80	Passed
Pm+Pl+Q (TOTAL)	161.90	3609.61	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: CH. OUT Noz N4

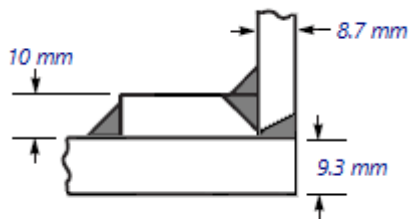
From: 20

Pressure for Reinforcement Calculations	P	10.199	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	1203.20	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1203.20	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	254.46	mm
Shell Finished (Minimum) Thickness	t	9.2700	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	65.0000	mm
Distance from Bottom/Left Tangent		11.60	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		61.56	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	203.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	3.9116	mm
Pad Material		SA-106 B	
Pad Allowable Stress at Temperature	Sp	1203.20	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1203.20	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: CH. OUT Noz N4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.2 \cdot 130.23 / (1203 \cdot 1.0 - 0.6 \cdot 10.2)$
 $= 1.1095 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.2 \cdot 24.42 / (1203 \cdot 1.0 - 0.6 \cdot 10.2)$
 $= 0.2081 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.4233 mm
Parallel to Vessel Wall, opening length	d	57.2117 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.6750 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: CH. OUT Noz N4.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.645, tr = 0.208, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.645, tr = 0.208, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 7.645, tr = 0.208, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9.27, tr = 1.11, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.177$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.645, tr = 0.208, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $10.20/52.01 = 0.196$

Weld Size Calculations, Description: CH. OUT Noz N4

Intermediate Calc. for nozzle/shell Welds Tmin 5.7376 mm
Intermediate Calc. for pad/shell Welds TminPad 6.2700 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.0163 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$3.1350 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: CH. OUT Noz N4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P * R / (S_v * E - 0.6 * P)$ per UG-27 (c)(1)
= $10.2 * 130.23 / (1203 * 1.0 - 0.6 * 10.2)$
= 1.1095 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P * R / (S_n * E - 0.6 * P)$ per UG-27 (c)(1)
= $10.2 * 24.42 / (1203 * 1.0 - 0.6 * 10.2)$
= 0.2081 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	97.6996 mm
Parallel to Vessel Wall, opening length	d	48.8498 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.6750 mm



The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: CH. OUT Noz N4.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.2081$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 4.5000$ mm
 Wall Thickness, shell/head, internal pressure $tr_{b1} = 4.1095$ mm
 Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 4.5000$ mm
 Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000$ mm
 Wall Thickness per table UG-45 $tb_3 = 6.4200$ mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.42, \max(4.5, 4.5)]$
 $= 4.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(3.2081, 4.5)$
 $= 4.5000$ mm

Available Nozzle Neck Thickness = $0.875 * 8.738 = 7.645$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	6.4,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	3001.6 kgf/cm ²	Passed
Occasional	:	19.4,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	65.8,	Allowable	:	842.2 kgf/cm ²	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
 the expansion stress allowable.*

Weld Size Calculations, Description: CH. OUT Noz N4

Intermediate Calc. for nozzle/shell Welds	T_{min}	5.7376	mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	6.2700	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.0163 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$3.1350 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 24.9245 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 239.0816 mm

Warning:

*The equations for nozzle local stress analysis pertain to radial nozzles only
 and this nozzle is not specified as radial. Please check the specific
 analysis code for all assumptions and limitations.*

Input Echo, WRC297 Item 1, Description: CH. OUT Noz N4:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	254.460	mm
Shell Thickness	9.2700	mm
Shell Stress Concentration Factor	1.000	



Vessel Material		SA-106 B	
Vessel Cold S.I. Allowable	Smc	1203.20	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	42.850	mm	
Nozzle Thickness	8.7376	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-106 B		
Nozzle Cold S.I. Allowable	SNmc	1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	160.325	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	10.20	kgf/cm ²
Radial Load	P	-132.57	kgf
Circumferential Shear	Vc	203.95	kgf
Longitudinal Shear	Vl	203.95	kgf
Circumferential Moment	Mc	0.03	kgf-m.
Longitudinal Moment	Ml	0.04	kgf-m.
Torsional Moment	Mt	0.05	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (6.904) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(138.365(9.27 - 3.0))}$$

$$= 157.524 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	276.730	mm
Nozzle Outside Diameter	(d) =	60.325	mm
Vessel Thickness used	(T) =	16.270	mm
Nozzle Thickness used	(t) =	5.738	mm
T / t	=	2.836	
d / t	=	10.514	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.899$$

Nr/P	=	0.156
Mr/P	=	0.125
M0/P	=	0.126
N0/P	=	0.184
MrD/Mc	=	0.187
NrDL/Mc	=	0.181
M0d/Mc	=	0.194
N0DL/Mc	=	0.074
MrD/Ml	=	0.154
NrDL/Ml	=	0.155
M0D/Ml	=	0.168



NØDL/M1 = 0.149

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	76	86	76	86
Outplane Stress Summary	123	58	123	58

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	83	8	83	8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	13	13	-13	-13
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	13	13	-13	-13

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	127	61	127	61

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	85	9	85	9

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	76	86	76	86

Inplane Stress Summary	121	57	122	57
------------------------	-----	----	-----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-13	-13	13	13
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-13	-13	13	13
----------------------	-----	-----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	126	60	126	60
------------------------	-----	----	-----	----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	46	58	46	58

Hoop Stress Summary	55	67	55	67
---------------------	----	----	----	----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	113	-113	113	-113
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	23	23	23	23

Axial Stress Summary	149	-76	151	-77
----------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	37	37	-37	-37
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	37	37	-37	-37
----------------------	----	----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	162	162	164	163

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	46	58	46	58
Hoop Stress Summary	55	67	55	67

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	113	-113	113	-113
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	23	23	23	23
Axial Stress Summary	149	-76	151	-77

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-37	-37	37	37
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-37	-37	37	37

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	162	162	164	163

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{133.365(9.27 - 3.0)}$$

$$= 155.751 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	266.730 mm
Nozzle Outside Diameter	(d) =	160.325 mm
Vessel Thickness used	(T) =	6.270 mm
Nozzle Thickness used	(t) =	55.738 mm
T / t	=	0.112
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.920$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.



Nr/P = 0.052
 Mr/P = 0.085
 MØ/P = 0.025
 NØ/P = 0.037
 MrD/Mc = 0.254
 NrDL/Mc = 0.102
 MØd/Mc = 0.076
 NØDL/Mc = 0.064
 MrD/Ml = 0.063
 NrDL/Ml = 0.060
 MØD/Ml = 0.019
 NØDL/Ml = 0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	206	217	206	217
Outplane Stress Summary	269	179	269	179

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	171	-171	171	-171
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	103	103	103	103
Inplane Stress Summary	292	-50	293	-51

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	12	12	-12	-12

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	298	232	298	232

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0



Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	103	103	103	103

Outplane Stress Summary	165	66	165	66
-------------------------	-----	----	-----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	171	-171	171	-171
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	206	217	206	217

Inplane Stress Summary	395	63	397	62
------------------------	-----	----	-----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-12	-12	12	12
----------------------	-----	-----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	396	77	397	77
------------------------	-----	----	-----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	76	86	76	86	76	86	76	86
Circ. Pl (SUS)	9	9	9	9	7	7	7	7
Circ. Q (SUS)	37	-37	37	-37	37	-37	37	-37
Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	7	7	7	7	9	9	9	9
Long. Q (SUS)	37	-37	37	-37	37	-37	37	-37
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	13	13	-13	-13	-13	-13	13	13
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	76.0	86.0	76.0	86.0	76.0	86.0	76.0	86.0
Pm+Pl (SUS)	88.9	98.2	88.9	98.2	87.2	96.4	87.2	96.4
Pm+Pl+Q (Total)	125.9	61.2	125.9	61.2	124.2	59.4	124.2	59.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of	Max. S.I.	S.I. Allowable	Result
---------	-----------	----------------	--------

Stress Int.			
Pm (SUS)	86.00	1203.20	Passed
Pm+Pl (SUS)	98.18	1804.80	Passed
Pm+Pl+Q (TOTAL)	125.85	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	206	217	206	217	206	217	206	217
Circ. Pl (SUS)	12	12	12	12	17	17	17	17
Circ. Q (SUS)	49	-49	49	-49	171	-171	171	-171
Long. Pm (SUS)	103	103	103	103	103	103	103	103
Long. Pl (SUS)	17	17	17	17	12	12	12	12
Long. Q (SUS)	171	-171	171	-171	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	206.0	217.0	206.0	217.0	206.0	217.0	206.0	217.0
Pm+Pl (SUS)	219.4	230.3	219.4	230.3	224.3	235.2	224.3	235.2
Pm+Pl+Q (Total)	296.0	232.2	296.0	232.2	394.6	76.6	394.6	76.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	217.00	1203.20	Passed
Pm+Pl (SUS)	235.20	1804.80	Passed
Pm+Pl+Q (TOTAL)	394.62	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	58	46	58	46	58	46	58
Circ. Pl (SUS)	9	9	9	9	9	9	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	13	13	13	13	13	13	13	13
Long. Q (SUS)	113	-113	113	-113	113	-113	113	-113
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	37	37	-37	-37	-37	-37	37	37
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	46.0	58.0	46.0	58.0	46.0	58.0	46.0	58.0
Pm+Pl (SUS)	83.7	91.6	83.7	91.6	83.7	91.6	83.7	91.6
Pm+Pl+Q (Total)	161.8	161.9	161.8	161.9	161.8	161.9	161.8	161.9



Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.00	1203.20	Passed
Pm+Pl (SUS)	91.62	1804.80	Passed
Pm+Pl+Q (TOTAL)	161.90	3609.61	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: SHELL IN Noz N1

From: 50

Pressure for Reinforcement Calculations	P	5.609	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	180	°C

Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	1203.20	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1203.20	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	254.46	mm
Design Length of Section	L	2312.0002	mm
Shell Finished (Minimum) Thickness	t	9.2700	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		248.85	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

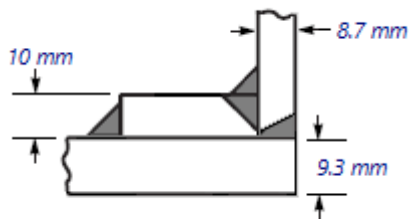
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	203.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	3.9116	mm

Pad Material		SA-106 B	
Pad Allowable Stress at Temperature	Sp	1203.20	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1203.20	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SHELL IN Noz N1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.61 \cdot 130.23 / (1203 \cdot 1.0 - 0.6 \cdot 5.61)$
 $= 0.6088$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.61 \cdot 24.42 / (1203 \cdot 1.0 - 0.6 \cdot 5.61)$
 $= 0.1142$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3231 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	97.6996 mm
Parallel to Vessel Wall, opening length	d	48.8498 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.6750 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: SHELL IN Noz N1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3231 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 3.6088 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 4.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(4.5, 4.5)]$
 $= 4.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3231, 4.5)$
 $= 4.5000$ mm

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**



Sustained :	-2.4,	Allowable :	1203.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3010.4 kgf/cm ²	Passed
Occasional :	10.7,	Allowable :	1600.3 kgf/cm ²	Passed
Shear :	65.8,	Allowable :	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.645, tr = 0.114, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.645, tr = 0.114, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 7.645, tr = 0.114, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9.27, tr = 0.609, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.645, tr = 0.114, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $5.61/104.01 = 0.054$

Weld Size Calculations, Description: SHELL IN Noz N1

Intermediate Calc. for nozzle/shell Welds	Tmin	5.7376	mm
Intermediate Calc. for pad/shell Welds	TminPad	6.2700	mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 4.0163 = $0.7 * t_{min}$ 7.0700 = $0.7 * W_o$ mm
 Pad Weld 3.1350 = $0.5 * T_{minPad}$ 5.6560 = $0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6270 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.3970 mm

Input Echo, WRC297 Item 1, Description: SHELL IN Noz N:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	254.460	mm
Shell Thickness	9.2700	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-106 B	
Vessel Cold S.I. Allowable	Smc	1203.20 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1203.20 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	42.850	mm
Nozzle Thickness	8.7376	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1203.20 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	160.325	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	5.61	kgf/cm ²
Radial Load	P	-132.57	kgf
Circumferential Shear	Vc	203.95	kgf
Longitudinal Shear	Vl	203.95	kgf
Circumferential Moment	Mc	0.03	kgf-m.
Longitudinal Moment	Ml	0.04	kgf-m.
Torsional Moment	Mt	0.05	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (6.904) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{mean}(t - ca)}$
 $= 60.325 + 2 \cdot 1.65 \cdot \sqrt{138.365(9.27 - 3.0)}$
 $= 157.524 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	276.730 mm
Nozzle Outside Diameter	(d) =	60.325 mm
Vessel Thickness used	(T) =	16.270 mm



Nozzle Thickness used (t) = 5.738 mm
 T / t = 2.836
 d / t = 10.514

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.899$$

Nr/P = 0.156
 Mr/P = 0.125
 M0/P = 0.126
 N0/P = 0.184
 MrD/Mc = 0.187
 NrDL/Mc = 0.181
 M0d/Mc = 0.194
 N0DL/Mc = 0.074
 MrD/M1 = 0.154
 NrDL/M1 = 0.155
 M0D/M1 = 0.168
 N0DL/M1 = 0.149

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	47	42	47
Outplane Stress Summary	88	19	89	18

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	21	21	21	21
Inplane Stress Summary	66	-8	66	-8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	13	13	-13	-13
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	13	13	-13	-13

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	94	38	95	38

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21
Outplane Stress Summary	67	-7	68	-7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	42	47	42	47
Inplane Stress Summary	87	18	87	17

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-13	-13	13	13
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-13	-13	13	13

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	94	36	94	36

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	25	32	25	32
Hoop Stress Summary	34	41	34	41

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	113	-113	113	-113
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0



Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	138	-86	141	-88

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	37	37	-37	-37
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	37	37	-37	-37

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	151	148	152	149

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	25	32	25	32
Hoop Stress Summary	34	41	34	41

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	113	-113	113	-113
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	138	-86	141	-88

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-37	-37	37	37
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-37	-37	37	37

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	150	148	153	149

Stress Computations at the Edge of the Reinforcing Pad:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(133.365(9.27 - 3.0))}$$

$$= 155.751 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	266.730 mm
Nozzle Outside Diameter	(d) =	160.325 mm
Vessel Thickness used	(T) =	6.270 mm
Nozzle Thickness used	(t) =	55.738 mm
T / t	=	0.112
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.920$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.052
Mr/P	=	0.085
M0/P	=	0.025
N0/P	=	0.037
MrD/Mc	=	0.254
NrDL/Mc	=	0.102
M0d/Mc	=	0.076
N0DL/Mc	=	0.064
MrD/M1	=	0.063
NrDL/M1	=	0.060
M0D/M1	=	0.019
N0DL/M1	=	0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	113	119	113	119
Outplane Stress Summary	175	82	176	82

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	171	-171	171	-171
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	246	-97	246	-97

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (V1)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0



Shear Stress Summary 12 12 -12 -12

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	248	181	248	181

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56

Outplane Stress Summary 118 19 119 19

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	171	-171	171	-171
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	113	119	113	119

Inplane Stress Summary 302 -34 303 -35

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary -12 -12 12 12

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	303	59	304	60

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	42	47	42	47	42	47	42	47
Circ. Pl (SUS)	9	9	9	9	7	7	7	7
Circ. Q (SUS)	37	-37	37	-37	37	-37	37	-37



Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	7	7	7	7	9	9	9	9
Long. Q (SUS)	37	-37	37	-37	37	-37	37	-37
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	13	13	-13	-13	-13	-13	13	13
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	42.0	47.0	42.0	47.0	42.0	47.0	42.0	47.0
Pm+Pl (SUS)	56.9	61.1	56.9	61.1	55.6	59.7	55.6	59.7
Pm+Pl+Q (Total)	93.9	38.2	93.9	38.2	92.6	35.4	92.6	35.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	47.00	1203.20	Passed
Pm+Pl (SUS)	61.10	1804.80	Passed
Pm+Pl+Q (TOTAL)	93.86	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	113	119	113	119	113	119	113	119
Circ. Pl (SUS)	12	12	12	12	17	17	17	17
Circ. Q (SUS)	49	-49	49	-49	171	-171	171	-171
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	17	17	17	17	12	12	12	12
Long. Q (SUS)	171	-171	171	-171	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	113.0	119.0	113.0	119.0	113.0	119.0	113.0	119.0
Pm+Pl (SUS)	127.6	133.4	127.6	133.4	132.2	138.1	132.2	138.1
Pm+Pl+Q (Total)	246.0	181.6	246.0	181.6	301.8	59.1	301.8	59.1

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119.00	1203.20	Passed
Pm+Pl (SUS)	138.06	1804.80	Passed
Pm+Pl+Q (TOTAL)	301.78	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	25	32	25	32	25	32	25	32
Circ. Pl (SUS)	9	9	9	9	9	9	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	13	13	13	13	13	13	13	13
Long. Q (SUS)	113	-113	113	-113	113	-113	113	-113
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	37	37	-37	-37	-37	-37	37	37
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	25.0	32.0	25.0	32.0	25.0	32.0	25.0	32.0
Pm+Pl (SUS)	74.5	75.7	74.5	75.7	74.5	75.7	74.5	75.7
Pm+Pl+Q (Total)	149.8	148.7	149.8	148.7	149.8	148.7	149.8	148.7

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	1203.20	Passed
Pm+Pl (SUS)	75.71	1804.80	Passed
Pm+Pl+Q (TOTAL)	149.82	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: SHELL OUT Noz N2

From: 50

Pressure for Reinforcement Calculations	P	5.634	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	180	°C

Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	1203.20	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1203.20	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	254.46	mm
Design Length of Section	L	2312.0002	mm
Shell Finished (Minimum) Thickness	t	9.2700	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		45.45	cm
-----------------------------------	--	-------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

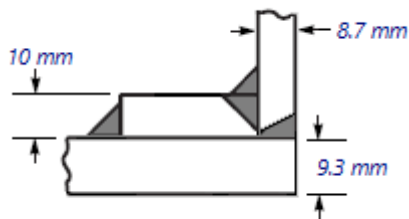
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	203.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	3.9116	mm

Pad Material		SA-106 B	
Pad Allowable Stress at Temperature	Sp	1203.20	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1203.20	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SHELL OUT Noz N2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.63 \cdot 130.23 / (1203 \cdot 1.0 - 0.6 \cdot 5.63)$
 $= 0.6115 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.63 \cdot 24.42 / (1203 \cdot 1.0 - 0.6 \cdot 5.63)$
 $= 0.1147 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3231 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	97.6996 mm
Parallel to Vessel Wall, opening length	d	48.8498 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.6750 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: SHELL OUT Noz N.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3231 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.6115 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(4.5, 4.5)]$
 $= 4.5000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3231, 4.5)$
 $= 4.5000 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645 \text{ mm}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):



Sustained :	-2.3,	Allowable :	1203.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3010.3 kgf/cm ²	Passed
Occasional :	10.7,	Allowable :	1600.3 kgf/cm ²	Passed
Shear :	65.8,	Allowable :	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.645, tr = 0.115, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.645, tr = 0.115, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 7.645, tr = 0.115, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9.27, tr = 0.612, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.098$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 7.645, tr = 0.115, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5.63/104.01 = 0.054$

Weld Size Calculations, Description: SHELL OUT Noz N

Intermediate Calc. for nozzle/shell Welds	Tmin	5.7376 mm
Intermediate Calc. for pad/shell Welds	TminPad	6.2700 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 4.0163 = $0.7 * t_{min}$ 7.0700 = $0.7 * W_o$ mm
 Pad Weld 3.1350 = $0.5 * T_{minPad}$ 5.6560 = $0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6270 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.3970 mm

Input Echo, WRC297 Item 1, Description: SHELL OUT Noz :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	254.460	mm
Shell Thickness	9.2700	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-106 B	
Vessel Cold S.I. Allowable	Smc	1203.20 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1203.20 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	42.850	mm
Nozzle Thickness	8.7376	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1203.20 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	160.325	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	5.63	kgf/cm ²
Radial Load	P	-132.57	kgf
Circumferential Shear	Vc	203.95	kgf
Longitudinal Shear	Vl	203.95	kgf
Circumferential Moment	Mc	0.03	kgf-m.
Longitudinal Moment	Ml	0.04	kgf-m.
Torsional Moment	Mt	0.05	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, D_n/T_n (6.904) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{mean}(t - ca)}$
 $= 60.325 + 2 \cdot 1.65 \cdot \sqrt{138.365(9.27 - 3.0)}$
 $= 157.524$ mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	276.730	mm
Nozzle Outside Diameter	(d) =	60.325	mm
Vessel Thickness used	(T) =	16.270	mm



Nozzle Thickness used (t) = 5.738 mm
 T / t = 2.836
 d / t = 10.514

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.899$$

Nr/P = 0.156
 Mr/P = 0.125
 M0/P = 0.126
 N0/P = 0.184
 MrD/Mc = 0.187
 NrDL/Mc = 0.181
 M0d/Mc = 0.194
 N0DL/Mc = 0.074
 MrD/Ml = 0.154
 NrDL/Ml = 0.155
 M0D/Ml = 0.168
 N0DL/Ml = 0.149

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	48	42	48
Outplane Stress Summary	89	19	89	19

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	21	21	21	21
Inplane Stress Summary	66	-8	66	-8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	13	13	-13	-13
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	13	13	-13	-13

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	95	38	95	38

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21
Outplane Stress Summary	67	-7	68	-7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	42	48	42	48
Inplane Stress Summary	87	18	87	18

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-13	-13	13	13
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-13	-13	13	13

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	94	36	94	36

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	25	32	25	32
Hoop Stress Summary	34	41	34	41

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	113	-113	113	-113
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0



Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	138	-86	141	-88

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	37	37	-37	-37
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	37	37	-37	-37

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	151	148	153	149

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	25	32	25	32
Hoop Stress Summary	34	41	34	41

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	113	-113	113	-113
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	139	-86	141	-88

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-37	-37	37	37
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-37	-37	37	37

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	151	148	153	149

Stress Computations at the Edge of the Reinforcing Pad:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(133.365(9.27 - 3.0))}$$

$$= 155.751 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 266.730 mm
 Nozzle Outside Diameter (d) = 160.325 mm
 Vessel Thickness used (T) = 6.270 mm
 Nozzle Thickness used (t) = 55.738 mm
 T / t = 0.112
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.920$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.052
 Mr/P = 0.085
 M0/P = 0.025
 N0/P = 0.037
 MrD/Mc = 0.254
 NrDL/Mc = 0.102
 M0d/Mc = 0.076
 N0DL/Mc = 0.064
 MrD/M1 = 0.063
 NrDL/M1 = 0.060
 M0D/M1 = 0.019
 N0DL/M1 = 0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	114	119	114	119
Outplane Stress Summary	176	82	176	82

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	171	-171	171	-171
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	246	-97	246	-97

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (V1)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0



Shear Stress Summary 12 12 -12 -12

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	248	181	249	181

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57

Outplane Stress Summary 119 20 119 19

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	171	-171	171	-171
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	114	119	114	119

Inplane Stress Summary 303 -33 304 -35

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary -12 -12 12 12

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	303	59	305	60

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	42	48	42	48	42	48	42	48
Circ. Pl (SUS)	9	9	9	9	7	7	7	7
Circ. Q (SUS)	37	-37	37	-37	37	-37	37	-37

Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	7	7	7	7	9	9	9	9
Long. Q (SUS)	37	-37	37	-37	37	-37	37	-37
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	13	13	-13	-13	-13	-13	13	13
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	42.0	48.0	42.0	48.0	42.0	48.0	42.0	48.0
Pm+Pl (SUS)	56.9	62.0	56.9	62.0	55.6	60.5	55.6	60.5
Pm+Pl+Q (Total)	93.9	38.9	93.9	38.9	92.6	36.1	92.6	36.1

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	48.00	1203.20	Passed
Pm+Pl (SUS)	61.97	1804.80	Passed
Pm+Pl+Q (TOTAL)	93.86	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	114	119	114	119	114	119	114	119
Circ. Pl (SUS)	12	12	12	12	17	17	17	17
Circ. Q (SUS)	49	-49	49	-49	171	-171	171	-171
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	17	17	17	17	12	12	12	12
Long. Q (SUS)	171	-171	171	-171	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	114.0	119.0	114.0	119.0	114.0	119.0	114.0	119.0
Pm+Pl (SUS)	128.6	133.4	128.6	133.4	133.2	138.1	133.2	138.1
Pm+Pl+Q (Total)	247.0	180.6	247.0	180.6	302.8	60.0	302.8	60.0

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119.00	1203.20	Passed
Pm+Pl (SUS)	138.08	1804.80	Passed
Pm+Pl+Q (TOTAL)	302.78	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	25	32	25	32	25	32	25	32
Circ. Pl (SUS)	9	9	9	9	9	9	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	13	13	13	13	13	13	13	13
Long. Q (SUS)	113	-113	113	-113	113	-113	113	-113
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	37	37	-37	-37	-37	-37	37	37
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	25.0	32.0	25.0	32.0	25.0	32.0	25.0	32.0
Pm+Pl (SUS)	74.5	75.7	74.5	75.7	74.5	75.7	74.5	75.7
Pm+Pl+Q (Total)	149.8	148.7	149.8	148.7	149.8	148.7	149.8	148.7

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	1203.20	Passed
Pm+Pl (SUS)	75.71	1804.80	Passed
Pm+Pl+Q (TOTAL)	149.82	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: Vent Noz V

From: 80

Pressure for Reinforcement Calculations	P	10.197	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	40	°C

Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	1203.20	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1203.20	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	254.46	mm
Design Length of Section	L	253.2050	mm
Shell Finished (Minimum) Thickness	t	9.2700	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		284.50	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

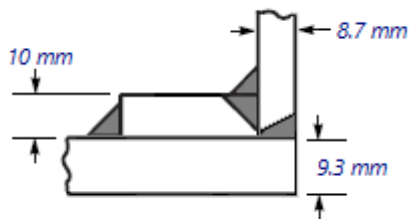
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	203.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	3.9116	mm

Pad Material		SA-106 B	
Pad Allowable Stress at Temperature	Sp	1203.20	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1203.20	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Vent Noz V

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.2 \cdot 130.23 / (1203 \cdot 1.0 - 0.6 \cdot 10.2)$
 $= 1.1094$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.2 \cdot 24.42 / (1203 \cdot 1.0 - 0.6 \cdot 10.2)$
 $= 0.2081$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3207 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	97.6996 mm
Parallel to Vessel Wall, opening length	d	48.8498 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.6750 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: Vent Noz V.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3207 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 4.1094 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 4.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(4.5, 4.5)]$
 $= 4.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3207, 4.5)$
 $= 4.5000$ mm

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)



Govrn. thk, tg = 7.645, tr = 0.208, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.645, tr = 0.208, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 7.645, tr = 0.208, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9.27, tr = 1.109, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.177$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.645, tr = 0.208, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $10.20/52.01 = 0.196$

Weld Size Calculations, Description: Vent Noz V

Intermediate Calc. for nozzle/shell Welds	Tmin	5.7376 mm
Intermediate Calc. for pad/shell Welds	TminPad	6.2700 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.0163 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$3.1350 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 3.6270 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.3970 mm



Input, Nozzle Desc: Drain Noz D

From: 80

Pressure for Reinforcement Calculations	P	10.223	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	40	°C

Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	1203.20	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1203.20	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	254.46	mm
Design Length of Section	L	253.2050	mm
Shell Finished (Minimum) Thickness	t	9.2700	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		284.50	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

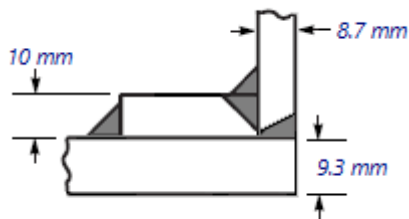
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	203.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	3.9116	mm

Pad Material		SA-106 B	
Pad Allowable Stress at Temperature	Sp	1203.20	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1203.20	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Drain Noz D

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.22 \cdot 130.23 / (1203 \cdot 1.0 - 0.6 \cdot 10.22)$
 $= 1.1122$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.22 \cdot 24.42 / (1203 \cdot 1.0 - 0.6 \cdot 10.22)$
 $= 0.2086$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3207 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	97.6996 mm
Parallel to Vessel Wall, opening length	d	48.8498 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.6750 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: Drain Noz D.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3207 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 4.1122 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 4.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(4.5, 4.5)]$
 $= 4.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3207, 4.5)$
 $= 4.5000$ mm

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)



Govrn. thk, tg = 7.645, tr = 0.209, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.645, tr = 0.209, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 7.645, tr = 0.209, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9.27, tr = 1.112, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.177$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.645, tr = 0.209, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $10.22/52.01 = 0.197$

Weld Size Calculations, Description: Drain Noz D

Intermediate Calc. for nozzle/shell Welds	Tmin	5.7376 mm
Intermediate Calc. for pad/shell Welds	TminPad	6.2700 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.0163 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$3.1350 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6270 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.3970 mm



Input Echo, Tubesheet Number 1, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Maximum Design Pressure	Psd,max	5.61 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	5.61 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05 kgf/cm ²
Shell Thickness	ts	9.2700 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	254.460 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	180.00 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	1203.20 kgf/cm ²
Shell Allowable Stress at Ambient		1203.20 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	10.20 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	10.20 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	9.2700 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	254.460 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	1203.20 kgf/cm ²
Channel Allowable Stress at Ambient		1203.20 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	52
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2394.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2314.00 mm
Design Temperature of the Tubes		180.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1586.61 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type	Full Strength	
Method for Tube-Tubesheet Jt. Allowable	UW-20.3	
Is the Tube Inset	No	

Radius to Outermost Tube Hole Center	ro	118.170 mm
Largest Center-to-Center Tube Distance	Ul	31.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:



Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 180.00 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03017
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 40.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 400.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
 Is the Tubesheet Flat on Tubeside? No
 Step 1 Diameter on the Tubeside dt1 260.46 mm
 Step 1 Depth on the Tubeside ht1 5.00 mm
 Is the Tubesheet Flat on Shellside? No
 Calculated Tubesheet Diameter per Div 2, 4.18.4 400.00 mm

Note: Tubesheet diameter is now: 400.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 168.1 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 419.000 mm
 Tube End condition corresponding to Span (l) k 1.00
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Mean Shell Metal Temp. along Shell len. Tsm 0.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 0.00 °C
 Junction Stress Reduction option Analyze as Simply Supported

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
 Flange Outside Diameter A 400.000 mm
 Flange Inside Diameter B 254.460 mm
 Flange Face Outside Diameter Fod 309.460 mm
 Flange Face Inside Diameter Fid 254.460 mm
 Gasket Outside Diameter Go 306.460 mm
 Gasket Inside Diameter Gi 260.460 mm
 Small end Hub thk. g0 9.2700 mm
 Large end Hub thk. g1 15.0000 mm
 Gasket Factor, m 0.00
 Gasket Design Seating Stress y 0.00 kgf/cm²
 Flange Facing Sketch Code Sketch 1b
 Gasket Thickness tg 4.5000 mm
 Full face Gasket Flange Option Program Selects
 Length of Partition Gasket lp 26.460 mm
 Width of Partition Gasket wp 6.0000 mm
 Partition Gasket Factor, mPart 3.7500
 Partition Gasket Design Seating Stress yPart 534.33 kgf/cm²
 Partition Gasket Facing Sketch Code Sketch 1a
 Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

Diameter of Bolt Circle C 345.000 mm
 Nominal Bolt Diameter dB 20.0000 mm
 Thread Series TEMA Metric Thread
 Number of Bolts n 16
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 1753.82 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1753.82 kgf/cm²

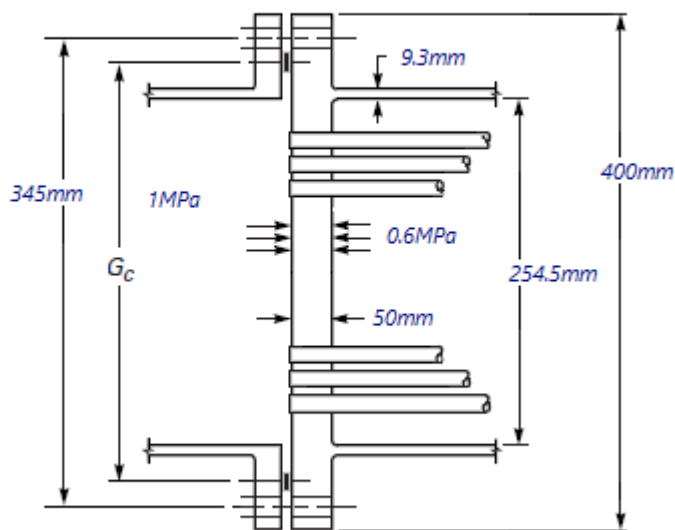


Weld between Flange and Shell/Channel 0.0000 mm

Note: The full bolt load $S_a \cdot A_b$, was used for all load cases.

Tubesheet Integral with Shell
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 35.0000 mm
 Is Bolt Load Transferred to the Tubesheet Yes
 Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20812E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20414E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20812E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with $C \leq 0.3\%$



Elastic Mod. at Design Temperature 180.0 °C 0.19739E+07 kgf/cm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(10.2 \cdot (19.05/2 - 0)) / (942.17 \cdot 1.0 + 0.4 \cdot 10.2)$
= $0.1026 + 0.0000 = 0.1026$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 180.00 °C
Elastic Modulus for Material 1999551.12 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2314.00	9.03	50.0000	0.0134948	961.53

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 150.6899$ kgf/cm²

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2979	19.05	2314.00	63.94	50.0000	0.0002690	268.98

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 268.9805) / (3 \cdot 63.9422) = 5.6088$ kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2979 mm
Allowable Internal Pressure at Corroded thickness	229.00 kgf/cm ²
Required Internal Design Pressure	10.20 kgf/cm ²
Allowable External Pressure at Corroded thickness	150.69 kgf/cm ²
Required External Design Pressure	5.61 kgf/cm ²
Required Thickness due to Shell Side pressure	0.2979 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	23.000 mm
Basic Gasket Width,	$b_0 = N / 2$	11.500 mm
Effective Gasket Width,	$b = \sqrt{b_0} \cdot 2.5$	8.542 mm
Gasket Reaction Diameter,	$G = G_o$ (Self-Energizing)	306.460 mm

Note:

The Tubesheet stress reduction option was Simply Supported. Performing
the 1st pass by including the stiffness of integral shell/channel to
ensure that the junction stress is < SPS.

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:
= l_{tx} / h (modified for corrosion if present)
= 0/40.0
= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d^*]:



$$\begin{aligned} &= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (\rho); dt - 2 \cdot tt) \\ &= \max(19.05 - 2 \cdot 2.11 (1973947/1973947) \cdot \\ &\quad (942/1407) \cdot (0); 19.05 - 2 \cdot 2.11) \\ &= 19.0500 \text{ mm} \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot ro + dt \\ &= 2 \cdot 118.17 + 19.05 \\ &= 255.390 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (25.0 - 19.05) / 25.0 \\ &= 0.238 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 255.39 / 2 \\ &= 127.6950 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 25.0 / \sqrt{ (1 - 4 \cdot \min(168.06 \cdot 100.0 ; 4 \cdot 255.39 \cdot 25.0) / (\pi \cdot 255.39^2)) } \\ &= 30.4985 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (30.4985 - 19.05) / 30.4985 \\ &= 0.3754 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= as / ao \\ &= 127.23 / 127.695 \\ &= 0.9964 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 153.23 / 127.695 \\ &= 1.2000 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\ &= 1 - 52 ((19.05 - 2 \cdot 2.11) / (2 \cdot 127.695))^2 \\ &= 0.8247 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot ao))^2 \\ &= 1 - 52 (19.05 / (2 \cdot 127.695))^2 \\ &= 0.7107 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max((hg - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((6.0 - 0); 0) \\ &= 6.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned} &= \pi \cdot ts(Ds + ts) Es / L \\ &= \pi \cdot 9.27 (254.46 + 9.27) 1973947 / 2314.0 \\ &= 65518.0195 \text{ kgf/mm} \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned} &= \pi \cdot tt(Dt - tt) Et / L \\ &= \pi \cdot 2.11 (19.05 - 2.11) 1973947 / 2314.0 \\ &= 957.8937 \text{ kgf/mm} \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned} &= Ks / (Nt \cdot Kt) \\ &= 65518.02 / (52 \cdot 957.894) \end{aligned}$$



$$= 1.31535$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s/K_j)$$

$$= 1 / (1 + 65518.02/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - v_s^2))^{1/4}) / ((D_s + t_s)t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((254.46 + 9.27)9.27)$$

$$= 0.0368 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$$

$$= 0.037 \cdot 1973947 \cdot 9.27^3 / (6(1 - 0.3^2))$$

$$= 10588103.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(254.46 + 9.27) \cdot 10588103 / 40.0^3 \cdot (1 + 40.0 \cdot 0.037 + (40.0 \cdot 0.037)^2 / 2)$$

$$= 929854.7500 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 254.46^2 / (4 \cdot 1973947 \cdot 9.27) \cdot (1 - 254.46 \cdot 0.3 / (2(254.46 + 9.27)))$$

$$= 0.75660\text{E-}03 \text{ mm/kgf/cm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 1.600000 ; \mu^* = 0.375378 ; E^* = 783107 \text{ kgf/cm}^2$$

$$E^*/E = 0.396721 ; \nu^* = 0.307135$$

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^{*2})Nt \cdot E_t \cdot t_t(dt - t_t)a_0^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.307^2)52 \cdot 1973947 \cdot 2.11$$

$$(19.05 - 2.11)127.695^2) / (783107 \cdot 2314.0 \cdot 40.0^3))^{1/4}$$

$$= 1.8299$$

Values from Table 4.18.3

$$Z_d = 0.229966 ; Z_v = 0.171315 ; Z_m = 0.704850$$

$$Z_a = 0.687159\text{E}+00 ; Z_w = 0.171315$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 400.0/255.39$$

$$= 1.5662$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.31) / 783107 \cdot (929854 + 0 +$$

$$1973947 \cdot \ln(1.57))$$

$$= 1.6063$$

Compute parameter [ϕ]:

$$= (1 + \nu^*) F$$

$$= (1 + 0.3071) 1.6063$$

$$= 2.0996$$

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (0.9964 - 1 - 2.0996 \cdot 0.1713) / (1 + 2.0996 \cdot 0.7048)$$

$$= -0.146513$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$



$$= (0.22997 + -0.14651 \cdot 0.17131) / 2 \cdot 1.82989^4$$

$$= 1.1485$$

Compute parameter [Qz2]:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.17131 + -0.14651 \cdot 0.70485) / 2 \cdot 1.82989^4$$

$$= 0.3815$$

Compute parameter [U]:

$$= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.1713 + (0.9964 - 1) \cdot 0.7048) \cdot 1.82989^4 / (1 + 2.0996 \cdot 0.7048)$$

$$= 0.7630$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= (G_c - C) / D_o \text{ (config b)}$$

$$= (306.46 - 345.0) / 255.39 = -0.15091$$

Compute Parameter [y]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 0.9964 \cdot 10588103 \cdot 0.0368 \cdot 0.000757 (1 + 40.0 \cdot 0.0368)$$

$$= 725.0055 \text{ mm}^2$$

Calculate parameter [w*s]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - w_s$$

$$= 127.695^2 (0.996^2 - 1) \cdot (0.996 - 1) / 4 - 725.005$$

$$= -724.8976 \text{ mm}^2$$

Parameter [wc]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.2 \cdot 0 [0 (0 + 0 \cdot 0 \cdot 40.0) -$$

$$0 (0/40.0 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [w*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - w_c$$

$$= 127.695^2 [(1.19997^2 + 1) \cdot (1.19997 - 1) / 4 - (0.99636 - 1) / 2] - 0$$

$$= 2018.6500 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s -$$

$$[(\rho_s^2 - 1) / (J \cdot K_{st})] - [(1 - J) / (2J \cdot K_{st})] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 5.609 \{ 0.711 + 2(1 - 0.711) \cdot 0.3 + [2/1.315 \cdot$$

$$(254.46/255.39)^2] \cdot 0.3 - [(0.996^2 - 1) / (1.0 \cdot 1.315)] -$$

$$[(1 - 1.0) / (2 \cdot 1.0 \cdot 1.315)] [(0^2 - (254.46)^2) / 255.39^2] \}$$

$$= 7.5303 \text{ kgf/cm}^2$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t$$

$$= [0.825 + 2(1 - 0.825) \cdot 0.3 + 1/(1 \cdot 1.315)] 10.197$$

$$= 17.2348 \text{ kgf/cm}^2$$

Pressure [P_v]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a o^2)$$

$$= 52 \cdot 957.894 \cdot 0 / (3.142 \cdot 127.695^2)$$

$$= 0.0000 \text{ kgf/cm}^2$$



Pressure [Pw]:

$$\begin{aligned} &= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2) \\ &= -0.151 \cdot 0.763 \cdot 60906.84 / (2 \cdot \pi \cdot 127.695^2) \\ &= 6.8445 \text{ kgf/cm}^2 \end{aligned}$$

Calculate the pressure [Prim]:

$$\begin{aligned} &= -(U/a_o^2) \cdot (\omega^*s \cdot P_s - \omega^*c \cdot P_t) \\ &= -(0.763/127.695^2) \cdot (-724.898 \cdot 5.609 - 2018.65 \cdot 10.197) \\ &= 1.1534 \text{ kgf/cm}^2 \end{aligned}$$

Calculate the pressure [Pw]:

$$\begin{aligned} &= (U/a_o^2) \cdot (\omega_s \cdot P^*s - \omega_c \cdot P^*c) \\ &= (0.763/127.695^2) \cdot (725.0055 \cdot 0 - 0 \cdot 0) \\ &= 0.0000 \text{ kgf/cm}^2 \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot \\ &\quad (P^*s - P^*t + P_\gamma + P_\omega + P_w + P_{prim}) \\ &= 0.1000E+01 \cdot 1.315 / (1 + 1.0 \cdot 1.315 (1.149 + (0.996 - \\ &\quad 1) 0.381)) \cdot (7.53 - 17.235 + 0 + 0 + 6.845 + 1.153) \\ &= -0.8948 \text{ kgf/cm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned} &= [((\omega^*s \cdot P_s - \omega^*c \cdot P_t) - (\omega_s \cdot P^*s - \omega_c \cdot P^*c)) + \\ &\quad W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\ &= [((-724.898 \cdot 5.609 - 2018.65 \cdot 10.197) - \\ &\quad (725.005 \cdot 0 - 0 \cdot 0)) \cdot 0.01 + \\ &\quad 60906.8 \cdot -0.151 / (2 \cdot \pi)] / (1 + 2.09965 \cdot 0.70485) \\ &= -689.2664 \text{ kgf} \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned} &= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2) \\ &= -0.147 + 2 \cdot -689.266 / (-0.895 \cdot 127.695^2) \\ &= 9.30196 \end{aligned}$$

Fm Value from Table 4.18.3 = 4.678805

Tube sheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe \\ &= (1.5 \cdot 4.6788 / 0.3754) \cdot (2 \cdot 127.695 / (40.0 - 6.0))^2 \cdot -0.89 \\ &= -943.8763 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tube sheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 1407.13 = 2110.7 \text{ kgf/cm}^2$$

Tube sheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot Pe \\ &= (1.5 \cdot 1.1644 / 0.3754) \cdot (2 \cdot 127.695 / (27.296 - 6.0))^2 \cdot 3.15 \\ &= 2110.6294 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tube sheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 27.2964 + 0 + 0 \\ &= 27.2964 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y) \\ &= (2 \cdot 0.238 \cdot 40.0 / 127.695) \cdot 1125.707 \\ &= 167.8489 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= τ_{limit}

Tube sheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned} \text{Tube sheet thickness (Incl. Corr.)} &= 11.4332 \text{ mm} \\ \text{Tube sheet Shear Stress} &= 168.1635 \text{ kgf/cm}^2 \end{aligned}$$



Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :
= max(HreqB; HreqS)
= max(27.2964 ; 11.4332)
= 27.2964 mm

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:
Ftmin = -7.9296, Ftmax = 10.2231

First Extreme Tube Axial Stress from among all the tubes [σ_1]:
= ((Ps • xs - Pt • xt) - Pe • Ftmin) / (Xt - Xs)
= ((5.61 • 0.7107 - 10.2 • 0.8247) - (-0.895) • -7.93) /
(0.8247 - 0.7107)
= -101.0530 kgf/cm²

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:
= ((Ps • xs - Pt • xt) - Pe • Ftmax) / (Xt - Xs)
= ((5.61 • 0.7107 - 10.2 • 0.8247) - (-0.895) • 10.223) /
(0.8247 - 0.7107)
= 41.4410 kgf/cm²

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
= max(| $\sigma_{t,1}$ | ; | $\sigma_{t,2}$ |) = 101.053 kgf/cm²

Allowable Tube Stress [$\sigma_{t,A}$]:
= Sot = 942.1682 kgf/cm²

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
= max((3.25 - 0.25(Zd + Q3 • Zw)Xa⁴); 1.25)
= max((3.25 - 0.25(0.23 + 9.302 • 0.171)1.83⁴); 1.25)
= 1.2500 (Should be ≤ 2)

Determine the factor [rt]:
= ($\sqrt{(dt^2 + (dt - 2 \cdot tt)^2)}$)/4
= ($\sqrt{(19.05^2 + (19.05 - 2 \cdot 2.11)^2)}$)/4
= 6.0355 mm

Determine the factor [Ct]:
= $\sqrt{(2 \cdot \pi^2 \cdot Et/Syt)}$
= $\sqrt{(2 \cdot \pi^2 \cdot 1973947/1586)}$
= 156.7104

Determine the factor [Ft]:
= k • L/r
= 1.0 • 419.0/6.035
= 69.4229

Buckling Allowable Stress [Stb]:
= (Sy,t/Fs) • (1 - Ft/(2 • Ct))
= (1586/1.25) • (1 - 69.423/(2 • 156.71))
= 942.168 kgf/cm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
= $\sigma_{t,max}$ • Tube Area
= 101.05 • 1.1229
= 113.47 kgf

Tube Metal Area [At]:
= $\pi(do - t)t$
= $\pi(19.05 - 2.11)2.11$
= 1.123 cm²

Tube Weld Size Results per UW-20.3:



Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19.05 - 2.11) 942.168$$

$$= 1057.971 \text{ kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 942.17$$

$$= 1057.9706 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 942.168 / 942.168$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 942.17$$

$$= 702.9219 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm}$$

$$\text{Minimum Required Fillet Weld Leg [afr]: } 1.2810 \text{ mm}$$

$$\text{Minimum Required Groove Weld Leg [agr]: } 1.2810 \text{ mm}$$

Tube-Tubesheet Jt allowable, 1057.97 is $\geq$ tube strength 1057.97 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 1203.2 \cdot 1.0 = 1203.2 \text{ kgf/cm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 127.695^2 / ((254.46 + 9.27) 9.27) \cdot [-0.89 + (0.996^2 - 1)(5.61 - 10.2)] + 127.23^2 \cdot 10.2 / ((254.46 + 9.27) 9.27)$$

$$= 61.7742 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 127.695^2 / ((254.46 + 9.27) 9.27) [-0.89 + (0.996^2 - 1)(5.61 - 10.2)] + 127.23^2 \cdot 10.2 / ((254.46 + 9.27) 9.27)$$

$$= 61.7742 \text{ kgf/cm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^*) / (E^*) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 10588103 / 9.27^2 \{ 0.037 [0.000757 \cdot 5.609 + 127.23^2 \cdot 0 / (18298492)] + 6(1 - 0.31^2) / (783107) (127.69 / 40.0)^3 (1 + 40.0 \cdot 0.04 / 2) [-0.9(0.171 + 0.705 \cdot -0.147) + 2 / 127.69^2 \cdot 0.705 \cdot -689.266] \}$$

$$= -1628.1039 \text{ kgf/cm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_{s,m}| + |\sigma_{s,b}| \leq SPS_s$$

$$\therefore |61.77| + |-1628.1| \leq 4270.59 \text{ kgf/cm}^2$$



:: 1689.88, must be $\leq$ 4270.59 kgf/cm²

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Intermediate Shell side parameters for the Simply Supported case:
betas, ks, lambdas, deltas are all = 0

Intermediate Channel side parameters for the Simply Supported case:
betaC, kc, lambdaC, deltaC are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.31) / 783107 \cdot (0 + 0 + 1973947 \cdot \ln(1.57)) \\ &= 0.7836 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu) F \\ &= (1 + 0.3071) \cdot 0.7836 \\ &= 1.0243 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (0.9964 - 1 - 1.0243 \cdot 0.1713) / (1 + 1.0243 \cdot 0.7048) \\ &= -0.104018 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot Xa^4 \\ &= (0.22997 + -0.10402 \cdot 0.17131) / 2 \cdot 1.82989^4 \\ &= 1.1893 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot Xa^4 \\ &= (0.17131 + -0.10402 \cdot 0.70485) / 2 \cdot 1.82989^4 \\ &= 0.5494 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (p_s - 1) Z_m) Xa^4 / (1 + \phi \cdot Z_m) \\ &= (0.1713 + (0.9964 - 1) \cdot 0.7048) \cdot 1.82989^4 / (1 + 1.0243 \cdot 0.7048) \\ &= 1.0988 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$\begin{aligned} &= (G_c - C) / D_o \text{ (config b)} \\ &= (306.46 - 345.0) / 255.39 = -0.15091 \end{aligned}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ &= 0.9964 \cdot 0 \cdot 0 \cdot 0 (1 + 40.0 \cdot 0) \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s \\ &= 127.695^2 (0.996^2 - 1) \cdot (0.996 - 1) / 4 - 0 \\ &= 0.1079 \text{ mm}^2 \end{aligned}$$



Parameter [wc]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.2 \cdot 0 [0 (0 + 0 \cdot 0 \cdot 40.0) - 0 (0/40.0 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 127.695^2 [(1.19997^2 + 1) \cdot (1.19997 - 1) / 4 - (0.99636 - 1) / 2] - 0$$

$$= 2018.6500 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Calculate the pressure [Prim]:

$$= - (U / a_o^2) \cdot (\omega * s \cdot P_s - \omega * c \cdot P_t)$$

$$= - (1.099 / 127.695^2) \cdot (0.108 \cdot 5.609 - 2018.65 \cdot 10.197)$$

$$= 1.3871 \text{ kgf/cm}^2$$

Calculate the pressure [Pω]:

$$= (U / a_o^2) \cdot (\omega_s \cdot P * s - \omega_c \cdot P * c)$$

$$= (1.099 / 127.695^2) \cdot (0 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (\rho_s - 1) Q_{z2})) \cdot (P'_s - P'_t + P_\gamma + P_\omega + P_w + P_{prim})$$

$$= 0.1000E+01 \cdot 1.315 / (1 + 1.0 \cdot 1.315 (1.189 + (0.996 - 1) 0.549)) \cdot (7.53 - 17.235 + 0 + 0 + 9.857 + 1.387)$$

$$= 0.7907 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega_s \cdot P * s - \omega_c \cdot P * c)) + W * \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((0.108 \cdot 5.609 - 2018.65 \cdot 10.197) - (0 \cdot 0 - 0 \cdot 0)) \cdot 0.01 + 60906.8 \cdot -0.151 / (2 \cdot \pi)] / (1 + 1.02427 \cdot 0.70485)$$

$$= -969.0588 \text{ kgf}$$

Calculate factor [Q3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$$

$$= -0.104 + 2 \cdot -969.059 / (0.791 \cdot 127.695^2)$$

$$= -15.13660$$

Fm Value from Table 4.18.3 = 7.568298

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 7.5683 / 0.3754) \cdot (2 \cdot 127.695 / (40.0 - 6.0))^2 \cdot 0.79$$

$$= 1349.1790 \text{ kgf/cm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 1407.13 = 2110.7 \text{ kgf/cm}^2$$

Tubesheet Bending Stress - final thickness [σf]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 1.6437 / 0.3754) \cdot (2 \cdot 127.695 / (33.847 - 6.0))^2 \cdot 3.82$$

$$= 2110.6965 \text{ kgf/cm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 33.8466 + 0 + 0$$

$$= 33.8466 \text{ mm}$$



ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y) \\ &= (2 \cdot 0.238 \cdot 40.0 / 127.695) \cdot 1125.707 \\ &= 167.8489 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; $[Pe] \leq \tau_{limit}$

Tubesheet Shear Stress - final thickness [τ_f]:

$$\begin{aligned} &= (1 / (2 \cdot \mu)) \cdot (a_o / h) \cdot Pe \\ &= (1 / (2 \cdot 0.238)) \cdot (127.695 / 10.827) \cdot 43.278 \\ &= 1072.2653 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [H_{reqS}]:

$$\begin{aligned} &= H + C_{ats} + C_{atc} \\ &= 10.8275 + 0 + 0 \\ &= 10.8275 \text{ mm} \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [H_{reqd}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(33.8466; 10.8275) \\ &= 33.8466 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -13.2483, F_{tmax} = 14.6504$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - Pe \cdot F_{tmin}) / (X_t - X_s) \\ &= ((5.61 \cdot 0.7107 - 10.2 \cdot 0.8247) - (0.791) \cdot -13.248) / \\ &\quad (0.8247 - 0.7107) \\ &= 53.0909 \text{ kgf/cm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - Pe \cdot F_{tmax}) / (X_t - X_s) \\ &= ((5.61 \cdot 0.7107 - 10.2 \cdot 0.8247) - (0.791) \cdot 14.65) / \\ &\quad (0.8247 - 0.7107) \\ &= -140.4321 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 140.432 \text{ kgf/cm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 942.1682 \text{ kgf/cm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [F_s]:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ &= \max((3.25 - 0.25(0.23 + -15.137 \cdot 0.171)1.83^4); 1.25) \\ &= 2.0000 \text{ (Should be } \leq 2) \end{aligned}$$

Determine the factor [τ_t]:

$$\begin{aligned} &= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4 \\ &= (\sqrt{19.05^2 + (19.05 - 2 \cdot 2.11)^2}) / 4 \\ &= 6.0355 \text{ mm} \end{aligned}$$

Determine the factor [C_t]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot Et / S_{yt})} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 1973947 / 1586)} \\ &= 156.7104 \end{aligned}$$

Determine the factor [F_t]:

$$\begin{aligned} &= k \cdot L / r \\ &= 1.0 \cdot 419.0 / 6.035 \\ &= 69.4229 \end{aligned}$$



Buckling Allowable Stress [Stb]:
= (Sy,t/Fs) • (1 - Ft/(2 • Ct))
= (1586/2.0) • (1 - 69.423/(2 • 156.71))
= 617.587 kgf/cm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
= σ_{t,max} • Tube Area
= 140.43 • 1.1229
= 157.69 kgf

Tube Metal Area [At]:
= π(do - t)t
= π(19.05 - 2.11)2.11
= 1.123 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
= π • t(do - t)S
= π • 2.11 (19.05 - 2.11)942.168
= 1057.971 kgf

Groove Weld Strength [Fg]:
= 0.85 • π • ag(do + 0.67•ag)Sw (but not > Ft)
= 0.85 • π • 2.11 (19.05 + 0.67•2.11)942.17
= 1057.9706 kgf

Weld Strength Factor [fw]:
= S / Sw where Sw = min(S and St)
= 942.168/942.168
= 1.000

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= 0.55 • π • af(do + 0.67 • af)Sw (but not > Ft)
= 0.55 • π • 2.11 (19.05 + 0.67 • 2.11)942.17
= 702.9219 kgf

Minimum Length of Weld [ar]:
= $2(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd} - 0.75 \cdot do)$
= $2(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11)1.0 \cdot 1.0} - 0.75 \cdot 19.05)$
= 2.562 mm

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm
Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1057.97 is >= tube strength 1057.97 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
= Ss • Esw = 1203.2 • 1.0 = 1203.2 kgf/cm²

Axial Membrane Stress in Shell [σs,m]:
= $ao^2 / ((Ds+ts)ts) \cdot [Pe + (ps^2-1)(Ps-Pt)] + as^2 \cdot Pt / ((Ds+ts)ts)$
= $127.695^2 / ((254.46 + 9.27)9.27) \cdot [0.79 + (0.996^2-1)$
 $(5.61 - 10.2)] + 127.23^2 \cdot 10.2 / ((254.46 + 9.27)9.27)$
= 73.0156 kgf/cm²

ASME Section VIII-2 Part 4.18.8.4 Step 11:



Shell Membrane Stress due to Joint Interaction [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1) (P_s - P_t)] + a_s^2 P_t / ((D_s + t_s) t_s)$
 $= 127.695^2 / ((254.46 + 9.27) 9.27) [0.79 + (0.996^2 - 1) (5.61 - 10.2)] + 127.23^2 \cdot 10.2 / ((254.46 + 9.27) 9.27)$
 $= 73.0156 \text{ kgf/cm}^2$

Shell Bending Stress due to Joint Interaction [σ_s, b]:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s P_s + a_s^2 P_s / (E_s t_s)] + 6(1 - \nu^2) / (E_s) (a_o / h)^3 (1 + h \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 0 / 9.27^2 \{ 0 [0 \cdot 5.609 + 127.23^2 \cdot 0 / (18298492)] + 6(1 - 0.31^2) / (783107) (127.69 / 40.0)^3 (1 + 40.0 \cdot 0 / 2) [0.8 (0.171 + 0.705 \cdot -0.104) + 2 / 127.69^2 \cdot 0.705 \cdot -969.059] \}$
 $= -0.0000 \text{ kgf/cm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_s, m| + |\sigma_s, b| \leq \text{SPS}, s$
 $:: |73.02| + |0| \leq 4270.59 \text{ kgf/cm}^2$
 $:: 73.02, \text{ must be } \leq 4270.59 \text{ kgf/cm}^2$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	1349.2 <=	2110.7 kgf/cm ²	Pass
Tubesheet Shear Stress	5.3 <=	1125.7 kgf/cm ²	Pass
Maximum Tube Stress	140.4 <=	942.2 kgf/cm ²	Pass
Minimum Tube Stress (Buckling)	-140.4 <=	-617.6 kgf/cm ²	Pass
Maximum Force on any one Tube	157.7 <=	1058.0 kgf	Pass
Axial Membrane Stress in Shell	73.0 <=	1203.2 kgf/cm ²	Pass
Shell Stress (jt. inter.)	73.0 <=	4270.6 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	33.847	40.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
			Bend	Allwd	Shear	Allwd		
D1uc	33.498	22.740	1332	2111	21	1126	Ps+Pt-Th	D1 Ok
D2uc	19.050	...	116	2111	26	1126	Ps+Pt-Th	D2 Ok
D3uc	33.847	...	1349	2111	5	1126	Ps+Pt-Th	D3 Ok
O1uc	5.931	...	669	4393	33	1126	Ps+Pt+Th	O1 Ok
O2uc	10.214	...	697	4393	51	1126	Ps+Pt+Th	O2 Ok
O3uc	8.779	...	682	4393	6	1126	Ps+Pt+Th	O3 Ok
O4uc	8.486	...	632	4393	24	1126	Ps+Pt+Th	O4 Ok
D1c	35.142	22.740	1468	2111	12	1126	Ps+Pt-Th-c	D1 Ok
D2c	9.463	...	117	2111	27	1126	Ps+Pt-Th-c	D2 Ok
D3c	35.407	...	1495	2111	15	1126	Ps+Pt-Th-c	D3 Ok
O1c	19.992	...	975	4393	23	1126	Ps+Pt+Th-c	O1 Ok
O2c	20.196	...	1002	4393	67	1126	Ps+Pt+Th-c	O2 Ok
O3c	20.497	...	1014	4393	5	1126	Ps+Pt+Th-c	O3 Ok
O4c	19.677	...	938	4393	38	1126	Ps+Pt+Th-c	O4 Ok
Max:	35.4072	22.740 mm		0.708		0.059 (Str. Ratio)		

Load Case Definitions:



[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	47	1203	...	...	:	...	...	...	...	Ok
D2uc	26	1203	...	...	:	...	...	...	...	Ok
D3uc	73	1203	...	...	:	...	...	...	...	Ok
O1uc	35	4270	...	...	:	...	...	...	...	Ok
O2uc	51	4270	...	...	:	...	...	...	...	Ok
O3uc	62	4270	...	...	:	...	...	...	...	Ok
O4uc	24	4270	...	...	:	...	...	...	...	Ok
D1c	83	1203	...	...	:	...	...	...	...	Ok
D2c	37	1203	...	...	:	...	...	...	...	Ok
D3c	120	1203	...	...	:	...	...	...	...	Ok
O1c	69	4270	...	...	:	...	...	...	...	Ok
O2c	88	4270	...	...	:	...	...	...	...	Ok
O3c	108	4270	...	...	:	...	...	...	...	Ok
O4c	49	4270	...	...	:	...	...	...	...	Ok
Max RATIO	0.100				:	...				...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	150	942	-150	-942	168	1057	47	4270	...	...	Ok
D2uc	9	942	-5	-617	10	1057	26	4270	...	...	Ok
D3uc	140	942	-140	-617	158	1057	73	4270	...	...	Ok
O1uc	106	1884	-106	-659	119	2115	1817	4270	...	...	Ok
O2uc	79	1884	-79	-617	89	2115	1303	4270	...	...	Ok
O3uc	101	1884	-101	-942	113	2115	1690	4270	...	...	Ok
O4uc	84	1884	-84	-617	94	2115	1430	4270	...	...	Ok
D1c	194	942	-194	-942	217	1057	83	4270	...	...	Ok
D2c	14	942	-5	-617	16	1057	37	4270	...	...	Ok
D3c	179	942	-179	-617	201	1057	120	4270	...	...	Ok
O1c	158	1884	-158	-942	178	2115	2417	4270	...	...	Ok
O2c	112	1884	-112	-617	126	2115	1700	4270	...	...	Ok
O3c	148	1884	-148	-617	166	2115	2212	4270	...	...	Ok
O4c	123	1884	-123	-617	138	2115	1905	4270	...	...	Ok
Max RATIO	0.205		0.290		0.205		0.566		...		...

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	35.407	40.000	Ok



Tubesheet thickness flanged extension :	22.740	35.000	Ok
Tube thickness :	0.298	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (9.270) adj. to tubesheet: 87.422 mm

For simply supported geometries that meet stress and thickness requirements, the above minimum length calculations are for information only and are not an ASME requirement. See VIII-2 sections 4.18.7.5, 4.18.8.8 and 4.18.9.7 for more information.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.008 mm
 Due to Pressure + Thermal Shell Compresses by : -0.008 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.716	1.000	101.248	1.000
Tubesheet Shear Stress	192.283	1.000	215.496	1.000
Tube Tensile Stress	155.652	1.000	368.466	1.000
Tube Compressive Stress	102.029	1.000	644.451	1.000
Tube-Tubesheet Joint load	155.652	1.000	368.466	1.000
Shell Stress (Axial, Junction)	47.099	1.000	84.793	1.000
Tube Pressure Stress	229.002	1.000	150.689	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAWP	44.716		84.793	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.741	1.000	101.808	1.000
Tubesheet Shear Stress	228.003	1.000	243.478	1.000
Tube Tensile Stress	208.623	1.000	577.337	1.000
Tube Compressive Stress	155.896	1.000	734.674	1.000
Tube-Tubesheet Joint load	208.623	1.000	577.338	1.000
Shell Stress (Axial, Junction)	88.318	1.000	151.842	1.000
Tube Pressure Stress	229.002	1.000	153.156	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAPnc	44.741		101.808	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0040	0.0040	...	...	...	...
2	-0.0078	-0.0078	...	...	...	...
3	-0.0038	-0.0038	...	...	...	...
4	...	...	...	...	...	...
5	0.0040	0.0040	...	...	...	...
6	-0.0078	-0.0078	...	...	...	...
7	-0.0038	-0.0038	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:



Governing thickness on the shell side per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, shell thickness))
= max(40.0/4, min(40.0, 9.27))
= 10.000 mm

Thickness Ratio = 0.228, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Channel Side MDMT calculation:

Governing thickness for the channel side:
= tubesheet thickness/4
= 40.0/4
= 10.000 mm

Thickness Ratio = 0.228, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
= max(10.2/44.74, 5.61/101.81)
= 0.228

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input Echo, Tubesheet Number 2, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Maximum Design Pressure	Psd,max	5.61 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	4.08 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05 kgf/cm ²
Shell Thickness	ts	9.2700 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	254.460 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	180.00 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	1203.20 kgf/cm ²
Shell Allowable Stress at Ambient		1203.20 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	10.20 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	4.59 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	9.2700 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	254.460 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	1203.20 kgf/cm ²
Channel Allowable Stress at Ambient		1203.20 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	52
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2394.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2314.00 mm
Design Temperature of the Tubes		180.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1586.61 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type	Full Strength	
Method for Tube-Tubesheet Jt. Allowable	UW-20.3	
Is the Tube Inset	No	

Radius to Outermost Tube Hole Center	ro	118.170 mm
Largest Center-to-Center Tube Distance	Ul	31.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:



Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 180.00 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03017
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 40.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 400.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
 Is the Tubesheet Flat on Tubeside? No
 Step 1 Diameter on the Tubeside dt1 260.46 mm
 Step 1 Depth on the Tubeside ht1 5.00 mm
 Is the Tubesheet Flat on Shellside? No
 Calculated Tubesheet Diameter per Div 2, 4.18.4 400.00 mm

Note: Tubesheet diameter is now: 400.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 168.1 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 419.000 mm
 Tube End condition corresponding to Span (l) k 1.00
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Mean Shell Metal Temp. along Shell len. Tsm 81.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 44.00 °C
 Junction Stress Reduction option Analyze as Simply Supported

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
 Flange Outside Diameter A 400.000 mm
 Flange Inside Diameter B 254.460 mm
 Flange Face Outside Diameter Fod 309.460 mm
 Flange Face Inside Diameter Fid 254.460 mm
 Gasket Outside Diameter Go 306.460 mm
 Gasket Inside Diameter Gi 260.460 mm
 Small end Hub thk. g0 9.2700 mm
 Large end Hub thk. g1 15.0000 mm
 Gasket Factor, m 0.00
 Gasket Design Seating Stress y 0.00 kgf/cm²
 Flange Facing Sketch Code Sketch 1b
 Gasket Thickness tg 4.5000 mm
 Full face Gasket Flange Option Program Selects
 Length of Partition Gasket lp 26.460 mm
 Width of Partition Gasket wp 6.0000 mm
 Partition Gasket Factor, mPart 3.7500
 Partition Gasket Design Seating Stress yPart 534.33 kgf/cm²
 Partition Gasket Facing Sketch Code Sketch 1a
 Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

Diameter of Bolt Circle C 345.000 mm
 Nominal Bolt Diameter dB 20.0000 mm
 Thread Series TEMA Metric Thread
 Number of Bolts n 16
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 1753.82 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1753.82 kgf/cm²

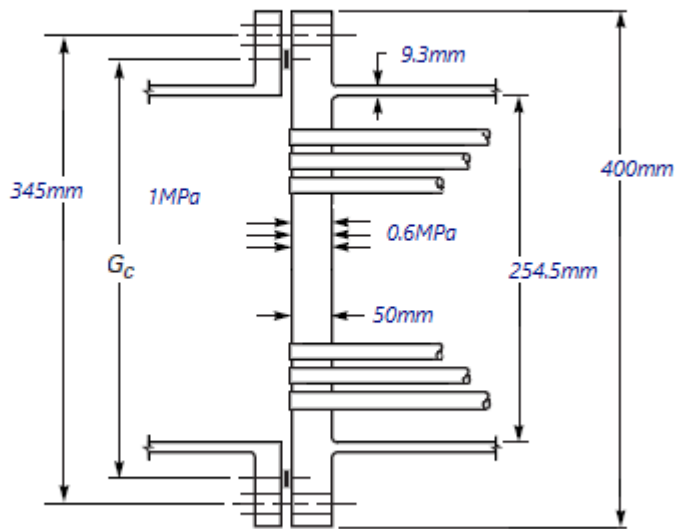


Weld between Flange and Shell/Channel 0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with Shell
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 35.0000 mm
 Is Bolt Load Transferred to the Tubesheet Yes
 Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	81.0 °C	0.0000119791 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	81.0 °C	0.20320E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20414E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	44.0 °C	0.0000117393 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	44.0 °C	0.20537E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with $C \leq 0.3\%$



Elastic Mod. at Design Temperature 180.0 °C 0.19739E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (10.2 \cdot (19.05/2 - 0)) / (942.17 \cdot 1.0 + 0.4 \cdot 10.2)$
 $= 0.1026 + 0.0000 = 0.1026 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 180.00 °C
 Elastic Modulus for Material 1999551.12 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2314.00	9.03	50.0000	0.0134948	961.53

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 150.6899 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2979	19.05	2314.00	63.94	50.0000	0.0002690	268.98

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 268.9805) / (3 \cdot 63.9422) = 5.6088 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2979 mm
Allowable Internal Pressure at Corroded thickness	229.00 kgf/cm ²
Required Internal Design Pressure	10.20 kgf/cm ²
Allowable External Pressure at Corroded thickness	150.69 kgf/cm ²
Required External Design Pressure	5.61 kgf/cm ²
Required Thickness due to Shell Side pressure	0.2979 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	33.498	22.740	1332	2111	21	1126	Ps+Pt-Th	D1 Ok
D2uc	19.050	...	116	2111	26	1126	Ps+Pt-Th	D2 Ok
D3uc	33.847	...	1349	2111	5	1126	Ps+Pt-Th	D3 Ok
O1uc	5.926	...	1518	4393	357	1126	Ps+Pt+Th	O1 Ok
O2uc	5.172	...	1369	4393	311	1126	Ps+Pt+Th	O2 Ok
O3uc	5.492	...	1472	4393	337	1126	Ps+Pt+Th	O3 Ok
O4uc	5.614	...	1415	4393	331	1126	Ps+Pt+Th	O4 Ok
D1c	35.142	22.740	1468	2111	12	1126	Ps+Pt-Th-c	D1 Ok
D2c	9.463	...	117	2111	27	1126	Ps+Pt-Th-c	D2 Ok
D3c	35.407	...	1495	2111	15	1126	Ps+Pt-Th-c	D3 Ok
O1c	15.920	...	1736	4393	313	1126	Ps+Pt+Th-c	O1 Ok
O2c	15.525	...	1548	4393	264	1126	Ps+Pt+Th-c	O2 Ok
O3c	16.138	...	1678	4393	292	1126	Ps+Pt+Th-c	O3 Ok
O4c	15.262	...	1606	4393	285	1126	Ps+Pt+Th-c	O4 Ok
Max:	35.4072	22.740 mm		0.708		0.317 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	47	1203	...	...		...	...	...	...			Ok
D2uc	26	1203	...	...		...	...	...	...			Ok
D3uc	73	1203	...	...		...	...	...	...			Ok
O1uc	324	4270	-324	-1194		...	...	...	...			Ok
O2uc	310	4270	-310	-1194		...	...	...	...			Ok
O3uc	305	4270	-305	-1194		...	...	...	...			Ok
O4uc	329	4270	-329	-1194		...	...	...	...			Ok
D1c	83	1203	...	...		...	...	...	...			Ok
D2c	37	1203	...	...		...	...	...	...			Ok
D3c	120	1203	...	...		...	...	...	...			Ok
O1c	359	4270	-359	-1157		...	...	...	...			Ok
O2c	339	4270	-339	-1157		...	...	...	...			Ok
O3c	330	4270	-330	-1157		...	...	...	...			Ok
O4c	367	4270	-367	-1157		...	...	...	...			Ok
Max RATIO 0.100 0.318 ...												

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	150	942	-150	-942	168	1057	47	4270	...	...	Ok
D2uc	9	942	-5	-617	10	1057	26	4270	...	...	Ok
D3uc	140	942	-140	-617	158	1057	73	4270	...	...	Ok
O1uc	567	1884	...	...	637	2115	2943	4270	...	...	Ok
O2uc	555	1884	...	...	623	2115	2647	4270	...	...	Ok
O3uc	564	1884	...	...	634	2115	2811	4270	...	...	Ok
O4uc	558	1884	...	...	627	2115	2778	4270	...	...	Ok
D1c	194	942	-194	-942	217	1057	83	4270	...	...	Ok
D2c	14	942	-5	-617	16	1057	37	4270	...	...	Ok
D3c	179	942	-179	-617	201	1057	120	4270	...	...	Ok
O1c	518	1884	...	...	582	2115	3678	4270	...	...	Ok
O2c	503	1884	...	...	565	2115	3259	4270	...	...	Ok
O3c	515	1884	...	...	579	2115	3472	4270	...	...	Ok
O4c	506	1884	...	...	568	2115	3465	4270	...	...	Ok
Max RATIO 0.301 0.290 0.301 0.861 ...											

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	35.407	40.000	Ok
Tubesheet thickness flanged extension :	22.740	35.000	Ok
Tube thickness :	0.298	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok



Min Shell length of thk, (9.270) adj. to tubesheet: 87.422 mm

For simply supported geometries that meet stress and thickness requirements, the above minimum length calculations are for information only and are not an ASME requirement. See VIII-2 sections 4.18.7.5, 4.18.8.8 and 4.18.9.7 for more information.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.040 mm
 Due to Pressure Shell Compresses by : -0.008 mm
 Due to Pressure + Thermal Shell Compresses by : -1.045 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.716	1.000	101.248	1.000
Tubesheet Shear Stress	138.920	1.000	233.738	1.000
Tube Tensile Stress	155.652	1.000	368.466	1.000
Tube Compressive Stress	98.466	0.332	644.451	1.000
Tube-Tubesheet Joint load	155.652	1.000	368.466	1.000
Shell Stress (Axial, Junction)	17.387	1.000	84.793	1.000
Tube Pressure Stress	229.002	1.000	150.689	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAWP	17.387		84.793	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.741	1.000	101.808	1.000
Tubesheet Shear Stress	228.003	1.000	243.478	1.000
Tube Tensile Stress	208.623	1.000	577.337	1.000
Tube Compressive Stress	155.896	1.000	734.674	1.000
Tube-Tubesheet Joint load	208.623	1.000	577.338	1.000
Shell Stress (Axial, Junction)	88.318	1.000	151.842	1.000
Tube Pressure Stress	229.002	1.000	153.156	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAPnc	44.741		101.808	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0040	0.0040	...	...	...	...
2	-0.0078	-0.0078	...	...	...	...
3	-0.0038	-0.0038	...	...	...	...
4	...	...	...	...	...	...
5	0.0018	0.0018	-1.0383	-1.0397	...	...
6	-0.0057	-0.0057	-1.0383	-1.0397	...	...
7	-0.0039	-0.0039	-1.0383	-1.0397	...	...
8	...	...	-1.0383	-1.0397	...	...



Input Echo, Tubesheet Number 3, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Maximum Design Pressure	Psd,max	5.61 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	4.08 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05 kgf/cm ²
Shell Thickness	ts	9.2700 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	254.460 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	180.00 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	1203.20 kgf/cm ²
Shell Allowable Stress at Ambient		1203.20 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	10.20 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	4.59 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	9.2700 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	254.460 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	1203.20 kgf/cm ²
Channel Allowable Stress at Ambient		1203.20 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	52
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2394.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2314.00 mm
Design Temperature of the Tubes		180.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1586.61 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type	Full Strength	
Method for Tube-Tubesheet Jt. Allowable	UW-20.3	
Is the Tube Inset	No	

Radius to Outermost Tube Hole Center	ro	118.170 mm
Largest Center-to-Center Tube Distance	Ul	31.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:



Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 180.00 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03017
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 40.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 400.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
 Is the Tubesheet Flat on Tubeside? No
 Step 1 Diameter on the Tubeside dt1 260.46 mm
 Step 1 Depth on the Tubeside ht1 5.00 mm
 Is the Tubesheet Flat on Shellside? No
 Calculated Tubesheet Diameter per Div 2, 4.18.4 400.00 mm

Note: Tubesheet diameter is now: 400.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 168.1 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 419.000 mm
 Tube End condition corresponding to Span (l) k 1.00
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Mean Shell Metal Temp. along Shell len. Tsm 79.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 44.00 °C
 Junction Stress Reduction option Analyze as Simply Supported

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
 Flange Outside Diameter A 400.000 mm
 Flange Inside Diameter B 254.460 mm
 Flange Face Outside Diameter Fod 309.460 mm
 Flange Face Inside Diameter Fid 254.460 mm
 Gasket Outside Diameter Go 306.460 mm
 Gasket Inside Diameter Gi 260.460 mm
 Small end Hub thk. g0 9.2700 mm
 Large end Hub thk. g1 15.0000 mm
 Gasket Factor, m 0.00
 Gasket Design Seating Stress y 0.00 kgf/cm²
 Flange Facing Sketch Code Sketch 1b
 Gasket Thickness tg 4.5000 mm
 Full face Gasket Flange Option Program Selects
 Length of Partition Gasket lp 26.460 mm
 Width of Partition Gasket wp 6.0000 mm
 Partition Gasket Factor, mPart 3.7500
 Partition Gasket Design Seating Stress yPart 534.33 kgf/cm²
 Partition Gasket Facing Sketch Code Sketch 1a
 Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

Diameter of Bolt Circle C 345.000 mm
 Nominal Bolt Diameter dB 20.0000 mm
 Thread Series TEMA Metric Thread
 Number of Bolts n 16
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 1753.82 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1753.82 kgf/cm²



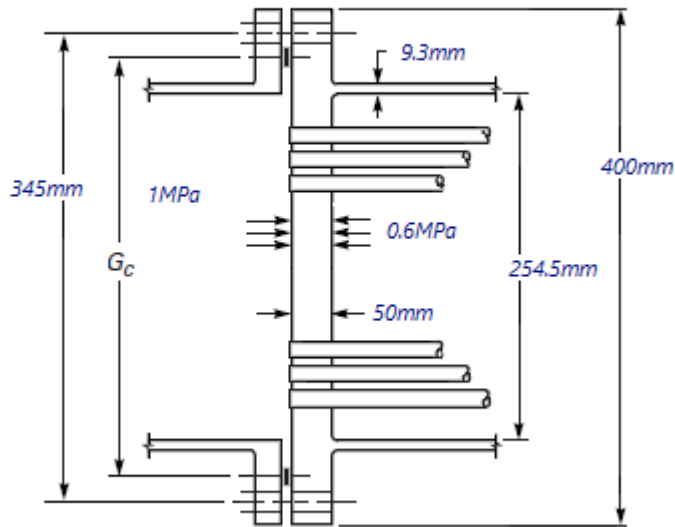
Weld between Flange and Shell/Channel

0.0000 mm

Note: The full bolt load $S_a \cdot A_b$, was used for all load cases.

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet Tf	35.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes
Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	79.0 °C	0.0000119661 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	79.0 °C	0.20332E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20414E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	44.0 °C	0.0000117393 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	44.0 °C	0.20537E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with $C \leq 0.3\%$



Elastic Mod. at Design Temperature 180.0 °C 0.19739E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(10.2 \cdot (19.05/2 - 0)) / (942.17 \cdot 1.0 + 0.4 \cdot 10.2)$

= 0.1026 + 0.0000 = 0.1026 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 180.00 °C
 Elastic Modulus for Material 1999551.12 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2314.00	9.03	50.0000	0.0134948	961.53

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 150.6899$ kgf/cm²

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2979	19.05	2314.00	63.94	50.0000	0.0002690	268.98

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 268.9805) / (3 \cdot 63.9422) = 5.6088$ kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2979 mm
Allowable Internal Pressure at Corroded thickness	229.00 kgf/cm ²
Required Internal Design Pressure	10.20 kgf/cm ²
Allowable External Pressure at Corroded thickness	150.69 kgf/cm ²
Required External Design Pressure	5.61 kgf/cm ²
Required Thickness due to Shell Side pressure	0.2979 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	33.498	22.740	1332	2111	21	1126	Ps+Pt-Th	D1 Ok
D2uc	19.050	...	116	2111	26	1126	Ps+Pt-Th	D2 Ok
D3uc	33.847	...	1349	2111	5	1126	Ps+Pt-Th	D3 Ok
O1uc	5.524	...	1463	4393	337	1126	Ps+Pt+Th	O1 Ok
O2uc	4.768	...	1314	4393	292	1126	Ps+Pt+Th	O2 Ok
O3uc	5.097	...	1417	4393	317	1126	Ps+Pt+Th	O3 Ok
O4uc	5.202	...	1360	4393	311	1126	Ps+Pt+Th	O4 Ok
D1c	35.142	22.740	1468	2111	12	1126	Ps+Pt-Th-c	D1 Ok
D2c	9.463	...	117	2111	27	1126	Ps+Pt-Th-c	D2 Ok
D3c	35.407	...	1495	2111	15	1126	Ps+Pt-Th-c	D3 Ok
O1c	16.009	...	1677	4393	295	1126	Ps+Pt+Th-c	O1 Ok
O2c	15.673	...	1489	4393	246	1126	Ps+Pt+Th-c	O2 Ok
O3c	16.231	...	1619	4393	274	1126	Ps+Pt+Th-c	O3 Ok
O4c	15.379	...	1547	4393	267	1126	Ps+Pt+Th-c	O4 Ok
Max:	35.4072	22.740 mm		0.708		0.299	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables



[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	47	1203	...	...		...	...	...	...			Ok
D2uc	26	1203	...	...		...	...	...	...			Ok
D3uc	73	1203	...	...		...	...	...	...			Ok
O1uc	305	4270	-305	-1194		...	...	...	...			Ok
O2uc	290	4270	-290	-1194		...	...	...	...			Ok
O3uc	285	4270	-285	-1194		...	...	...	...			Ok
O4uc	310	4270	-310	-1194		...	...	...	...			Ok
D1c	83	1203	...	...		...	...	...	...			Ok
D2c	37	1203	...	...		...	...	...	...			Ok
D3c	120	1203	...	...		...	...	...	...			Ok
O1c	336	4270	-336	-1157		...	...	...	...			Ok
O2c	316	4270	-316	-1157		...	...	...	...			Ok
O3c	307	4270	-307	-1157		...	...	...	...			Ok
O4c	344	4270	-344	-1157		...	...	...	...			Ok
Max RATIO 0.100 0.298 ...												

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	150	942	-150	-942	168	1057	47	4270	...	...	Ok
D2uc	9	942	-5	-617	10	1057	26	4270	...	...	Ok
D3uc	140	942	-140	-617	158	1057	73	4270	...	...	Ok
O1uc	538	1884	...	...	604	2115	2866	4270	...	...	Ok
O2uc	526	1884	...	...	590	2115	2570	4270	...	...	Ok
O3uc	535	1884	...	...	601	2115	2734	4270	...	...	Ok
O4uc	529	1884	...	...	594	2115	2702	4270	...	...	Ok
D1c	194	942	-194	-942	217	1057	83	4270	...	...	Ok
D2c	14	942	-5	-617	16	1057	37	4270	...	...	Ok
D3c	179	942	-179	-617	201	1057	120	4270	...	...	Ok
O1c	493	1884	...	...	553	2115	3586	4270	...	...	Ok
O2c	478	1884	...	...	536	2115	3168	4270	...	...	Ok
O3c	490	1884	...	...	550	2115	3381	4270	...	...	Ok
O4c	480	1884	...	...	539	2115	3374	4270	...	...	Ok
Max RATIO 0.286 0.290 0.286 0.840 ...											

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	35.407	40.000	Ok
Tubesheet thickness flanged extension :	22.740	35.000	Ok
Tube thickness :	0.298	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (9.270) adj. to tubesheet: 87.422 mm

For simply supported geometries that meet stress and thickness requirements, the above minimum length calculations are for information only and are not an ASME requirement. See VIII-2 sections 4.18.7.5, 4.18.8.8 and 4.18.9.7 for more information.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -0.982 mm
 Due to Pressure Shell Compresses by : -0.008 mm
 Due to Pressure + Thermal Shell Compresses by : -0.988 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.716	1.000	101.248	1.000
Tubesheet Shear Stress	141.858	1.000	233.738	1.000
Tube Tensile Stress	155.652	1.000	368.466	1.000
Tube Compressive Stress	100.543	0.367	644.451	1.000
Tube-Tubesheet Joint load	155.652	1.000	368.466	1.000
Shell Stress (Axial, Junction)	19.357	1.000	84.793	1.000
Tube Pressure Stress	229.002	1.000	150.689	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAWP	19.357		84.793	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.741	1.000	101.808	1.000
Tubesheet Shear Stress	228.003	1.000	243.478	1.000
Tube Tensile Stress	208.623	1.000	577.337	1.000
Tube Compressive Stress	155.896	1.000	734.674	1.000
Tube-Tubesheet Joint load	208.623	1.000	577.338	1.000
Shell Stress (Axial, Junction)	88.318	1.000	151.842	1.000
Tube Pressure Stress	229.002	1.000	153.156	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAPnc	44.741		101.808	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0040	0.0040	...	...	...	...
2	-0.0078	-0.0078	...	...	...	...
3	-0.0038	-0.0038	...	...	...	...
4	...	...	...	...	...	...
5	0.0018	0.0018	-0.9811	-0.9824	...	...
6	-0.0057	-0.0057	-0.9811	-0.9824	...	...
7	-0.0039	-0.0039	-0.9811	-0.9824	...	...
8	...	...	-0.9811	-0.9824	...	...



Input Echo, Tubesheet Number 4, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Maximum Design Pressure	Psd,max	5.61 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	4.08 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05 kgf/cm ²
Shell Thickness	ts	9.2700 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	254.460 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	180.00 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	1203.20 kgf/cm ²
Shell Allowable Stress at Ambient		1203.20 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	10.20 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	4.59 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	9.2700 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	254.460 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	1203.20 kgf/cm ²
Channel Allowable Stress at Ambient		1203.20 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	52
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2394.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2314.00 mm
Design Temperature of the Tubes		180.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1586.61 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type	Full Strength	
Method for Tube-Tubesheet Jt. Allowable	UW-20.3	
Is the Tube Inset	No	

Radius to Outermost Tube Hole Center	ro	118.170 mm
Largest Center-to-Center Tube Distance	Ul	31.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:



Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 180.00 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03017
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 40.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 400.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
 Is the Tubesheet Flat on Tubeside? No
 Step 1 Diameter on the Tubeside dt1 260.46 mm
 Step 1 Depth on the Tubeside ht1 5.00 mm
 Is the Tubesheet Flat on Shellside? No
 Calculated Tubesheet Diameter per Div 2, 4.18.4 400.00 mm

Note: Tubesheet diameter is now: 400.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 168.1 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 419.000 mm
 Tube End condition corresponding to Span (l) k 1.00
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Mean Shell Metal Temp. along Shell len. Tsm 69.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 35.00 °C
 Junction Stress Reduction option Analyze as Simply Supported

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
 Flange Outside Diameter A 400.000 mm
 Flange Inside Diameter B 254.460 mm
 Flange Face Outside Diameter Fod 309.460 mm
 Flange Face Inside Diameter Fid 254.460 mm
 Gasket Outside Diameter Go 306.460 mm
 Gasket Inside Diameter Gi 260.460 mm
 Small end Hub thk. g0 9.2700 mm
 Large end Hub thk. g1 15.0000 mm
 Gasket Factor, m 0.00
 Gasket Design Seating Stress y 0.00 kgf/cm²
 Flange Facing Sketch Code Sketch 1b
 Gasket Thickness tg 4.5000 mm
 Full face Gasket Flange Option Program Selects
 Length of Partition Gasket lp 26.460 mm
 Width of Partition Gasket wp 6.0000 mm
 Partition Gasket Factor, mPart 3.7500
 Partition Gasket Design Seating Stress yPart 534.33 kgf/cm²
 Partition Gasket Facing Sketch Code Sketch 1a
 Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

Diameter of Bolt Circle C 345.000 mm
 Nominal Bolt Diameter dB 20.0000 mm
 Thread Series TEMA Metric Thread
 Number of Bolts n 16
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 1753.82 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1753.82 kgf/cm²

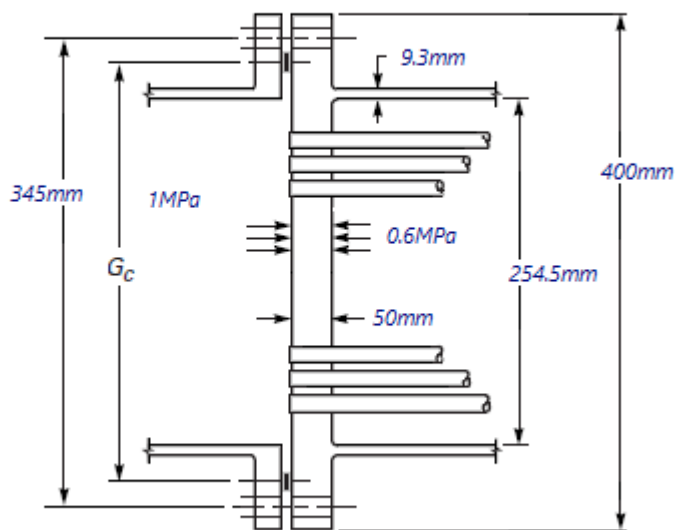


Weld between Flange and Shell/Channel 0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with Shell
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 35.0000 mm
 Is Bolt Load Transferred to the Tubesheet Yes
 Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	69.0 °C	0.0000119013 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	69.0 °C	0.20391E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20414E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	35.0 °C	0.0000116690 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	35.0 °C	0.20589E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with $C \leq 0.3\%$



Elastic Mod. at Design Temperature 180.0 °C 0.19739E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (10.2 \cdot (19.05/2 - 0)) / (942.17 \cdot 1.0 + 0.4 \cdot 10.2)$$

$$= 0.1026 + 0.0000 = 0.1026 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 180.00 °C
 Elastic Modulus for Material 1999551.12 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2314.00	9.03	50.0000	0.0134948	961.53

EMAWP = (2.167/(D/T) - 0.0833) * B = 150.6899 kgf/cm²

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2979	19.05	2314.00	63.94	50.0000	0.0002690	268.98

EMAWP = (4 * B) / (3 * (D/T)) = (4 * 268.9805) / (3 * 63.9422) = 5.6088 kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2979 mm
Allowable Internal Pressure at Corroded thickness	229.00 kgf/cm ²
Required Internal Design Pressure	10.20 kgf/cm ²
Allowable External Pressure at Corroded thickness	150.69 kgf/cm ²
Required External Design Pressure	5.61 kgf/cm ²
Required Thickness due to Shell Side pressure	0.2979 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	33.498	22.740	1332	2111	21	1126	Ps+Pt-Th	D1 Ok
D2uc	19.050	...	116	2111	26	1126	Ps+Pt-Th	D2 Ok
D3uc	33.847	...	1349	2111	5	1126	Ps+Pt-Th	D3 Ok
O1uc	5.263	...	1427	4393	324	1126	Ps+Pt+Th	O1 Ok
O2uc	4.508	...	1278	4393	279	1126	Ps+Pt+Th	O2 Ok
O3uc	4.841	...	1381	4393	305	1126	Ps+Pt+Th	O3 Ok
O4uc	4.937	...	1324	4393	299	1126	Ps+Pt+Th	O4 Ok
D1c	35.142	22.740	1468	2111	12	1126	Ps+Pt-Th-c	D1 Ok
D2c	9.463	...	117	2111	27	1126	Ps+Pt-Th-c	D2 Ok
D3c	35.407	...	1495	2111	15	1126	Ps+Pt-Th-c	D3 Ok
O1c	16.071	...	1638	4393	283	1126	Ps+Pt+Th-c	O1 Ok
O2c	15.772	...	1450	4393	235	1126	Ps+Pt+Th-c	O2 Ok
O3c	16.315	...	1580	4393	263	1126	Ps+Pt+Th-c	O3 Ok
O4c	15.477	...	1509	4393	255	1126	Ps+Pt+Th-c	O4 Ok
Max:	35.4072	22.740 mm		0.708		0.288 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables



[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	47	1203	...	...		...	...	...	...			Ok
D2uc	26	1203	...	...		...	...	...	...			Ok
D3uc	73	1203	...	...		...	...	...	...			Ok
O1uc	292	4270	-292	-1194		...	...	...	...			Ok
O2uc	278	4270	-278	-1194		...	...	...	...			Ok
O3uc	273	4270	-273	-1194		...	...	...	...			Ok
O4uc	297	4270	-297	-1194		...	...	...	...			Ok
D1c	83	1203	...	...		...	...	...	...			Ok
D2c	37	1203	...	...		...	...	...	...			Ok
D3c	120	1203	...	...		...	...	...	...			Ok
O1c	321	4270	-321	-1157		...	...	...	...			Ok
O2c	301	4270	-301	-1157		...	...	...	...			Ok
O3c	292	4270	-292	-1157		...	...	...	...			Ok
O4c	330	4270	-330	-1157		...	...	...	...			Ok
Max RATIO 0.100 0.285 ...												

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	150	942	-150	-942	168	1057	47	4270	...	...	Ok
D2uc	9	942	-5	-617	10	1057	26	4270	...	...	Ok
D3uc	140	942	-140	-617	158	1057	73	4270	...	...	Ok
O1uc	519	1884	...	...	583	2115	2816	4270	...	...	Ok
O2uc	507	1884	...	...	569	2115	2520	4270	...	...	Ok
O3uc	516	1884	...	...	579	2115	2684	4270	...	...	Ok
O4uc	510	1884	...	...	572	2115	2651	4270	...	...	Ok
D1c	194	942	-194	-942	217	1057	83	4270	...	...	Ok
D2c	14	942	-5	-617	16	1057	37	4270	...	...	Ok
D3c	179	942	-179	-617	201	1057	120	4270	...	...	Ok
O1c	476	1884	...	...	534	2115	3527	4270	...	...	Ok
O2c	461	1884	...	...	518	2115	3108	4270	...	...	Ok
O3c	473	1884	...	...	531	2115	3321	4270	...	...	Ok
O4c	464	1884	...	...	521	2115	3314	4270	...	...	Ok
Max RATIO 0.275 0.290 0.275 0.826 ...											

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	35.407	40.000	Ok
Tubesheet thickness flanged extension :	22.740	35.000	Ok
Tube thickness :	0.298	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok



Min Shell length of thk, (9.270) adj. to tubesheet: 87.422 mm

For simply supported geometries that meet stress and thickness requirements, the above minimum length calculations are for information only and are not an ASME requirement. See VIII-2 sections 4.18.7.5, 4.18.8.8 and 4.18.9.7 for more information.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -0.945 mm
 Due to Pressure Shell Compresses by : -0.008 mm
 Due to Pressure + Thermal Shell Compresses by : -0.951 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.716	1.000	101.248	1.000
Tubesheet Shear Stress	143.777	1.000	233.738	1.000
Tube Tensile Stress	155.652	1.000	368.466	1.000
Tube Compressive Stress	101.899	0.390	644.451	1.000
Tube-Tubesheet Joint load	155.652	1.000	368.466	1.000
Shell Stress (Axial, Junction)	20.644	1.000	84.793	1.000
Tube Pressure Stress	229.002	1.000	150.689	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAWP	20.644		84.793	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.741	1.000	101.808	1.000
Tubesheet Shear Stress	228.003	1.000	243.478	1.000
Tube Tensile Stress	208.623	1.000	577.337	1.000
Tube Compressive Stress	155.896	1.000	734.674	1.000
Tube-Tubesheet Joint load	208.623	1.000	577.338	1.000
Shell Stress (Axial, Junction)	88.318	1.000	151.842	1.000
Tube Pressure Stress	229.002	1.000	153.156	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAPnc	44.741		101.808	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0040	0.0040	...	...	...	...
2	-0.0078	-0.0078	...	...	...	...
3	-0.0038	-0.0038	...	...	...	...
4	...	...	...	...	...	...
5	0.0018	0.0018	-0.9438	-0.9450	...	...
6	-0.0057	-0.0057	-0.9438	-0.9450	...	...
7	-0.0039	-0.0039	-0.9438	-0.9450	...	...
8	...	...	-0.9438	-0.9450	...	...



Input Echo, Tubesheet Number 5, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Maximum Design Pressure	Psd,max	5.61 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	4.08 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05 kgf/cm ²
Shell Thickness	ts	9.2700 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	254.460 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	180.00 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	1203.20 kgf/cm ²
Shell Allowable Stress at Ambient		1203.20 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	10.20 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	5.10 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	9.2700 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	254.460 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	1203.20 kgf/cm ²
Channel Allowable Stress at Ambient		1203.20 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	52
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2394.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2314.00 mm
Design Temperature of the Tubes		180.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1586.61 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	118.170 mm
Largest Center-to-Center Tube Distance	Ul	31.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:



Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 180.00 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03017
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 40.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 400.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
 Is the Tubesheet Flat on Tubeside? No
 Step 1 Diameter on the Tubeside dt1 260.46 mm
 Step 1 Depth on the Tubeside ht1 5.00 mm
 Is the Tubesheet Flat on Shellside? No
 Calculated Tubesheet Diameter per Div 2, 4.18.4 400.00 mm

Note: Tubesheet diameter is now: 400.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 168.1 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 419.000 mm
 Tube End condition corresponding to Span (l) k 1.00
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Mean Shell Metal Temp. along Shell len. Tsm 36.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 36.00 °C
 Junction Stress Reduction option Analyze as Simply Supported

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
 Flange Outside Diameter A 400.000 mm
 Flange Inside Diameter B 254.460 mm
 Flange Face Outside Diameter Fod 309.460 mm
 Flange Face Inside Diameter Fid 254.460 mm
 Gasket Outside Diameter Go 306.460 mm
 Gasket Inside Diameter Gi 260.460 mm
 Small end Hub thk. g0 9.2700 mm
 Large end Hub thk. g1 15.0000 mm
 Gasket Factor, m 0.00
 Gasket Design Seating Stress y 0.00 kgf/cm²
 Flange Facing Sketch Code Sketch 1b
 Gasket Thickness tg 4.5000 mm
 Full face Gasket Flange Option Program Selects
 Length of Partition Gasket lp 26.460 mm
 Width of Partition Gasket wp 6.0000 mm
 Partition Gasket Factor, mPart 3.7500
 Partition Gasket Design Seating Stress yPart 534.33 kgf/cm²
 Partition Gasket Facing Sketch Code Sketch 1a
 Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

Diameter of Bolt Circle C 345.000 mm
 Nominal Bolt Diameter dB 20.0000 mm
 Thread Series TEMA Metric Thread
 Number of Bolts n 16
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 1753.82 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1753.82 kgf/cm²

0.0000 mm

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Diagram illustrating the geometry and dimensions of a vertical pipe section. The total height is 400mm. The distance from the top to the center of gravity (G_C) is 345mm. The distance from the center of gravity to the bottom is 254.5mm. The pipe has a diameter of 9.3mm. The internal pressure is 1MPa. The external pressure is 0.6MPa. The distance between the two pressure points is 50mm.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C ≤ 0,3%



Elastic Mod. at Design Temperature 180.0 °C 0.19739E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (10.2 \cdot (19.05/2 - 0)) / (942.17 \cdot 1.0 + 0.4 \cdot 10.2)$
 $= 0.1026 + 0.0000 = 0.1026 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 180.00 °C
 Elastic Modulus for Material 1999551.12 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2314.00	9.03	50.0000	0.0134948	961.53

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 150.6899 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2979	19.05	2314.00	63.94	50.0000	0.0002690	268.98

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 268.9805) / (3 \cdot 63.9422) = 5.6088 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2979 mm
Allowable Internal Pressure at Corroded thickness	229.00 kgf/cm ²
Required Internal Design Pressure	10.20 kgf/cm ²
Allowable External Pressure at Corroded thickness	150.69 kgf/cm ²
Required External Design Pressure	5.61 kgf/cm ²
Required Thickness due to Shell Side pressure	0.2979 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	33.498	22.740	1332	2111	21	1126	Ps+Pt-Th	D1 Ok
D2uc	19.050	...	116	2111	26	1126	Ps+Pt-Th	D2 Ok
D3uc	33.847	...	1349	2111	5	1126	Ps+Pt-Th	D3 Ok
O1uc	7.320	...	626	4393	5	1126	Ps+Pt+Th	O1 Ok
O2uc	9.794	...	679	4393	44	1126	Ps+Pt+Th	O2 Ok
O3uc	9.089	...	670	4393	15	1126	Ps+Pt+Th	O3 Ok
O4uc	8.486	...	632	4393	24	1126	Ps+Pt+Th	O4 Ok
D1c	35.142	22.740	1468	2111	12	1126	Ps+Pt-Th-c	D1 Ok
D2c	9.463	...	117	2111	27	1126	Ps+Pt-Th-c	D2 Ok
D3c	35.407	...	1495	2111	15	1126	Ps+Pt-Th-c	D3 Ok
O1c	19.836	...	944	4393	7	1126	Ps+Pt+Th-c	O1 Ok
O2c	20.056	...	984	4393	59	1126	Ps+Pt+Th-c	O2 Ok
O3c	20.210	...	991	4393	28	1126	Ps+Pt+Th-c	O3 Ok
O4c	19.677	...	938	4393	38	1126	Ps+Pt+Th-c	O4 Ok
Max:	35.4072	22.740 mm		0.708		0.052 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	47	1203	...	...		...	...	...	...	Ok
D2uc	26	1203	...	...		...	...	...	...	Ok
D3uc	73	1203	...	...		...	...	...	...	Ok
O1uc	29	4270	...	...		...	...	...	...	Ok
O2uc	43	4270	...	...		...	...	...	...	Ok
O3uc	49	4270	...	...		...	...	...	...	Ok
O4uc	24	4270	...	...		...	...	...	...	Ok
D1c	83	1203	...	...		...	...	...	...	Ok
D2c	37	1203	...	...		...	...	...	...	Ok
D3c	120	1203	...	...		...	...	...	...	Ok
O1c	59	4270	...	...		...	...	...	...	Ok
O2c	77	4270	...	...		...	...	...	...	Ok
O3c	87	4270	...	...		...	...	...	...	Ok
O4c	49	4270	...	...		...	...	...	...	Ok
Max RATIO 0.100 ...										

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	150	942	-150	-942	168	1057	47	4270	...	...	Ok
D2uc	9	942	-5	-617	10	1057	26	4270	...	...	Ok
D3uc	140	942	-140	-617	158	1057	73	4270	...	...	Ok
O1uc	95	1884	-95	-942	107	2115	1624	4270	...	...	Ok
O2uc	80	1884	-80	-617	90	2115	1338	4270	...	...	Ok
O3uc	91	1884	-91	-617	103	2115	1531	4270	...	...	Ok
O4uc	84	1884	-84	-617	94	2115	1430	4270	...	...	Ok
D1c	194	942	-194	-942	217	1057	83	4270	...	...	Ok
D2c	14	942	-5	-617	16	1057	37	4270	...	...	Ok
D3c	179	942	-179	-617	201	1057	120	4270	...	...	Ok
O1c	140	1884	-140	-617	158	2115	2161	4270	...	...	Ok
O2c	115	1884	-115	-617	129	2115	1756	4270	...	...	Ok
O3c	133	1884	-133	-617	149	2115	2012	4270	...	...	Ok
O4c	123	1884	-123	-617	138	2115	1905	4270	...	...	Ok
Max RATIO 0.205 0.290 0.205 0.506 ...											

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	35.407	40.000	Ok
Tubesheet thickness flanged extension :	22.740	35.000	Ok
Tube thickness :	0.298	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (9.270) adj. to tubesheet: 87.422 mm

For simply supported geometries that meet stress and thickness requirements, the above minimum length calculations are for information only and are not an ASME requirement. See VIII-2 sections 4.18.7.5, 4.18.8.8 and 4.18.9.7 for more information.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.008 mm
 Due to Pressure + Thermal Shell Compresses by : -0.008 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.716	1.000	101.248	1.000
Tubesheet Shear Stress	192.283	1.000	215.496	1.000
Tube Tensile Stress	155.652	1.000	368.466	1.000
Tube Compressive Stress	102.029	1.000	644.451	1.000
Tube-Tubesheet Joint load	155.652	1.000	368.466	1.000
Shell Stress (Axial, Junction)	47.099	1.000	84.793	1.000
Tube Pressure Stress	229.002	1.000	150.689	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAWP	44.716		84.793	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.741	1.000	101.808	1.000
Tubesheet Shear Stress	228.003	1.000	243.478	1.000
Tube Tensile Stress	208.623	1.000	577.337	1.000
Tube Compressive Stress	155.896	1.000	734.674	1.000
Tube-Tubesheet Joint load	208.623	1.000	577.338	1.000
Shell Stress (Axial, Junction)	88.318	1.000	151.842	1.000
Tube Pressure Stress	229.002	1.000	153.156	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAPnc	44.741		101.808	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0040	0.0040	...	...	...	...
2	-0.0078	-0.0078	...	...	...	...
3	-0.0038	-0.0038	...	...	...	...
4	...	...	...	...	...	...
5	0.0020	0.0020	...	...	...	...
6	-0.0057	-0.0057	...	...	...	...
7	-0.0037	-0.0037	...	...	...	...
8	...	...	...	...	...	...



Input Echo, Tubesheet Number 6, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Maximum Design Pressure	Psd,max	5.61 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	5.61 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05 kgf/cm ²
Shell Thickness	ts	9.2700 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	254.460 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	180.00 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	1203.20 kgf/cm ²
Shell Allowable Stress at Ambient		1203.20 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	10.20 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	10.20 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	9.2700 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	254.460 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	1203.20 kgf/cm ²
Channel Allowable Stress at Ambient		1203.20 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	52
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2394.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2314.00 mm
Design Temperature of the Tubes		180.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1586.61 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	118.170 mm
Largest Center-to-Center Tube Distance	Ul	31.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:



Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 180.00 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03017
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 40.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 400.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
 Is the Tubesheet Flat on Tubeside? No
 Step 1 Diameter on the Tubeside dt1 260.46 mm
 Step 1 Depth on the Tubeside ht1 5.00 mm
 Is the Tubesheet Flat on Shellside? No
 Calculated Tubesheet Diameter per Div 2, 4.18.4 400.00 mm

Note: Tubesheet diameter is now: 400.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 168.1 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 419.000 mm
 Tube End condition corresponding to Span (l) k 1.00
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Mean Shell Metal Temp. along Shell len. Tsm 81.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 44.00 °C
 Junction Stress Reduction option Analyze as Simply Supported

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
 Flange Outside Diameter A 400.000 mm
 Flange Inside Diameter B 254.460 mm
 Flange Face Outside Diameter Fod 309.460 mm
 Flange Face Inside Diameter Fid 254.460 mm
 Gasket Outside Diameter Go 306.460 mm
 Gasket Inside Diameter Gi 260.460 mm
 Small end Hub thk. g0 9.2700 mm
 Large end Hub thk. g1 15.0000 mm
 Gasket Factor, m 0.00
 Gasket Design Seating Stress y 0.00 kgf/cm²
 Flange Facing Sketch Code Sketch 1b
 Gasket Thickness tg 4.5000 mm
 Full face Gasket Flange Option Program Selects
 Length of Partition Gasket lp 26.460 mm
 Width of Partition Gasket wp 6.0000 mm
 Partition Gasket Factor, mPart 3.7500
 Partition Gasket Design Seating Stress yPart 534.33 kgf/cm²
 Partition Gasket Facing Sketch Code Sketch 1a
 Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

Diameter of Bolt Circle C 345.000 mm
 Nominal Bolt Diameter dB 20.0000 mm
 Thread Series TEMA Metric Thread
 Number of Bolts n 16
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 1753.82 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1753.82 kgf/cm²

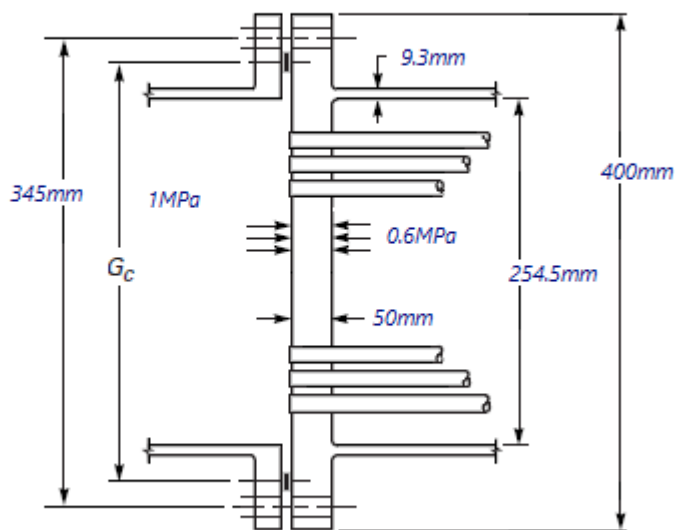


Weld between Flange and Shell/Channel 0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with Shell
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 35.0000 mm
 Is Bolt Load Transferred to the Tubesheet Yes
 Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	81.0 °C	0.0000119791 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	81.0 °C	0.20320E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20414E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	44.0 °C	0.0000117393 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	44.0 °C	0.20537E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with $C \leq 0.3\%$



Elastic Mod. at Design Temperature 180.0 °C 0.19739E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(10.2 \cdot (19.05/2 - 0)) / (942.17 \cdot 1.0 + 0.4 \cdot 10.2)$

= 0.1026 + 0.0000 = 0.1026 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 180.00 °C
 Elastic Modulus for Material 1999551.12 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2314.00	9.03	50.0000	0.0134948	961.53

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 150.6899$ kgf/cm²

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2979	19.05	2314.00	63.94	50.0000	0.0002690	268.98

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 268.9805) / (3 \cdot 63.9422) = 5.6088$ kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2979 mm
Allowable Internal Pressure at Corroded thickness	229.00 kgf/cm ²
Required Internal Design Pressure	10.20 kgf/cm ²
Allowable External Pressure at Corroded thickness	150.69 kgf/cm ²
Required External Design Pressure	5.61 kgf/cm ²
Required Thickness due to Shell Side pressure	0.2979 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	33.498	22.740	1332	2111	21	1126	Ps+Pt-Th	D1 Ok
D2uc	19.050	...	116	2111	26	1126	Ps+Pt-Th	D2 Ok
D3uc	33.847	...	1349	2111	5	1126	Ps+Pt-Th	D3 Ok
O1uc	6.289	...	1644	4393	388	1126	Ps+Pt+Th	O1 Ok
O2uc	5.009	...	1352	4393	304	1126	Ps+Pt+Th	O2 Ok
O3uc	5.706	...	1581	4393	361	1126	Ps+Pt+Th	O3 Ok
O4uc	5.614	...	1415	4393	331	1126	Ps+Pt+Th	O4 Ok
D1c	35.142	22.740	1468	2111	12	1126	Ps+Pt-Th-c	D1 Ok
D2c	9.463	...	117	2111	27	1126	Ps+Pt-Th-c	D2 Ok
D3c	35.407	...	1495	2111	15	1126	Ps+Pt-Th-c	D3 Ok
O1c	16.706	...	1894	4393	347	1126	Ps+Pt+Th-c	O1 Ok
O2c	15.635	...	1526	4393	256	1126	Ps+Pt+Th-c	O2 Ok
O3c	16.932	...	1814	4393	318	1126	Ps+Pt+Th-c	O3 Ok
O4c	15.262	...	1606	4393	285	1126	Ps+Pt+Th-c	O4 Ok
Max:	35.4072	22.740 mm		0.708		0.345	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	47	1203	...	...		...	...	...	...			Ok
D2uc	26	1203	...	...		...	...	...	...			Ok
D3uc	73	1203	...	...		...	...	...	...			Ok
O1uc	318	4270	-318	-1194		...	...	...	...			Ok
O2uc	302	4270	-302	-1194		...	...	...	...			Ok
O3uc	291	4270	-291	-1194		...	...	...	...			Ok
O4uc	329	4270	-329	-1194		...	...	...	...			Ok
D1c	83	1203	...	...		...	...	...	...			Ok
D2c	37	1203	...	...		...	...	...	...			Ok
D3c	120	1203	...	...		...	...	...	...			Ok
O1c	348	4270	-348	-1157		...	...	...	...			Ok
O2c	329	4270	-329	-1157		...	...	...	...			Ok
O3c	309	4270	-309	-1157		...	...	...	...			Ok
O4c	367	4270	-367	-1157		...	...	...	...			Ok
Max RATIO 0.100 0.318 ...												

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	150	942	-150	-942	168	1057	47	4270	...	...	Ok
D2uc	9	942	-5	-617	10	1057	26	4270	...	...	Ok
D3uc	140	942	-140	-617	158	1057	73	4270	...	...	Ok
O1uc	579	1884	...	...	650	2115	3143	4270	...	...	Ok
O2uc	554	1884	...	...	622	2115	2598	4270	...	...	Ok
O3uc	575	1884	...	...	645	2115	2963	4270	...	...	Ok
O4uc	558	1884	...	...	627	2115	2778	4270	...	...	Ok
D1c	194	942	-194	-942	217	1057	83	4270	...	...	Ok
D2c	14	942	-5	-617	16	1057	37	4270	...	...	Ok
D3c	179	942	-179	-617	201	1057	120	4270	...	...	Ok
O1c	533	1884	...	...	599	2115	3937	4270	...	...	Ok
O2c	502	1884	...	...	564	2115	3182	4270	...	...	Ok
O3c	529	1884	...	...	595	2115	3655	4270	...	...	Ok
O4c	506	1884	...	...	568	2115	3465	4270	...	...	Ok
Max RATIO 0.307 0.290 0.307 0.922 ...											

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	35.407	40.000	Ok
Tubesheet thickness flanged extension :	22.740	35.000	Ok
Tube thickness :	0.298	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok



Min Shell length of thk, (9.270) adj. to tubesheet: 87.422 mm

For simply supported geometries that meet stress and thickness requirements, the above minimum length calculations are for information only and are not an ASME requirement. See VIII-2 sections 4.18.7.5, 4.18.8.8 and 4.18.9.7 for more information.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.040 mm
 Due to Pressure Shell Compresses by : -0.008 mm
 Due to Pressure + Thermal Shell Compresses by : -1.047 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.716	1.000	101.248	1.000
Tubesheet Shear Stress	138.920	1.000	233.738	1.000
Tube Tensile Stress	155.652	1.000	368.466	1.000
Tube Compressive Stress	98.466	0.332	644.451	1.000
Tube-Tubesheet Joint load	155.652	1.000	368.466	1.000
Shell Stress (Axial, Junction)	17.387	1.000	84.793	1.000
Tube Pressure Stress	229.002	1.000	150.689	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAWP	17.387		84.793	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.741	1.000	101.808	1.000
Tubesheet Shear Stress	228.003	1.000	243.478	1.000
Tube Tensile Stress	208.623	1.000	577.337	1.000
Tube Compressive Stress	155.896	1.000	734.674	1.000
Tube-Tubesheet Joint load	208.623	1.000	577.338	1.000
Shell Stress (Axial, Junction)	88.318	1.000	151.842	1.000
Tube Pressure Stress	229.002	1.000	153.156	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAPnc	44.741		101.808	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0040	0.0040	...	...	...	...
2	-0.0078	-0.0078	...	...	...	...
3	-0.0038	-0.0038	...	...	...	...
4	...	...	...	...	...	...
5	0.0040	0.0040	-1.0383	-1.0397	...	...
6	-0.0078	-0.0078	-1.0383	-1.0397	...	...
7	-0.0038	-0.0038	-1.0383	-1.0397	...	...
8	...	...	-1.0383	-1.0397	...	...



Input Echo, Tubesheet Number 7, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Maximum Design Pressure	Psd,max	5.61 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	5.61 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05 kgf/cm ²
Shell Thickness	ts	9.2700 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	254.460 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	180.00 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	1203.20 kgf/cm ²
Shell Allowable Stress at Ambient		1203.20 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	10.20 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	5.10 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	9.2700 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	254.460 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	1203.20 kgf/cm ²
Channel Allowable Stress at Ambient		1203.20 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	52
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2394.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2314.00 mm
Design Temperature of the Tubes		180.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1586.61 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type	Full Strength	
Method for Tube-Tubesheet Jt. Allowable	UW-20.3	
Is the Tube Inset	No	

Radius to Outermost Tube Hole Center	ro	118.170 mm
Largest Center-to-Center Tube Distance	Ul	31.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:



Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 180.00 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03017
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 40.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 400.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
 Is the Tubesheet Flat on Tubeside? No
 Step 1 Diameter on the Tubeside dt1 260.46 mm
 Step 1 Depth on the Tubeside ht1 5.00 mm
 Is the Tubesheet Flat on Shellside? No
 Calculated Tubesheet Diameter per Div 2, 4.18.4 400.00 mm

Note: Tubesheet diameter is now: 400.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 168.1 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 419.000 mm
 Tube End condition corresponding to Span (l) k 1.00
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Mean Shell Metal Temp. along Shell len. Tsm 36.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 36.00 °C
 Junction Stress Reduction option Analyze as Simply Supported

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
 Flange Outside Diameter A 400.000 mm
 Flange Inside Diameter B 254.460 mm
 Flange Face Outside Diameter Fod 309.460 mm
 Flange Face Inside Diameter Fid 254.460 mm
 Gasket Outside Diameter Go 306.460 mm
 Gasket Inside Diameter Gi 260.460 mm
 Small end Hub thk. g0 9.2700 mm
 Large end Hub thk. g1 15.0000 mm
 Gasket Factor, m 0.00
 Gasket Design Seating Stress y 0.00 kgf/cm²
 Flange Facing Sketch Code Sketch 1b
 Gasket Thickness tg 4.5000 mm
 Full face Gasket Flange Option Program Selects
 Length of Partition Gasket lp 26.460 mm
 Width of Partition Gasket wp 6.0000 mm
 Partition Gasket Factor, mPart 3.7500
 Partition Gasket Design Seating Stress yPart 534.33 kgf/cm²
 Partition Gasket Facing Sketch Code Sketch 1a
 Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

Diameter of Bolt Circle C 345.000 mm
 Nominal Bolt Diameter dB 20.0000 mm
 Thread Series TEMA Metric Thread
 Number of Bolts n 16
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 1753.82 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1753.82 kgf/cm²

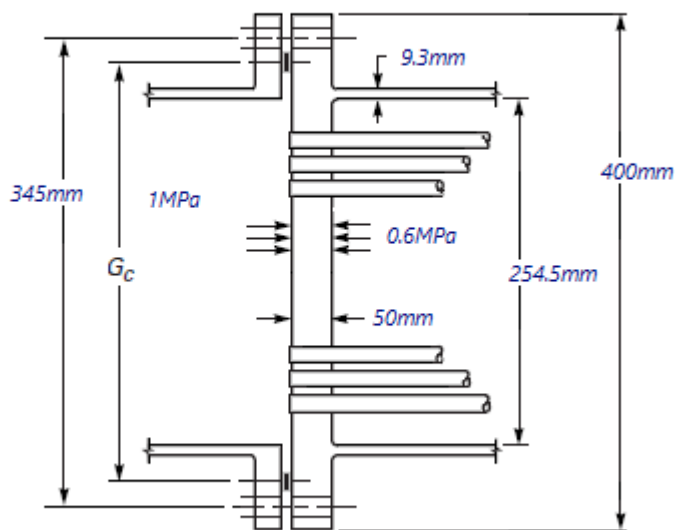


Weld between Flange and Shell/Channel 0.0000 mm

Note: The full bolt load $S_a \cdot A_b$, was used for all load cases.

Tubesheet Integral with Shell
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 35.0000 mm
 Is Bolt Load Transferred to the Tubesheet Yes
 Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	36.0 °C	0.0000116798 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	36.0 °C	0.20583E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20414E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	36.0 °C	0.0000116798 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	36.0 °C	0.20583E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with $C \leq 0.3\%$



Elastic Mod. at Design Temperature 180.0 °C 0.19739E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (10.2 \cdot (19.05/2 - 0)) / (942.17 \cdot 1.0 + 0.4 \cdot 10.2)$
 $= 0.1026 + 0.0000 = 0.1026 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 180.00 °C
 Elastic Modulus for Material 1999551.12 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2314.00	9.03	50.0000	0.0134948	961.53

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 150.6899 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2979	19.05	2314.00	63.94	50.0000	0.0002690	268.98

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 268.9805) / (3 \cdot 63.9422) = 5.6088 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2979 mm
Allowable Internal Pressure at Corroded thickness	229.00 kgf/cm ²
Required Internal Design Pressure	10.20 kgf/cm ²
Allowable External Pressure at Corroded thickness	150.69 kgf/cm ²
Required External Design Pressure	5.61 kgf/cm ²
Required Thickness due to Shell Side pressure	0.2979 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	33.498	22.740	1332	2111	21	1126	Ps+Pt-Th	D1 Ok
D2uc	19.050	...	116	2111	26	1126	Ps+Pt-Th	D2 Ok
D3uc	33.847	...	1349	2111	5	1126	Ps+Pt-Th	D3 Ok
O1uc	7.320	...	626	4393	5	1126	Ps+Pt+Th	O1 Ok
O2uc	10.214	...	697	4393	51	1126	Ps+Pt+Th	O2 Ok
O3uc	9.591	...	687	4393	23	1126	Ps+Pt+Th	O3 Ok
O4uc	8.486	...	632	4393	24	1126	Ps+Pt+Th	O4 Ok
D1c	35.142	22.740	1468	2111	12	1126	Ps+Pt-Th-c	D1 Ok
D2c	9.463	...	117	2111	27	1126	Ps+Pt-Th-c	D2 Ok
D3c	35.407	...	1495	2111	15	1126	Ps+Pt-Th-c	D3 Ok
O1c	19.836	...	944	4393	7	1126	Ps+Pt+Th-c	O1 Ok
O2c	20.196	...	1002	4393	67	1126	Ps+Pt+Th-c	O2 Ok
O3c	20.348	...	1008	4393	36	1126	Ps+Pt+Th-c	O3 Ok
O4c	19.677	...	938	4393	38	1126	Ps+Pt+Th-c	O4 Ok
Max:	35.4072	22.740 mm		0.708		0.059 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables



[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	47	1203	...	...		...	...	...	...	Ok
D2uc	26	1203	...	...		...	...	...	...	Ok
D3uc	73	1203	...	...		...	...	...	...	Ok
O1uc	29	4270	...	...		...	...	...	...	Ok
O2uc	51	4270	...	...		...	...	...	...	Ok
O3uc	56	4270	...	...		...	...	...	...	Ok
O4uc	24	4270	...	...		...	...	...	...	Ok
D1c	83	1203	...	...		...	...	...	...	Ok
D2c	37	1203	...	...		...	...	...	...	Ok
D3c	120	1203	...	...		...	...	...	...	Ok
O1c	59	4270	...	...		...	...	...	...	Ok
O2c	88	4270	...	...		...	...	...	...	Ok
O3c	98	4270	...	...		...	...	...	...	Ok
O4c	49	4270	...	...		...	...	...	...	Ok
Max RATIO	0.100		...			...		...		

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	150	942	-150	-942	168	1057	47	4270	...	...	Ok
D2uc	9	942	-5	-617	10	1057	26	4270	...	...	Ok
D3uc	140	942	-140	-617	158	1057	73	4270	...	...	Ok
O1uc	95	1884	-95	-942	107	2115	1624	4270	...	...	Ok
O2uc	79	1884	-79	-617	89	2115	1303	4270	...	...	Ok
O3uc	90	1884	-90	-617	101	2115	1496	4270	...	...	Ok
O4uc	84	1884	-84	-617	94	2115	1430	4270	...	...	Ok
D1c	194	942	-194	-942	217	1057	83	4270	...	...	Ok
D2c	14	942	-5	-617	16	1057	37	4270	...	...	Ok
D3c	179	942	-179	-617	201	1057	120	4270	...	...	Ok
O1c	140	1884	-140	-617	158	2115	2161	4270	...	...	Ok
O2c	112	1884	-112	-617	126	2115	1700	4270	...	...	Ok
O3c	130	1884	-130	-617	146	2115	1956	4270	...	...	Ok
O4c	123	1884	-123	-617	138	2115	1905	4270	...	...	Ok
Max RATIO	0.205		0.290		0.205		0.506		...		

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	35.407	40.000	Ok
Tubesheet thickness flanged extension :	22.740	35.000	Ok
Tube thickness :	0.298	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (9.270) adj. to tubesheet: 87.422 mm

For simply supported geometries that meet stress and thickness requirements, the above minimum length calculations are for information only and are not an ASME requirement. See VIII-2 sections 4.18.7.5, 4.18.8.8 and 4.18.9.7 for more information.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.008 mm
 Due to Pressure + Thermal Shell Compresses by : -0.008 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.716	1.000	101.248	1.000
Tubesheet Shear Stress	192.283	1.000	215.496	1.000
Tube Tensile Stress	155.652	1.000	368.466	1.000
Tube Compressive Stress	102.029	1.000	644.451	1.000
Tube-Tubesheet Joint load	155.652	1.000	368.466	1.000
Shell Stress (Axial, Junction)	47.099	1.000	84.793	1.000
Tube Pressure Stress	229.002	1.000	150.689	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAWP	44.716		84.793	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	44.741	1.000	101.808	1.000
Tubesheet Shear Stress	228.003	1.000	243.478	1.000
Tube Tensile Stress	208.623	1.000	577.337	1.000
Tube Compressive Stress	155.896	1.000	734.674	1.000
Tube-Tubesheet Joint load	208.623	1.000	577.338	1.000
Shell Stress (Axial, Junction)	88.318	1.000	151.842	1.000
Tube Pressure Stress	229.002	1.000	153.156	1.000
Tubesheet Extension Stress	81.909	...	No Calc	No Calc
Minimum MAPnc	44.741		101.808	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0040	0.0040	...	...	...	...
2	-0.0078	-0.0078	...	...	...	...
3	-0.0038	-0.0038	...	...	...	...
4	...	...	...	...	...	...
5	0.0020	0.0020	...	...	...	...
6	-0.0078	-0.0078	...	...	...	...
7	-0.0058	-0.0058	...	...	...	...
8	...	...	...	...	...	...